



UNIVERSITÀ DEGLI STUDI DI SALERNO
DIPARTIMENTO DI INFORMATICA

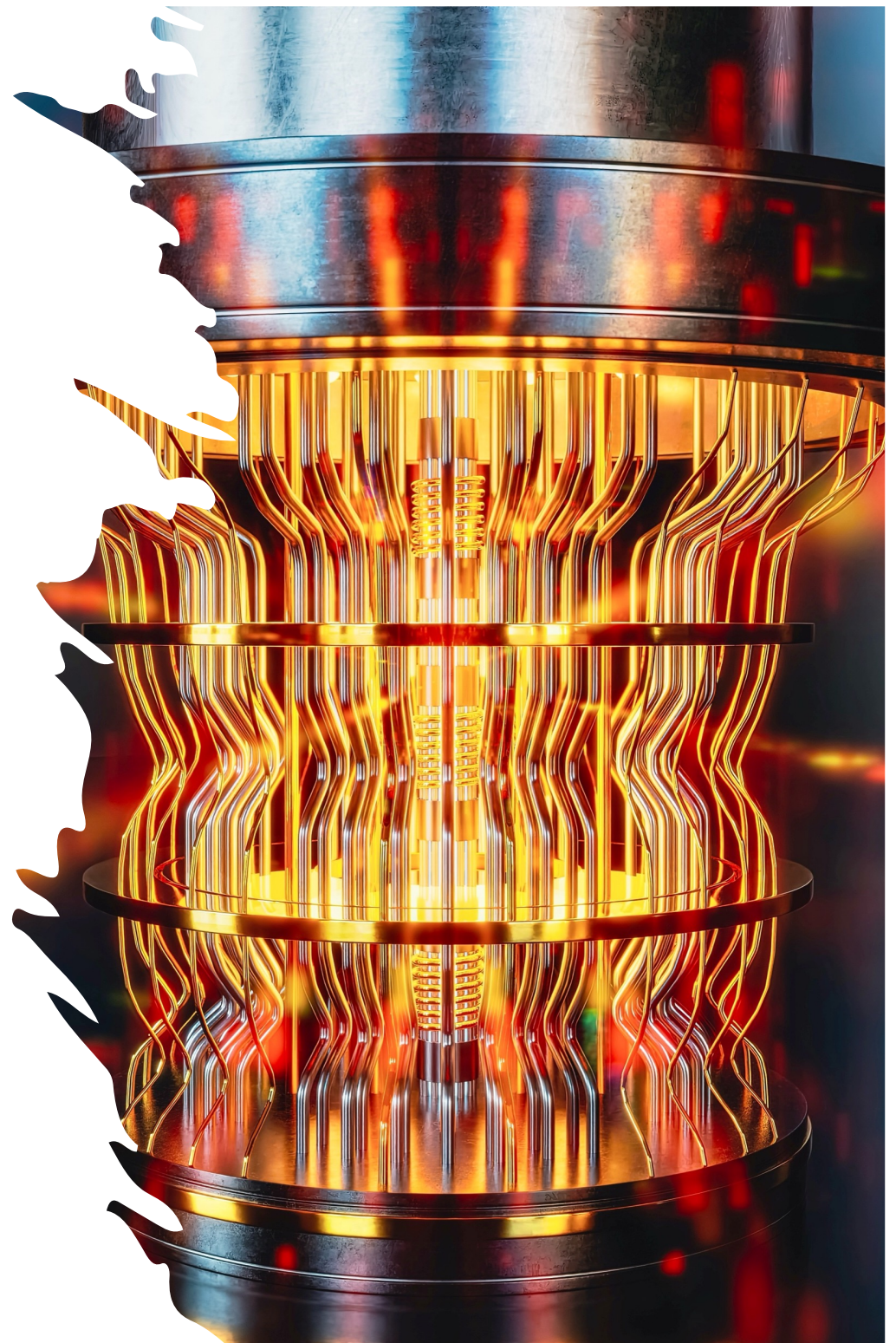
Course on Quantum Tech- nology for Security



Lecture 4 – Quantum Networking and Key Distribution

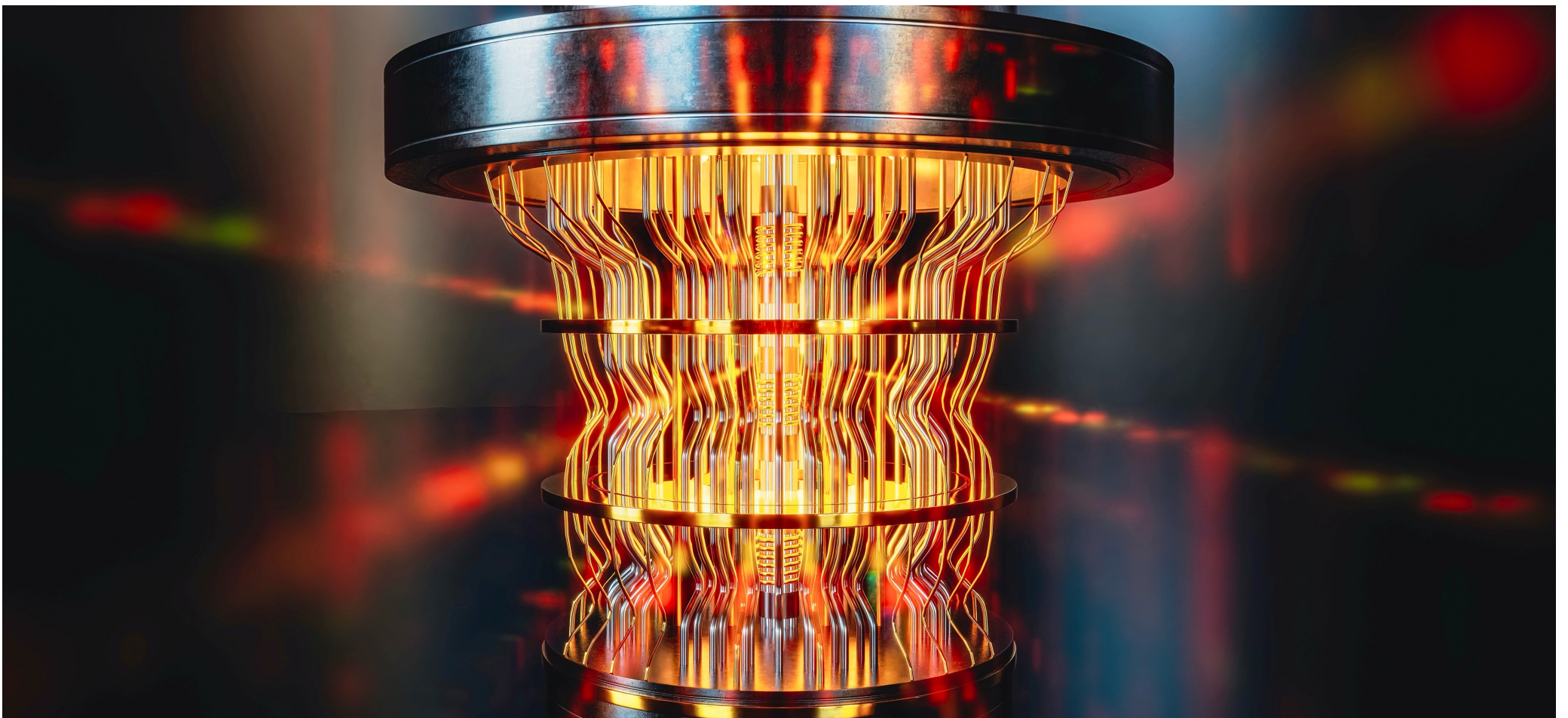


Prof. Esposito Christian



::.. Summary

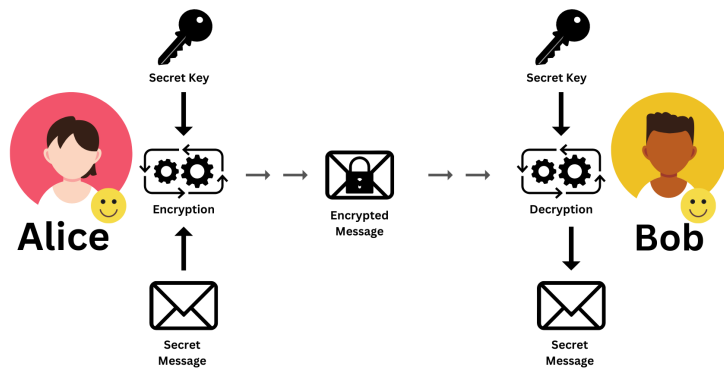
- Quantum Key Distribution
 - Introduction to QKD in theory and BB84, E91 and alternative protocols
- Quantum Communications
 - Quantum Entropy, Quantum-enabled communications, Quantum Information,
 - Entanglement, Superdense coding, Teleportation, Bell's experiment and EPR protocol



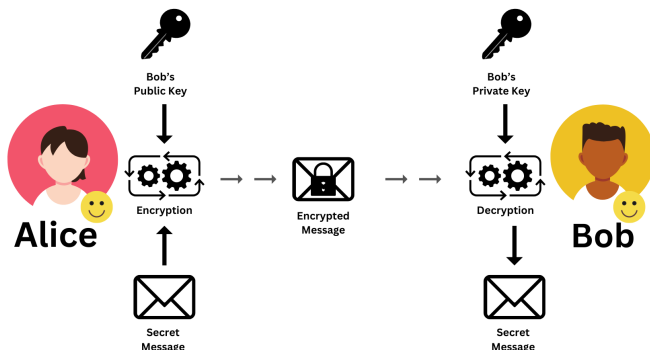
Introduction to Key Distribution

::... Introduction

The basic idea of **symmetric encryption** is that the two parties, i.e., Alice and Bob, share a secret key for encryption and decryption. This is completely different from **public-key** (or asymmetric) **encryption**, where each actor publicly announces their public key for use by anyone who intends to send them a message.



The other relative secret key is used to decrypt messages received from a source that used the public key for encryption.



In principle, knowledge of the public key might be used to reverse the encryption, but in practice, the cryptographic algorithms used are based on computationally very hard mathematical problems (such as factoring large integers) that make this reversal infeasible.

::... Introduction

There is no known algorithm that performs prime factorization efficiently, as the time required grows exponentially with the length of the integer to be factorized, and the computational resources required also grow exponentially.

Bob owns a private key, generated with the public one, that allows easy decryption of the received encrypted message. Such a key needs to be securely stored to prevent malicious entities from obtaining it.

The interest in quantum computing is that there is a means, at least on paper, to factorize efficiently. If a quantum device can be built large enough to implement Shor's algorithm, current encryption algorithms would no longer be safe. Current hardware is pretty far off from this scenario, but the theoretical possibility (and issue) exists, and we never know how fast technological progress will run.

As a result, there is growing interest in symmetric encryption, which has been proven secure when Alice and Bob do not reuse their private keys.

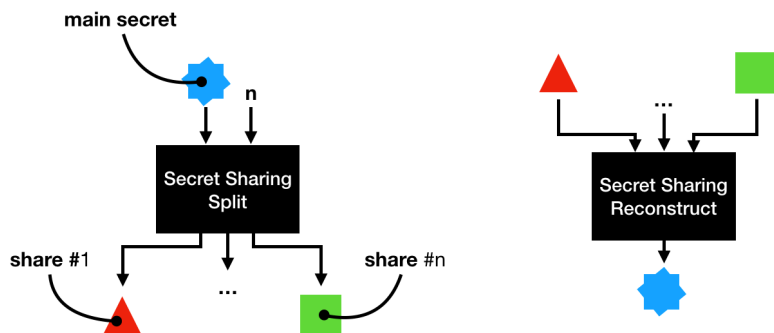
::... Introduction

It is important that the eavesdropper Eve does not accumulate sufficient data from past correspondence between Alice and Bob to learn the private key used. So, Alice and Bob change it regularly.

This, however, poses a practical problem: Alice and Bob need a way to securely share this key (or to change it) at the outset. Meeting in person in a secure room is one option, but it is impractical in most cases, so we need a better way to do it.

A naïve solution is to use public-key cryptography to establish a secure channel for exchanging the secret key, as in SSL/TLS.

An alternative is **secret sharing techniques**, where a secret s is being securely divided into n shares, so that reconstruction of s is possible only if any $t + 1$ parties exchange their information, but no less (in other words, for given $t < n$, no t parties can reconstruct the secret).



::... Quantum Key Distribution - BB84

Quantum mechanics can help devise a key distribution protocol with the proper secrecy and security guarantees. There is a number of approaches in the current literature, and one of the most basic ones is BB84, which works



with one qubit set to basis state $|0\rangle$ and two classical bits whose value is selected randomly.

These two classical bits can be used to make transformations to the qubit based on their values.

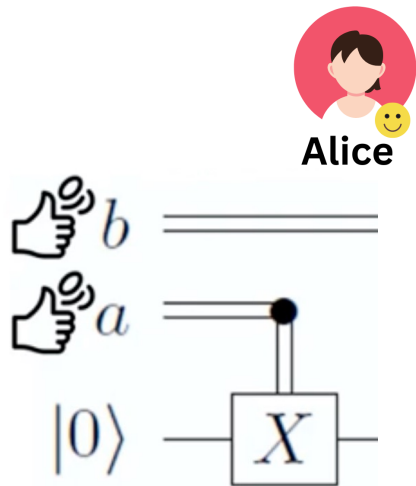
 $b =$

 $a =$

$|0\rangle -$

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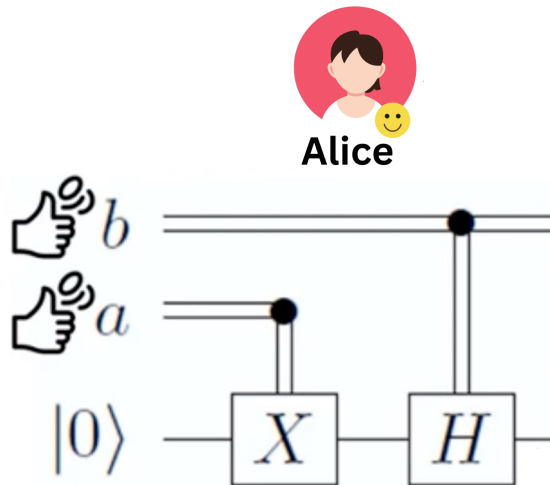


with one qubit set to basis state $|0\rangle$ and two classical bits whose value is selected randomly.

The first transformation is a Pauli-X controlled by the first classical bit a . If a has value 1, the qubit's state is flipped; while if a has value 0, the qubit's state is unchanged.

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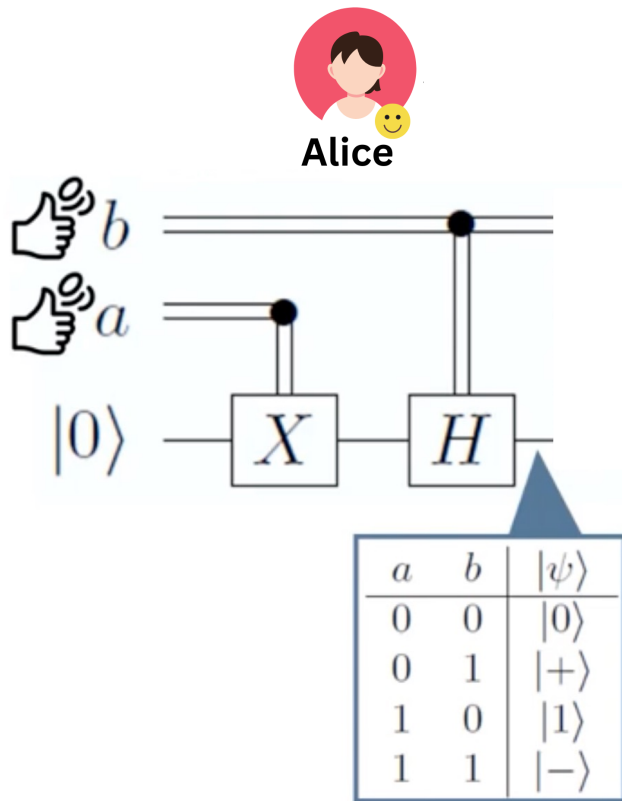
The second classical bit b controls a Hadamard gate applied to the qubit. If b has value 1, the qubit's state is altered by the Hadamard transformation; while if b has value 0, the qubit's state is unchanged.

Such a Hadamard transformation is a rotation on the Bloch sphere so that the basis states $|0\rangle$, $|1\rangle$ correspond to the $|+\rangle$, $|-\rangle$ ones. Let's see the possible combinations of the qubit state after these two controlled transformations, depending on the values of the two random bits.

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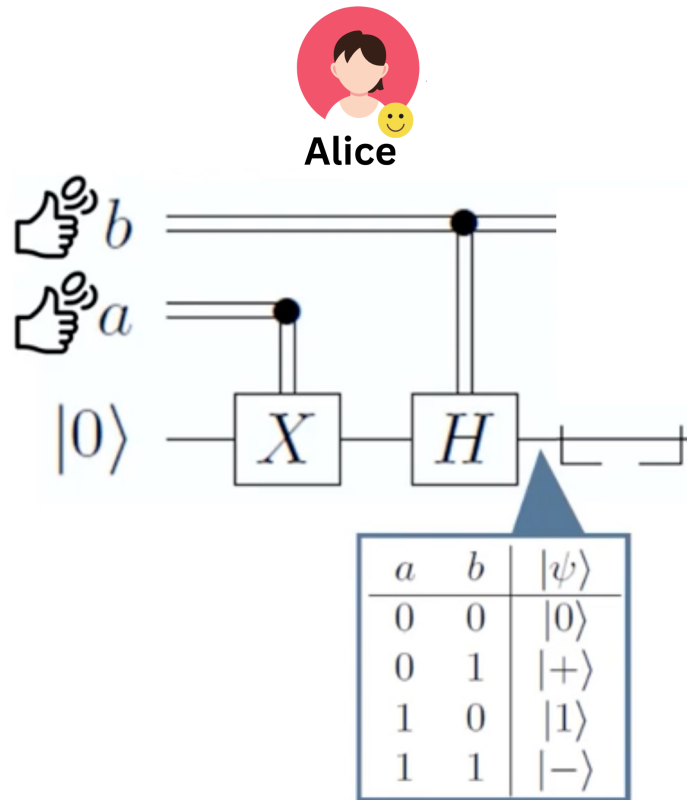
If both bits are 0, then no operation has been applied on the qubit, which remains in the basis state $|0\rangle$. If the first bit is 0 no Pauli-X is applied and if the second one is 1, then the Hadamard is applied, and the qubit is moved from $|0\rangle$ to $|+\rangle$. On the other hand, if the first bit is 1, then a Pauli-X is applied, and the qubit is flipped to the $|1\rangle$ state.

Depending on the value of the second bit, Hadamard operation is applied or not leaving the qubit to the $|1\rangle$ state or rotating it to the $|-\rangle$ state.

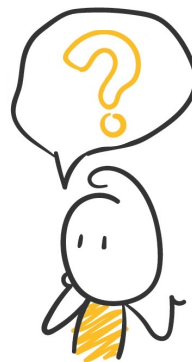
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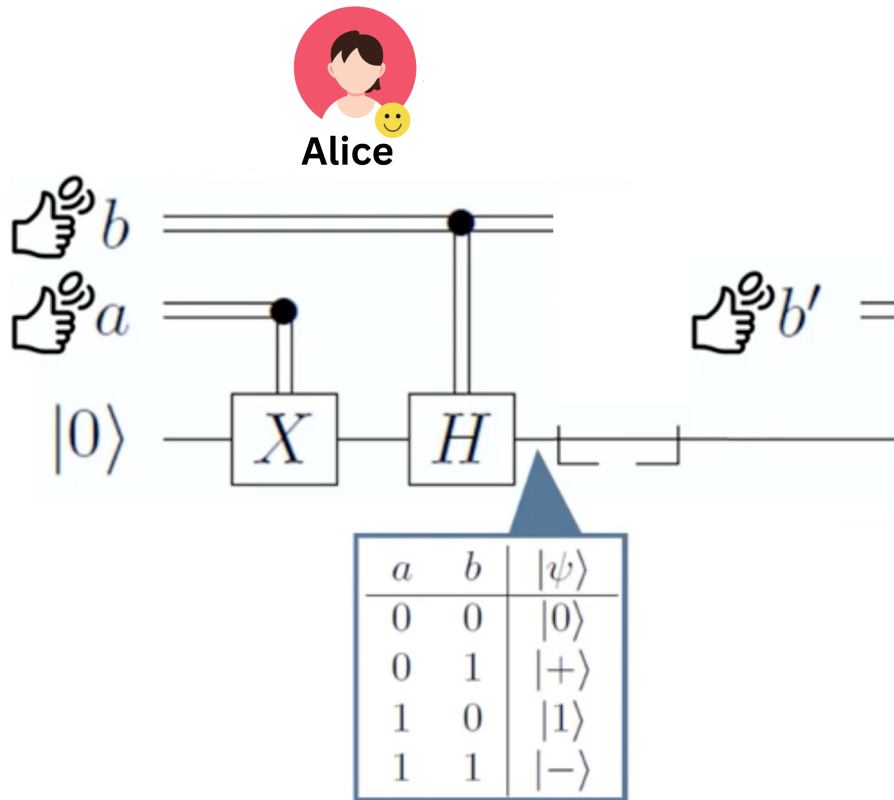
The qubit has four possible combinations and it is passed to Bob.



What does Bob behave?

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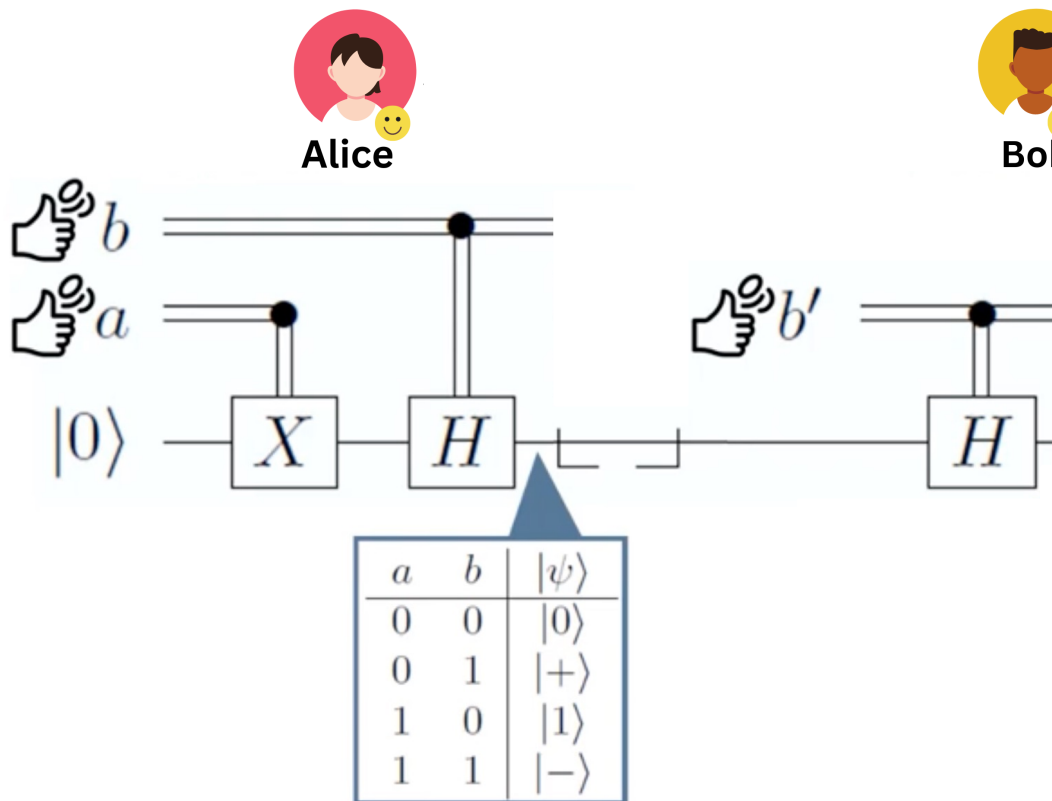
with one qubit set to basis state $|0\rangle$ and two classical bits whose value is selected randomly.

Bob creates a classic bit b' whose value is randomly chosen. This bit is also used to perform a controlled operation on the qubit Bob has just received.

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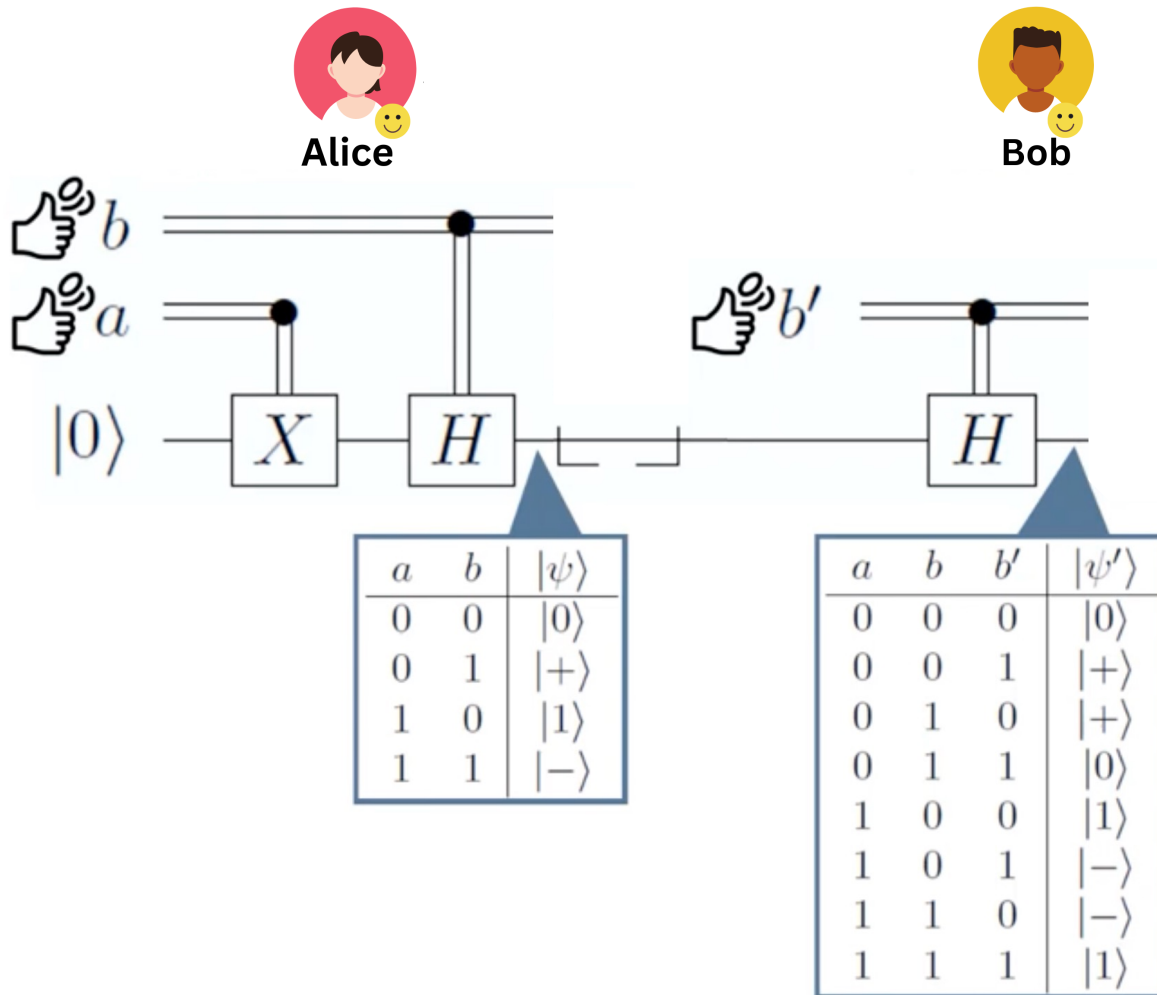


Bob uses b' to control another Hadamard gate, which further increases the number of possibilities in which the qubit state can be in.

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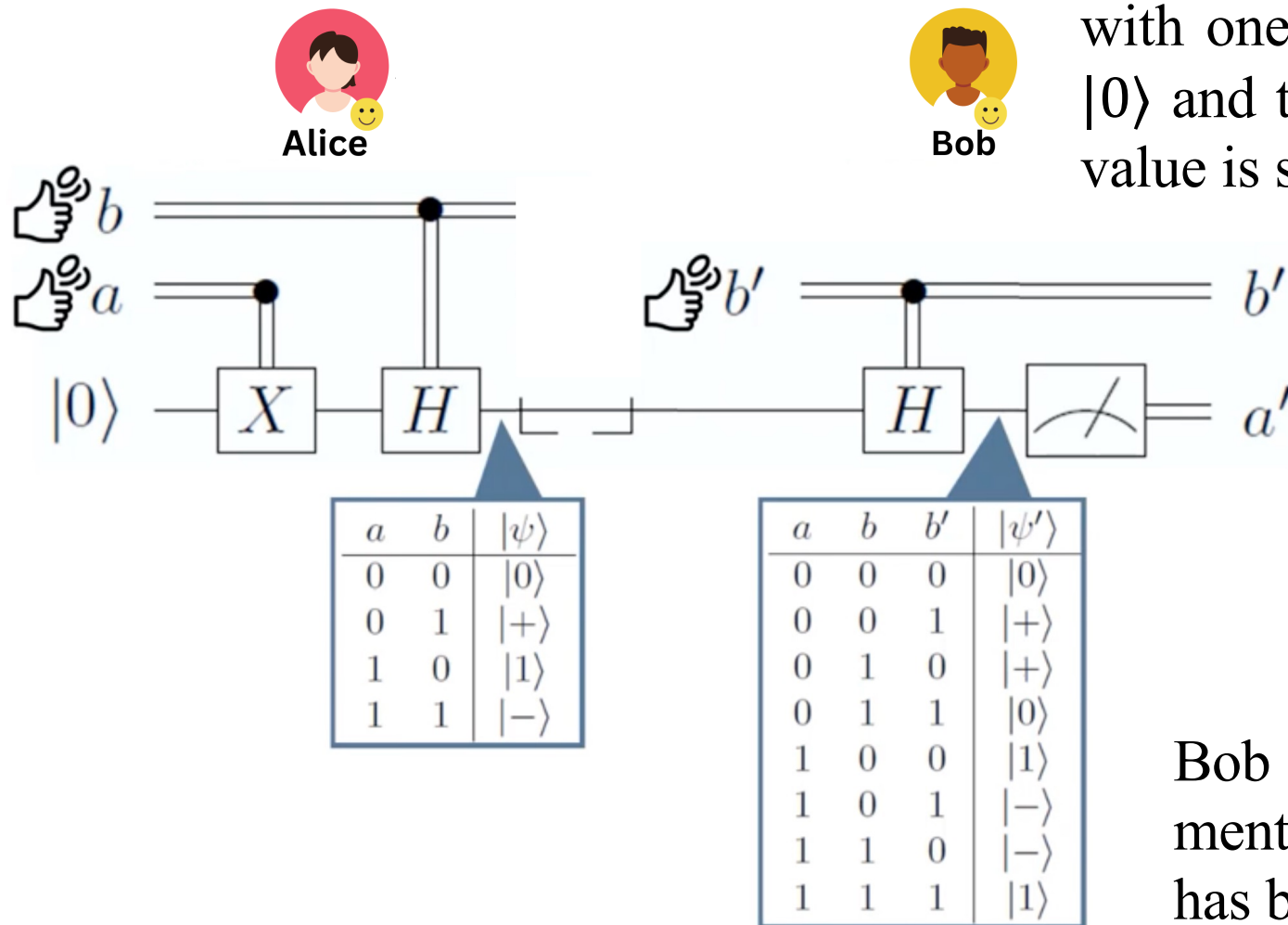


If b' has value equal to 0 then no operation is applied, but for b' equal to 1, then the operation is applied so that $|0\rangle \leftrightarrow |+\rangle$ and $|1\rangle \leftrightarrow |-\rangle$.

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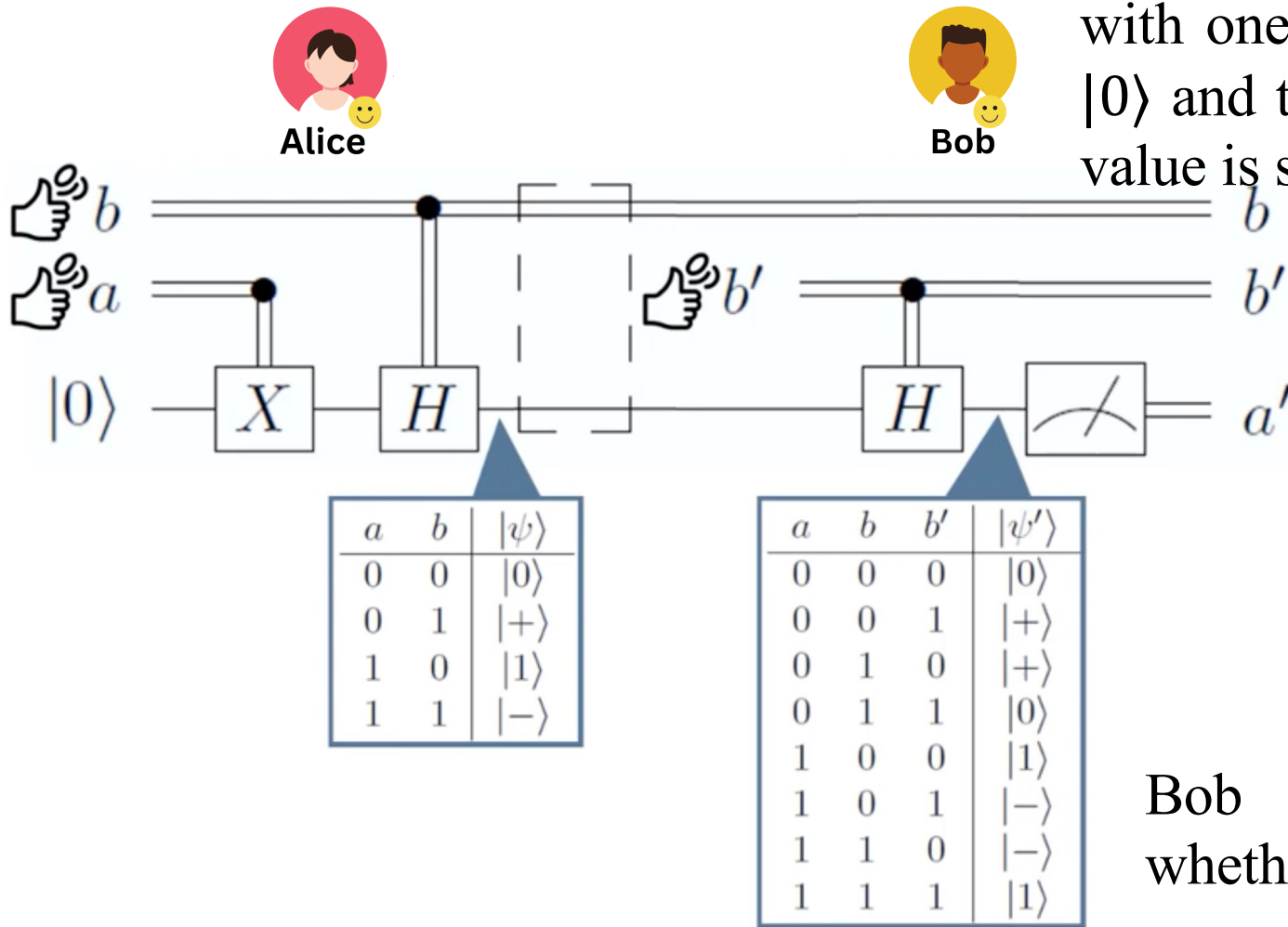
Now, Bob performs a measurement on the standard basis, and its measurement is 0 or 1, which is a classical bit a' .

Bob makes an announcement that the measurement has been done.

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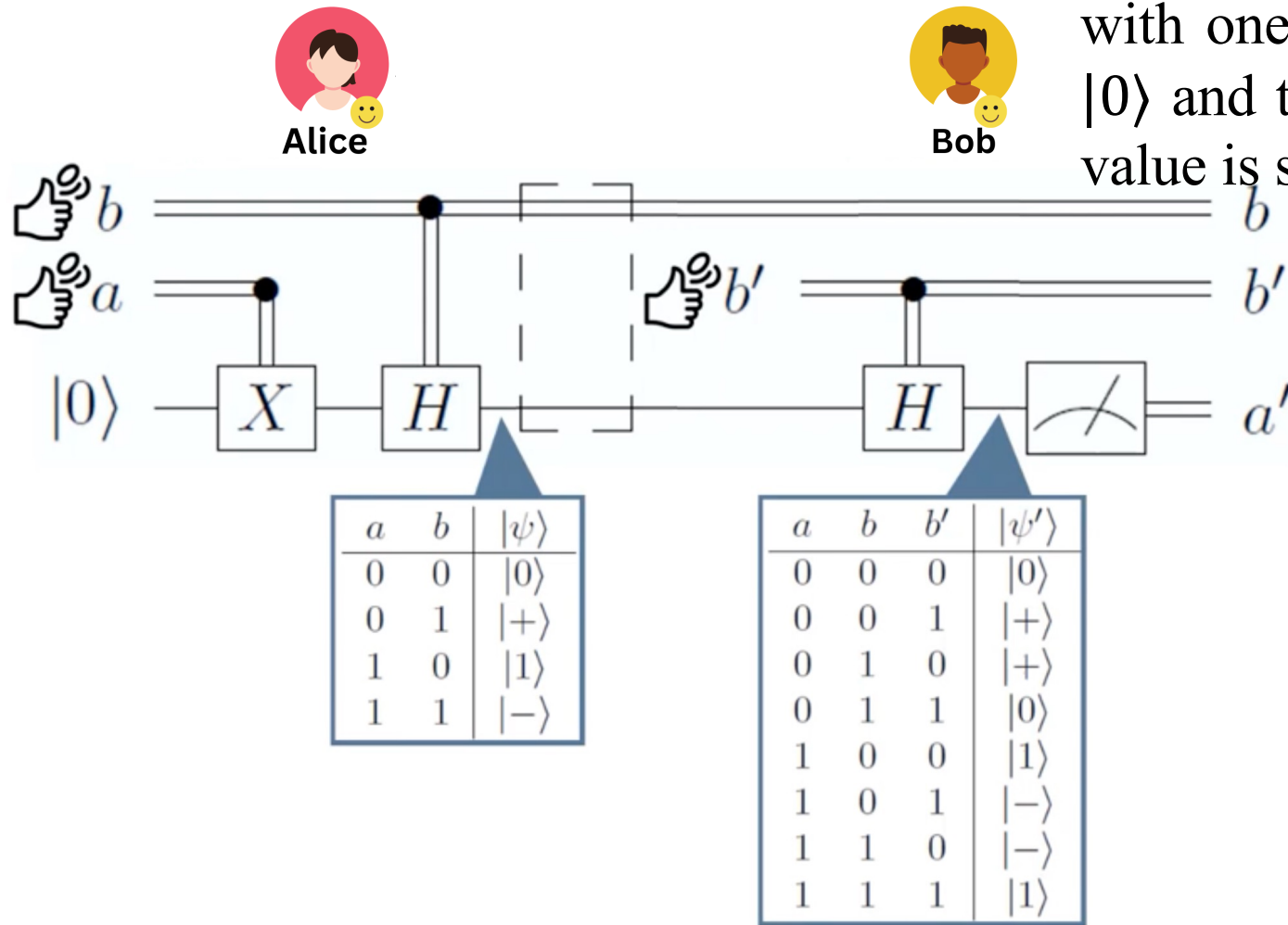
Alice sends to Bob her classic bit b , which makes a comparison between b and b' , checking if both are identical or not.

Bob announces to Alice whether they are equal or not.

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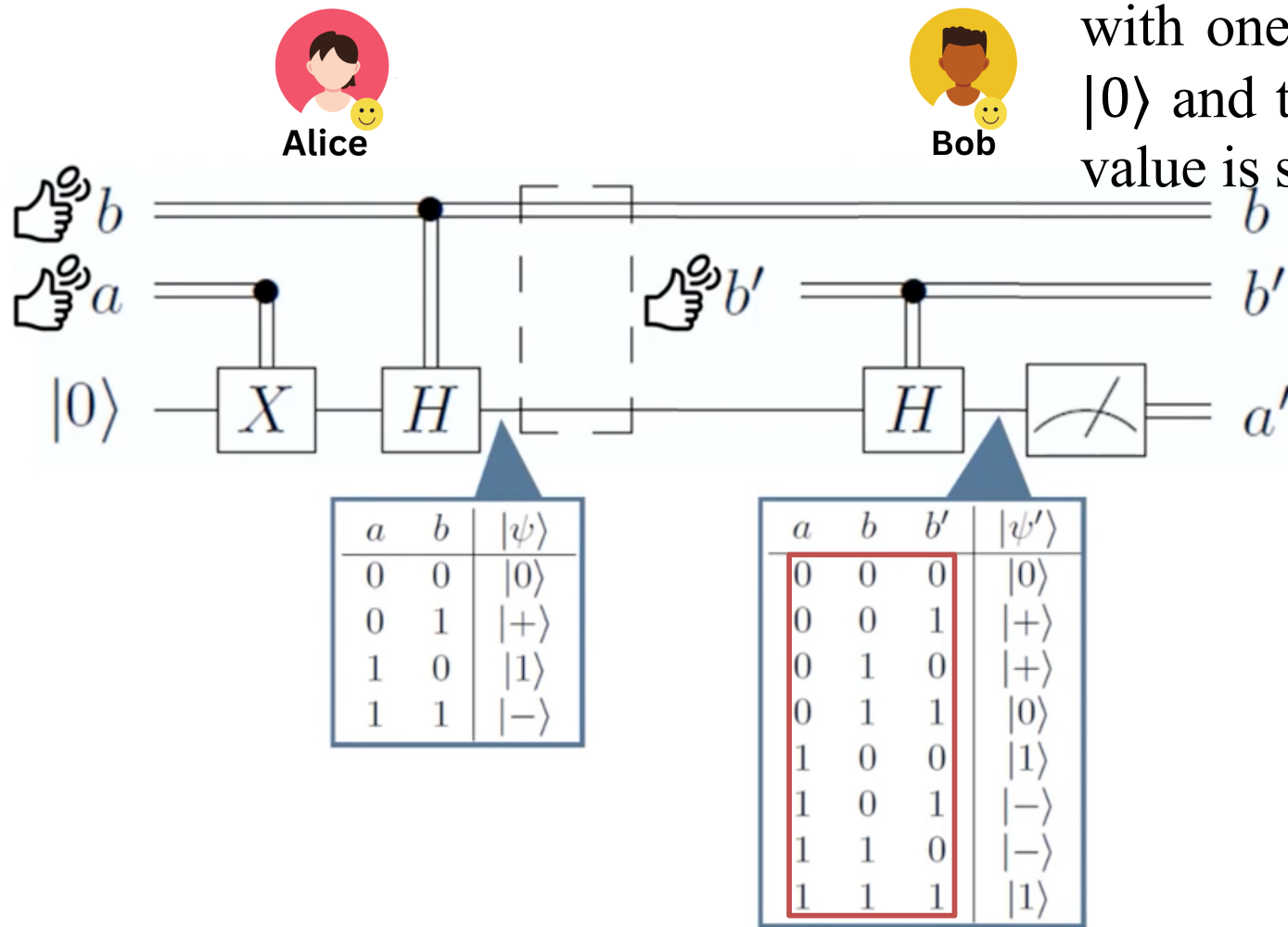


The announcements between Alice and Bob can be done publicly, without being confidential.

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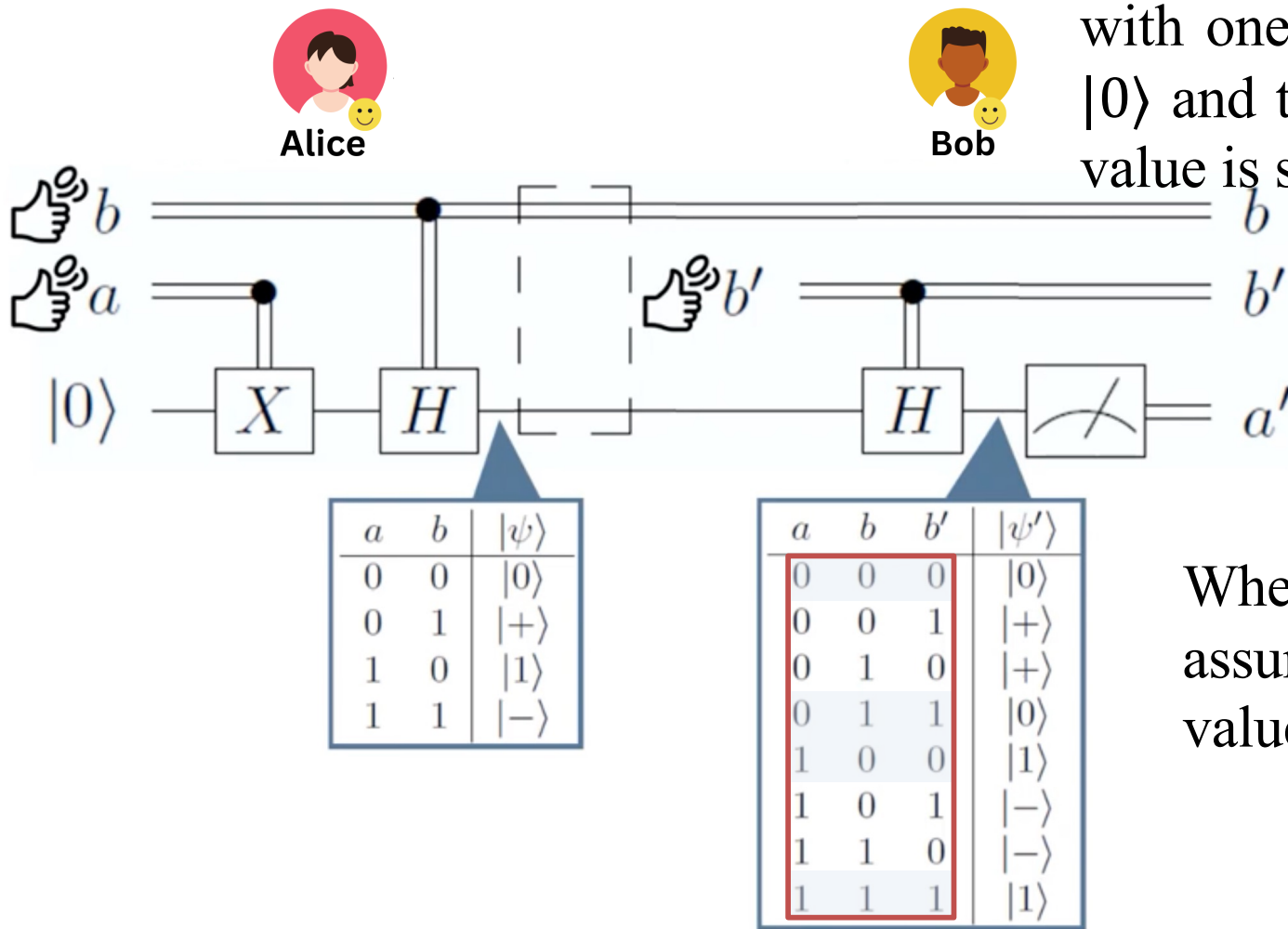


There is an interesting pattern in the table.

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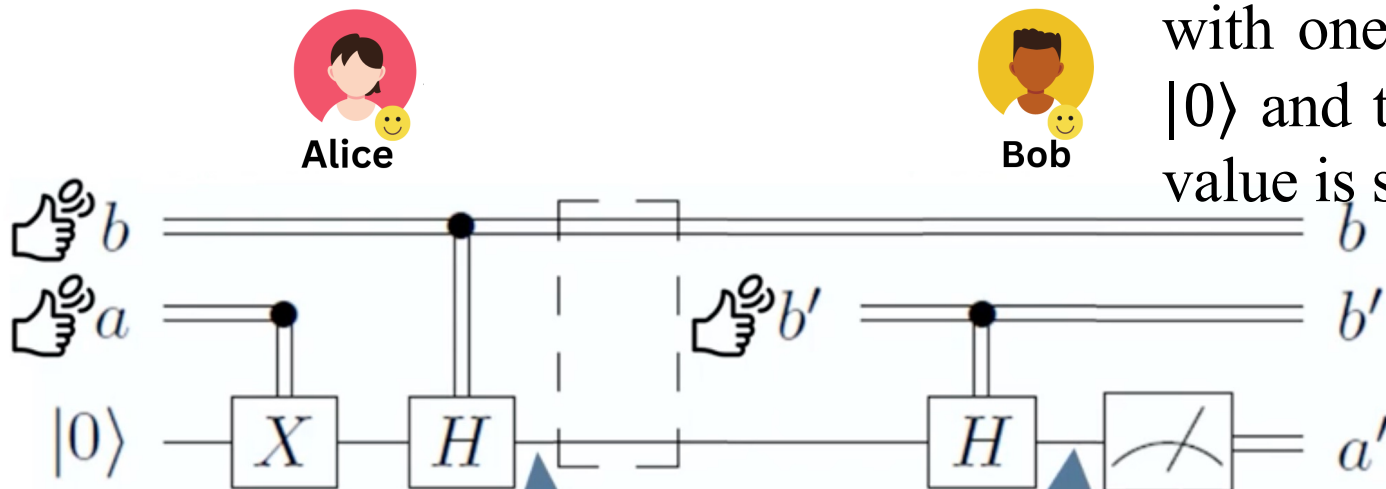
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When $b = b'$, then the qubit assumes a state coding the value of a , or $a = a'$.

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There is an interesting pattern in the table.

a	b	$ \psi\rangle$
0	0	$ 0\rangle$
0	1	$ +\rangle$
1	0	$ 1\rangle$
1	1	$ -\rangle$

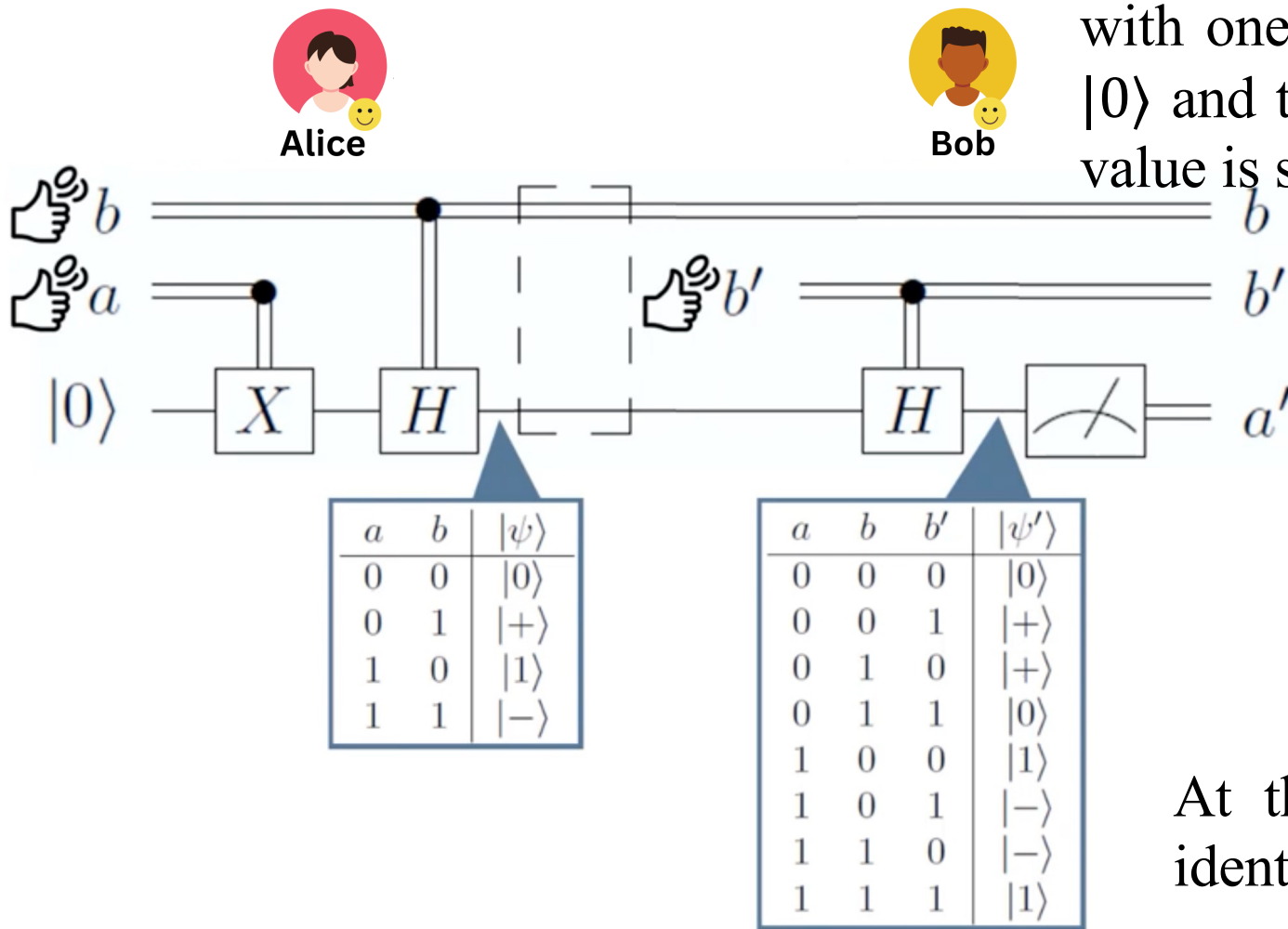
a	b	b'	$ \psi'\rangle$
0	0	0	$ 0\rangle$
0	0	1	$ +\rangle$
0	1	0	$ +\rangle$
0	1	1	$ 0\rangle$
1	0	0	$ 1\rangle$
1	0	1	$ -\rangle$
1	1	0	$ -\rangle$
1	1	1	$ 1\rangle$

When $b \neq b'$, there is no evident pattern, so the obtained classical bits and the qubit are thrown away and not registered to compose the key.

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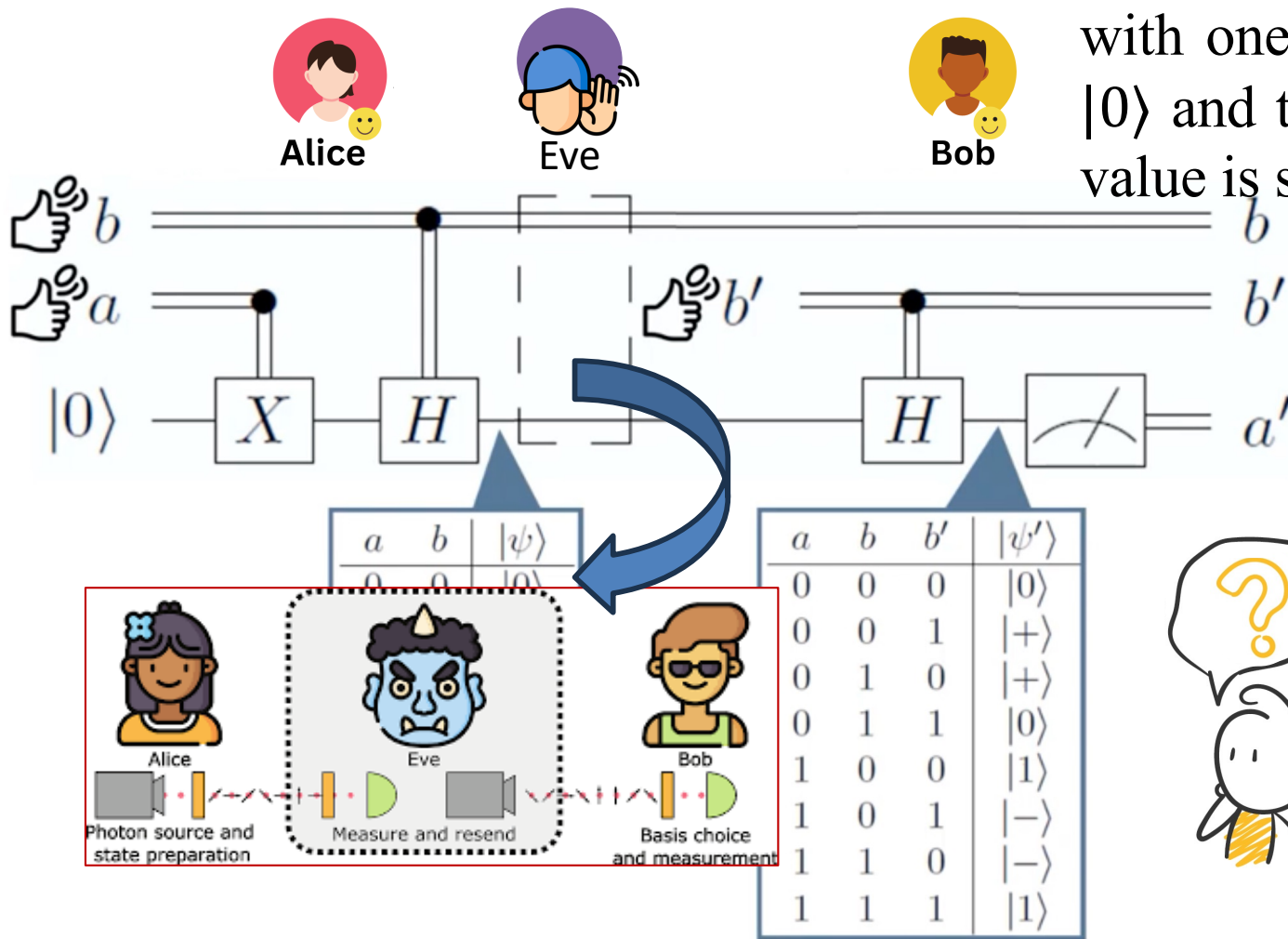


This procedure is repeated many times, until all the digits needed for the shared key are obtained and accumulated.

At the end both will have identical keys.

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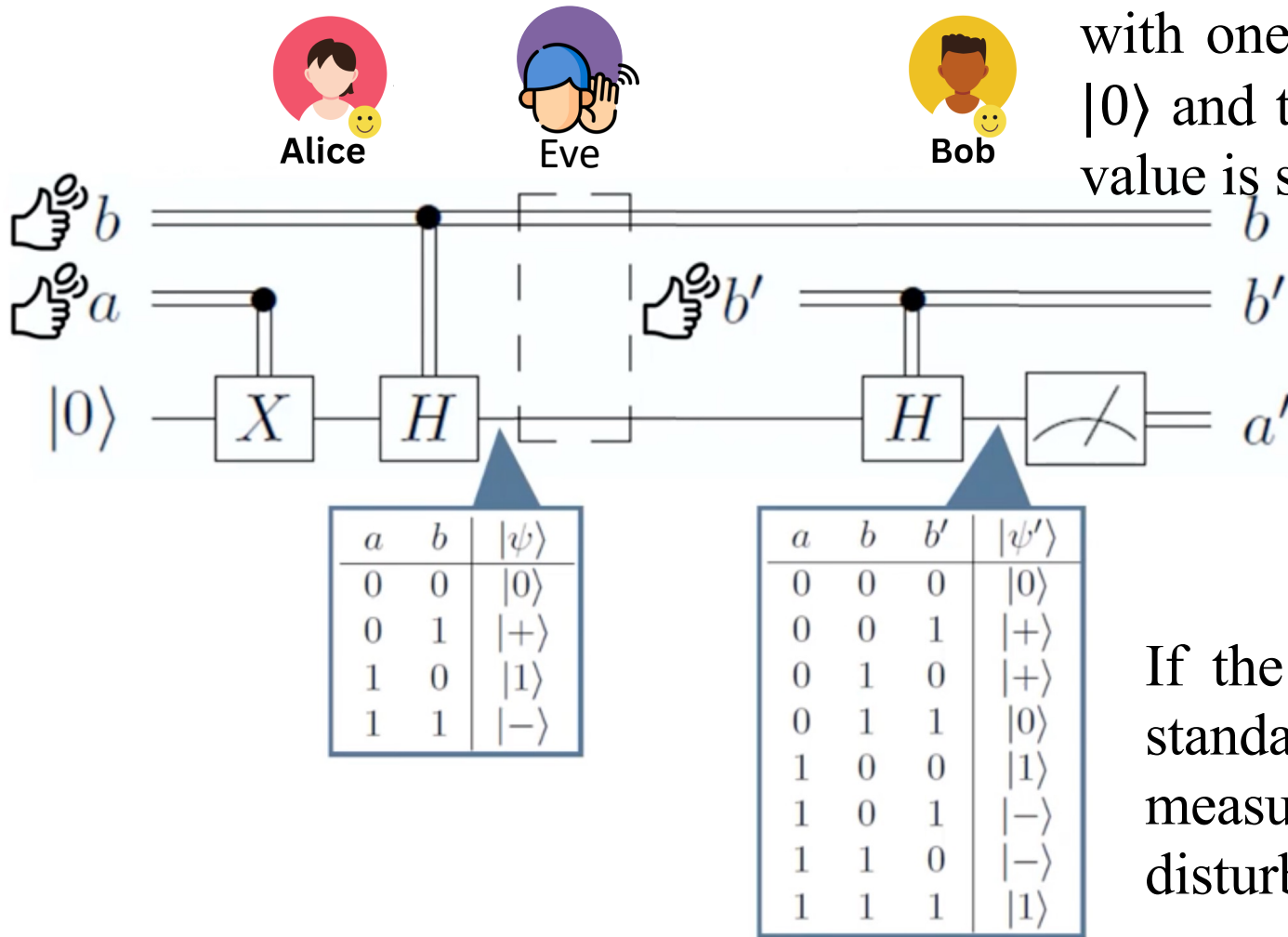


This approach is effective, but is it also secure? What happens if there is Eve?

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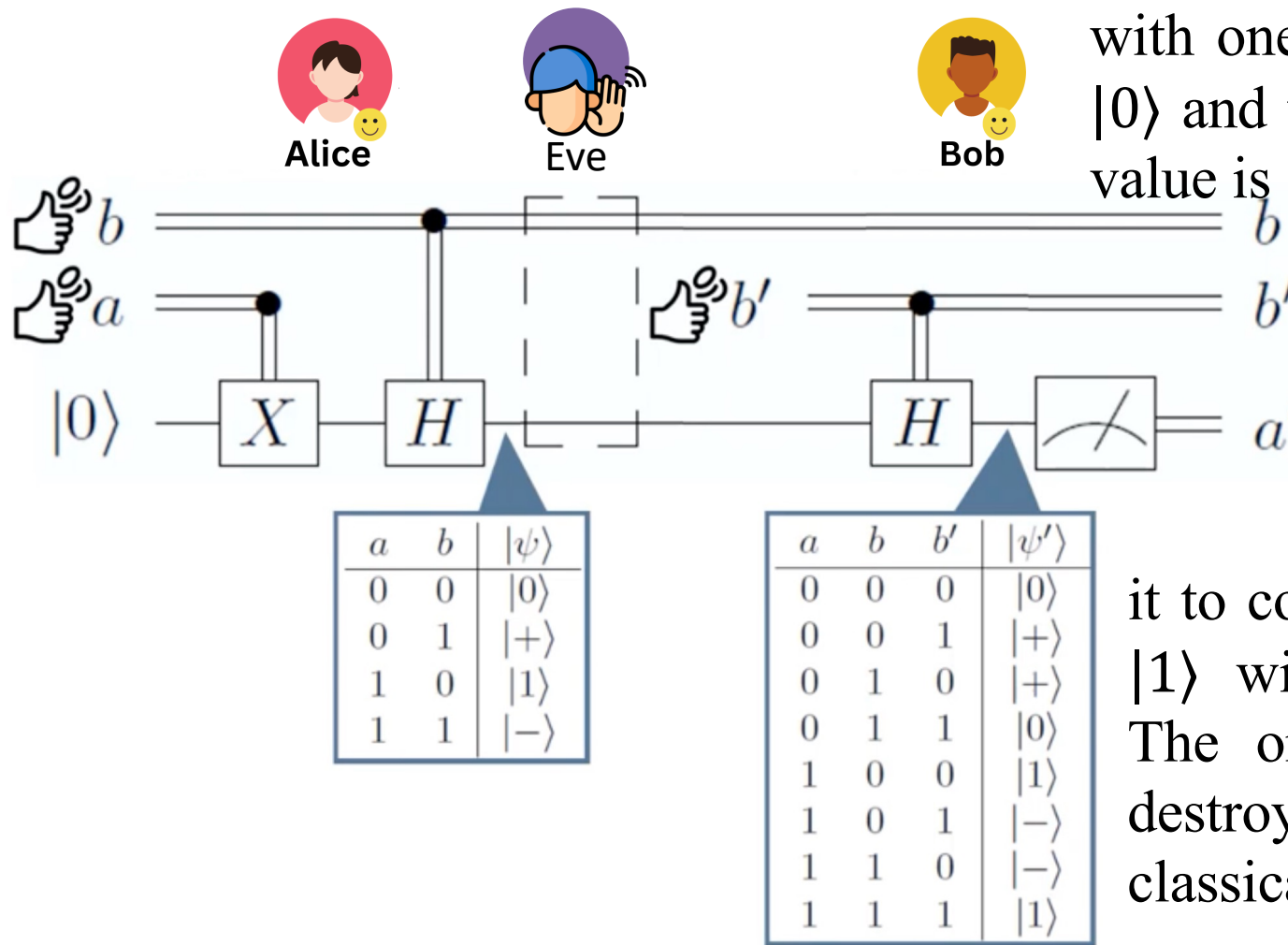


To extract information, Eve has to make a measurement on the qubit, which is in four possible states.

If the qubit had been in the standard basis only, the measurement would have not disturbed its state.

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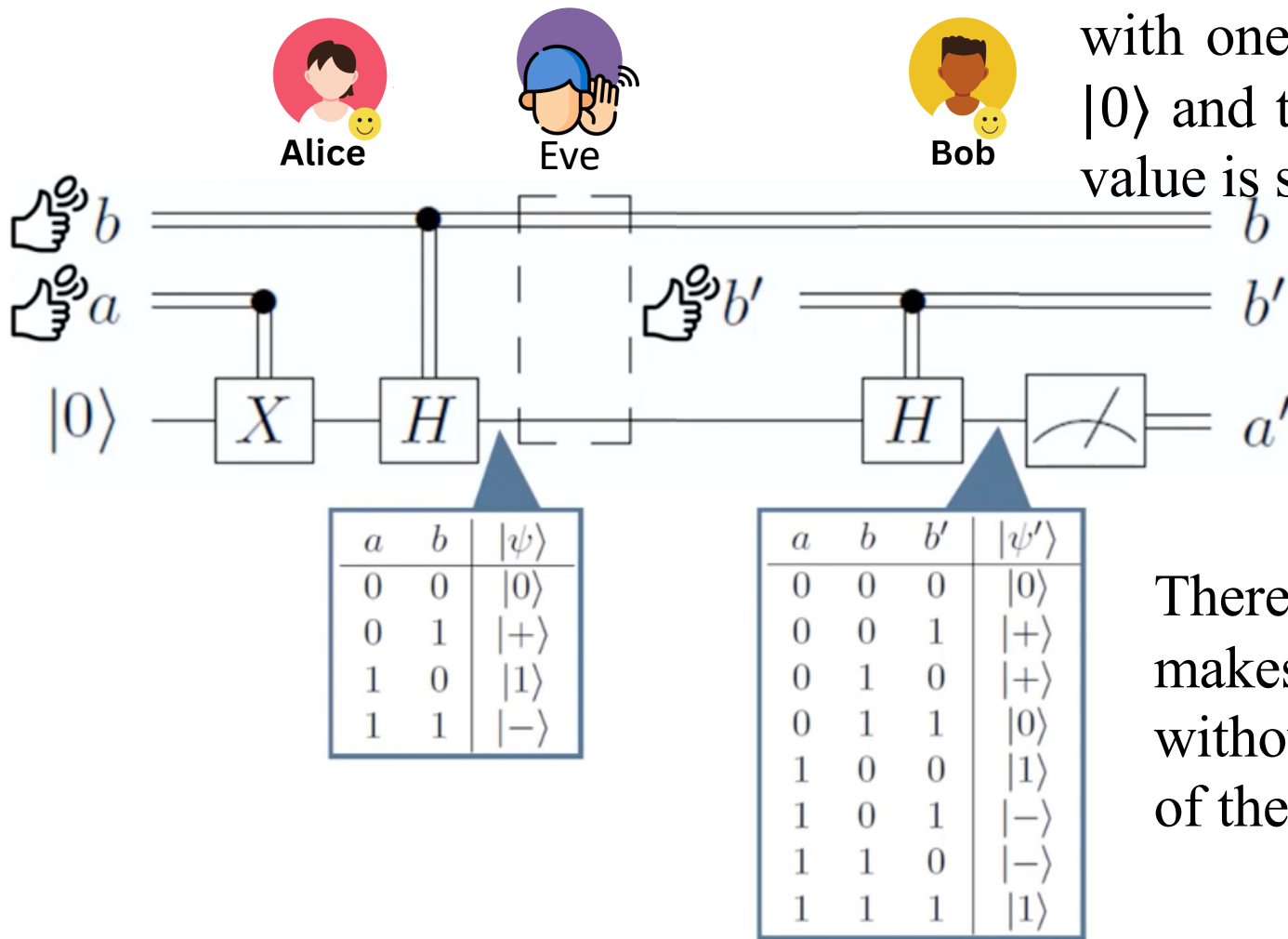
with one qubit set to basis state $|0\rangle$ and two classical bits whose value is selected randomly.

But, it can be in the Hadamard basis, and the measures in the standard basis cause it to collapse into either $|0\rangle$ or $|1\rangle$ with a 50/50 probability. The original superposition is destroyed, yielding a random classical outcome.

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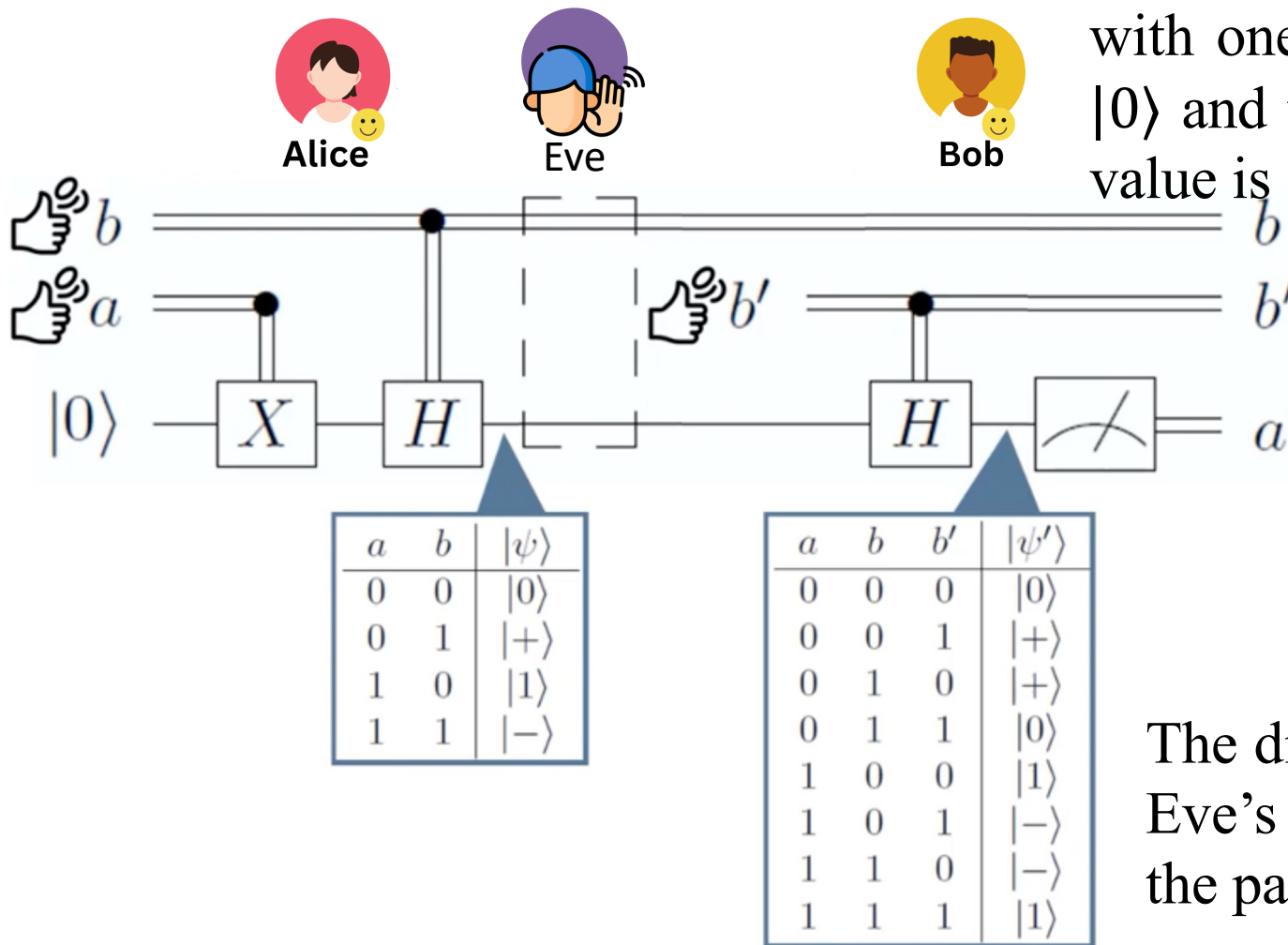
If we change the measurement basis, the vice versa happens.

There is no way that Eve makes a measurement without disturbing the state of the qubit.

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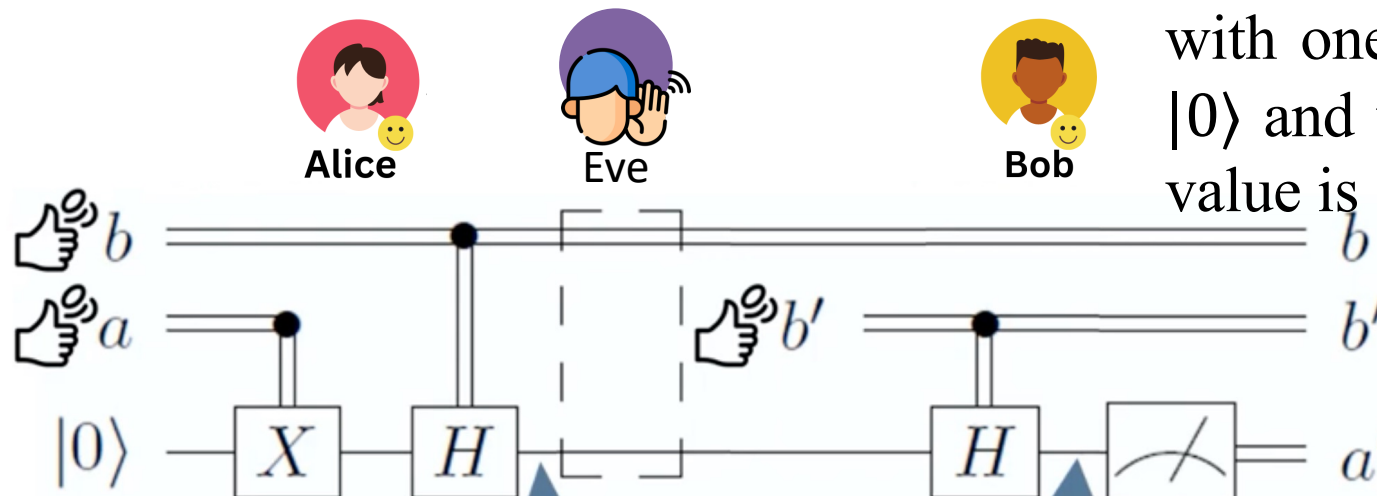


Since the transformations on Alice's side are decided randomly, the four cases happen evenly in probability.

The disturbance introduced by Eve's measurements breaks the pattern in the table.

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with one qubit set to basis state $|0\rangle$ and two classical bits whose value is selected randomly.

Let us have the qubit in state $|0\rangle$ and $a = 0$ and $b = 1$ so that the qubit is in state $|+\rangle$.

a	b	$ \psi\rangle$
0	0	$ 0\rangle$
0	1	$ +\rangle$
1	0	$ 1\rangle$
1	1	$ -\rangle$

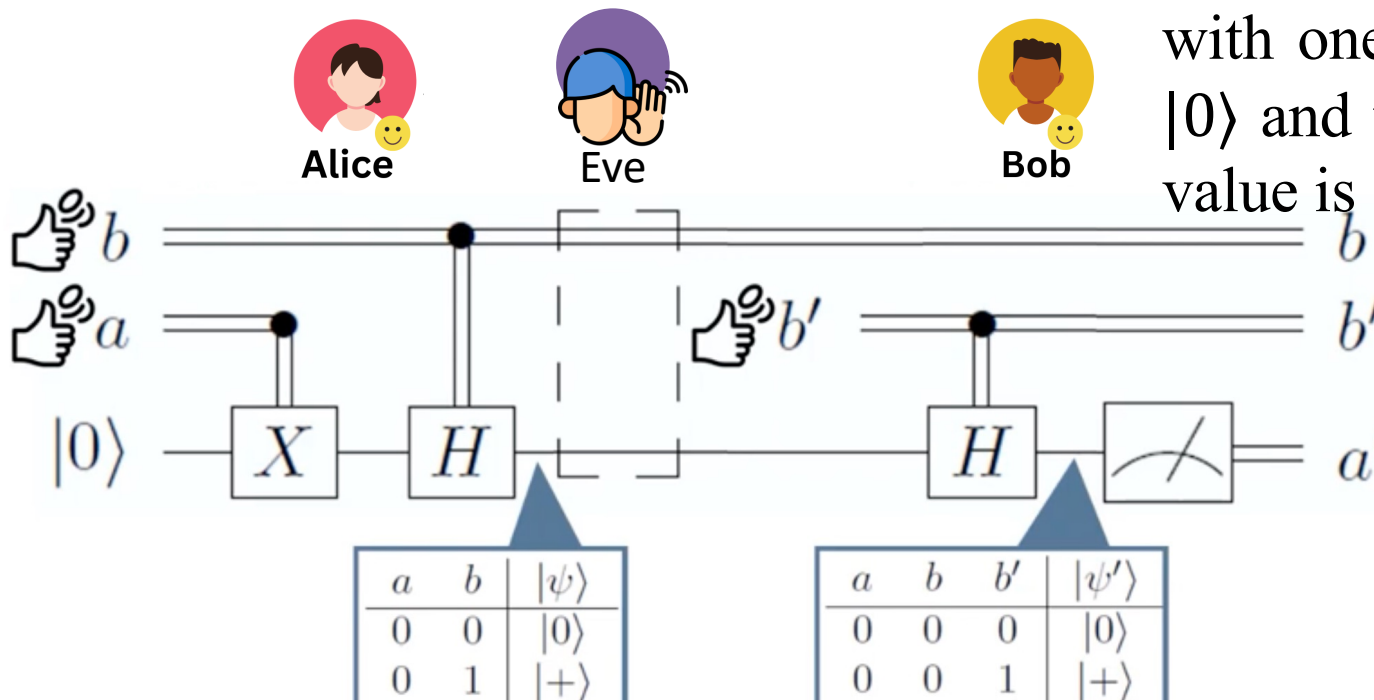
a	b	b'	$ \psi'\rangle$
0	0	0	$ 0\rangle$
0	0	1	$ +\rangle$
0	1	0	$ +\rangle$
0	1	1	$ 0\rangle$
1	0	0	$ 1\rangle$
1	0	1	$ -\rangle$
1	1	0	$ -\rangle$
1	1	1	$ 1\rangle$

The measurement in Pauli-Z collapses the state to one of the two basis, i.e., $|0\rangle$. Now Bob has $b' = 1$, so Hadamard is applied and the qubit has $|+\rangle$ state, different then $|0\rangle$.

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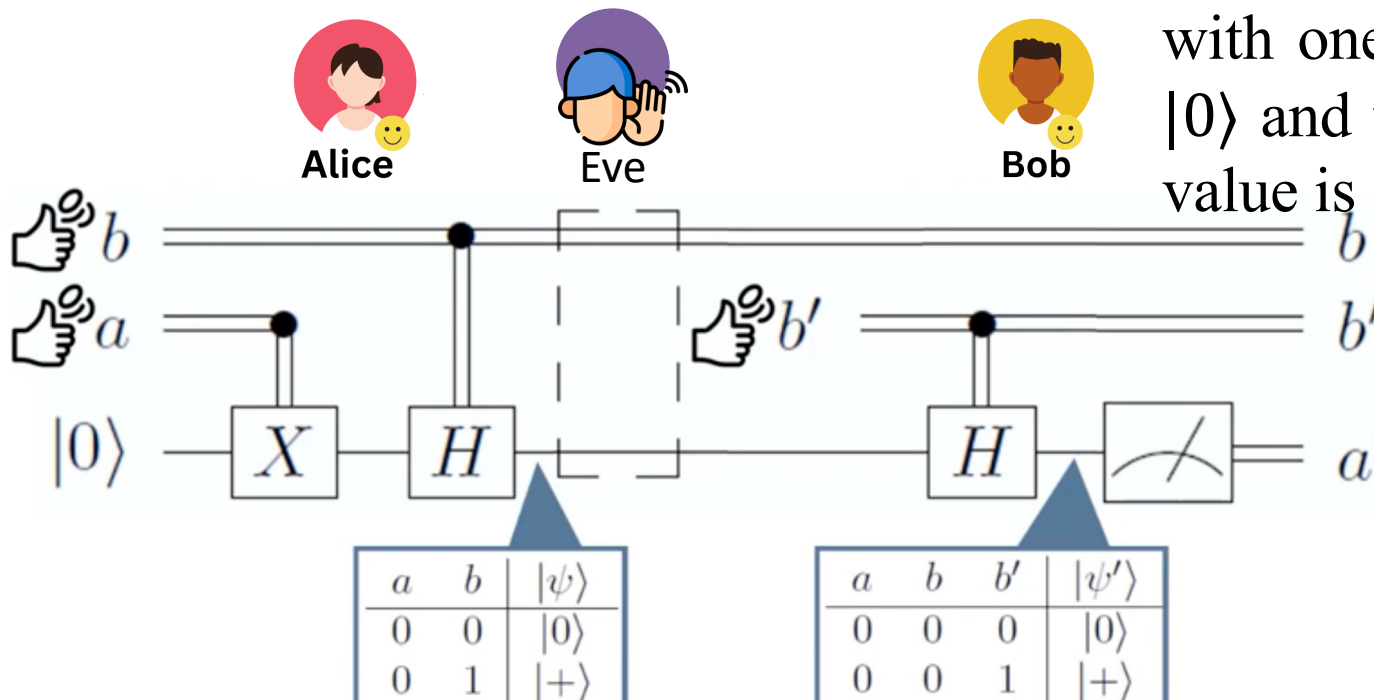
As the qubit is in the $|+\rangle$ state, there is a 50% chance that the output of the measurement is 1. We have a $\neq a'$ leading to issues.

This disturbance is a direct consequence of Heisenberg's uncertainty principle and the no-cloning theorem, preventing the perfect copy of unknown states. Alice and Bob can also select certain positions in the key, and compare the digits. If different, the presence of Eve is detected.

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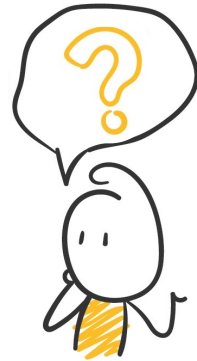
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As the qubit is in $|+\rangle$ state, there is a 50% chance that the output of the measurement is 1. We have $a \neq a'$ leading to issues.

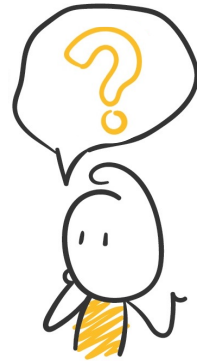
If the Quantum Bit Error Rate (QBER) is within tolerable limits, the disturbance still necessitates intensive post-processing, specifically privacy amplification, which reduces the length of the final key to eliminate any partial information gained by Eve.

::... Quantum Key Distribution - BB84



How can be implemented in a practical deployment the BB84 protocol?

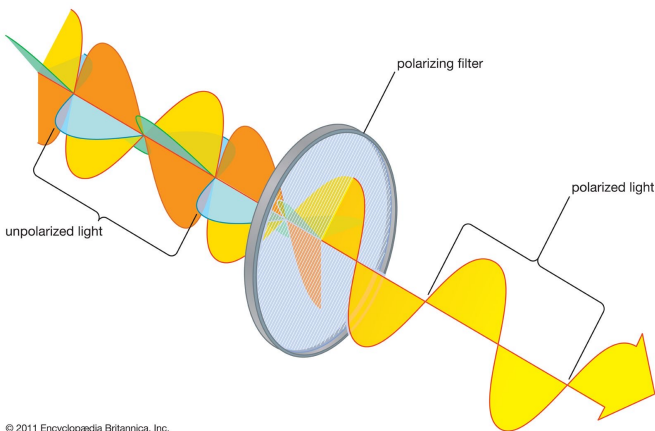
::... Quantum Key Distribution - BB84



How can be implemented in a practical deployment the BB84 protocol?

Let us consider implementing BB84 QKD with photons, which means encoding random bits into single-photon polarization states, sending them over an optical channel, then using classical communication to agree on a shared secret key.

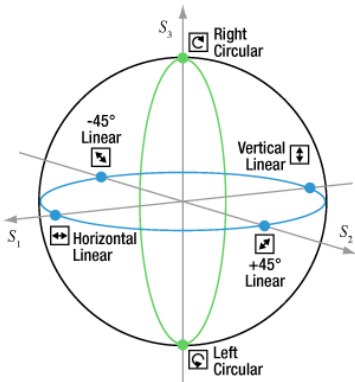
A photon has an electromagnetic field oscillating in a direction perpendicular to its motion, and its quantum description is its polarization: $|\psi\rangle = \alpha|H\rangle + \beta|V\rangle$, $|H\rangle$ is the horizontal polarization, $|V\rangle$ is vertical one, and α, β are complex amplitudes such that $|\alpha|^2 + |\beta|^2 = 1$. A single photon can be in a superposition: $|\psi\rangle = (|H\rangle + |V\rangle)/\sqrt{2}$.



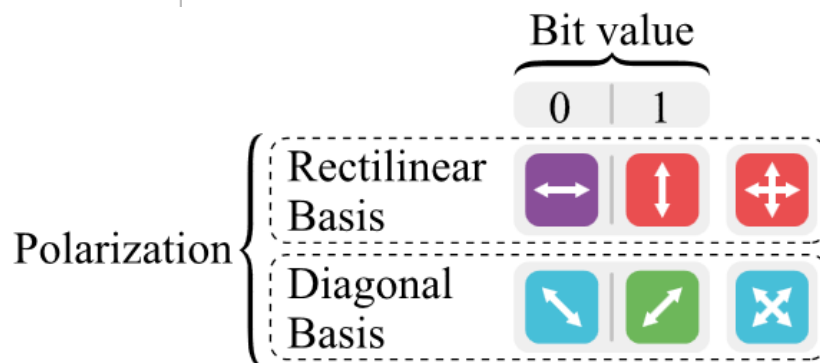
::... Quantum Key Distribution - BB84



There are four states of polarisation: (i) horizontal 0° , (ii) vertical 90° , (iii) diagonal 45° , and (iv) anti-diagonal 135° ; and two types of basis for encoding purposes: rectilinear (+), and diagonal (×), which also represents measurement basis.



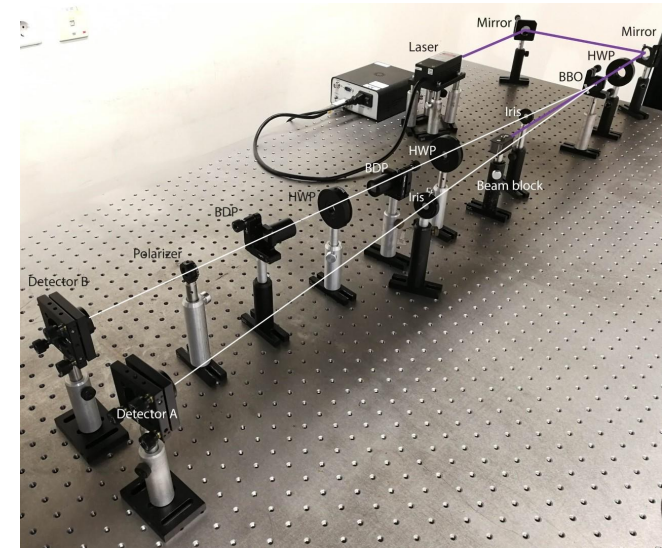
A polarization qubit can be visualized on a sphere: the North pole is the $|H\rangle$ state, the South pole is $|V\rangle$, diagonal states lie on the Equator, while the other points represent general superpositions. In optics, this is called the Poincaré sphere.



Bit-wise representation for polarization-coding is similar to what is seen for qubits, depending on the measurement basis being used.

::... Quantum Key Distribution - BB84

Creating single-photon polarization states means generating a photon (e.g., via a laser or Spontaneous parametric down-conversion), filtering it into a known polarization with a polarizer, and then precisely rotating its polarization using wave plates (half-wave or quarter-wave) to obtain the desired quantum state.

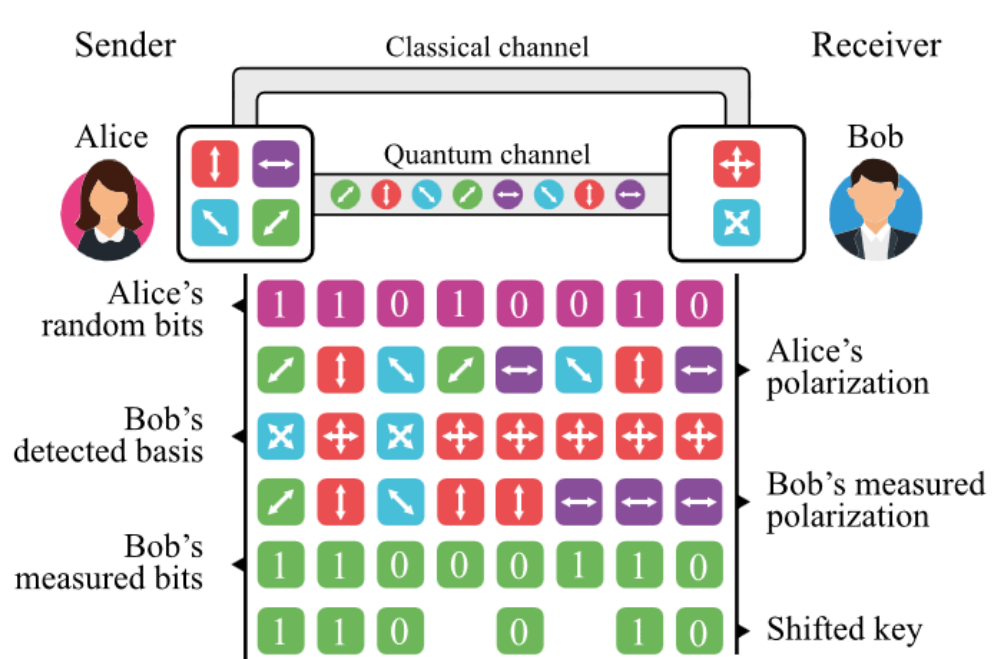


The BB84 protocol can have a dual, equivalent photon-based implementation rather than the one presented on quantum circuits.

- The two views are mathematically equivalent; they just use different languages. Specifically, the standard basis and the Hadamard one can be mapped into the rectilinear and diagonal bases.
- On Alice's side, the Pauli-X (bit flip) gate is physically implemented with a half-wave plate rotated to 45° , and the Hadamard one with a half-wave plate at 22.5° .

::... Quantum Key Distribution - BB84

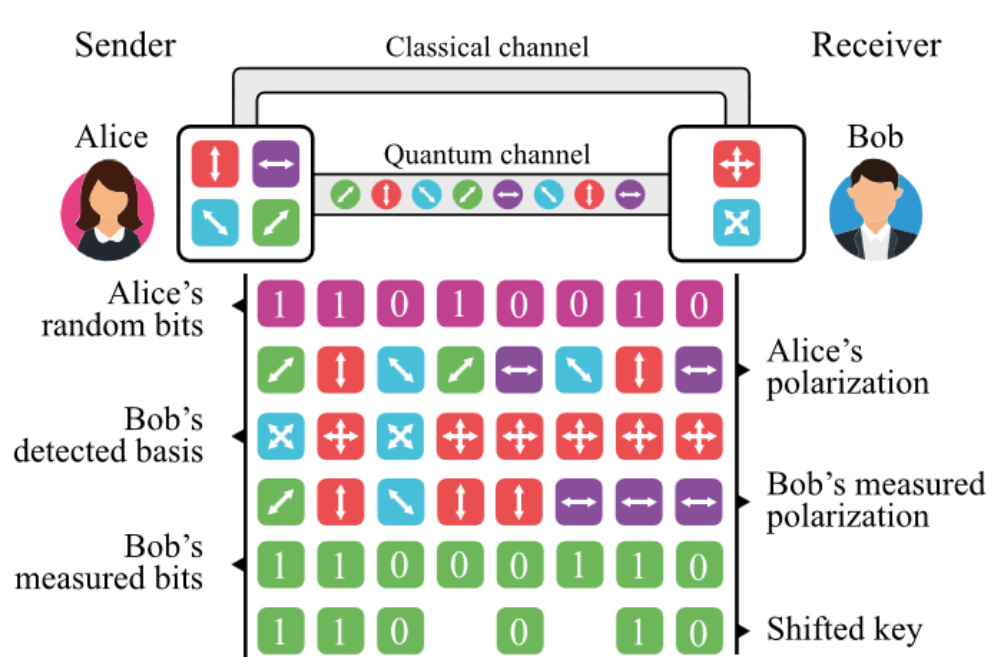
- On Bob's side, if a measurement is needed according to the Z basis, a direct measure is done; otherwise, if a measurement is needed according to the X basis, then a Hadamard gate is applied, followed by a direct measure. In the photon implementation, Bob uses optical elements that directly measure in a chosen basis:
 - for Z basis (H/V), a polarizing beam splitter (PBS) is used;
 - for X basis ($+45^\circ/-45^\circ$), a rotate polarization and PBS are used.



- Alice generates two random bit strings, i.e., basis sequence bits and key bits.
- She picks each key bit as the $|0\rangle$ or $|1\rangle$ state if the respective basis sequence bit is 0 otherwise, she picks the state $|+\rangle$ or $|-\rangle$.
- She transmits the encoded states to Bob.

... Quantum Key Distribution - BB84

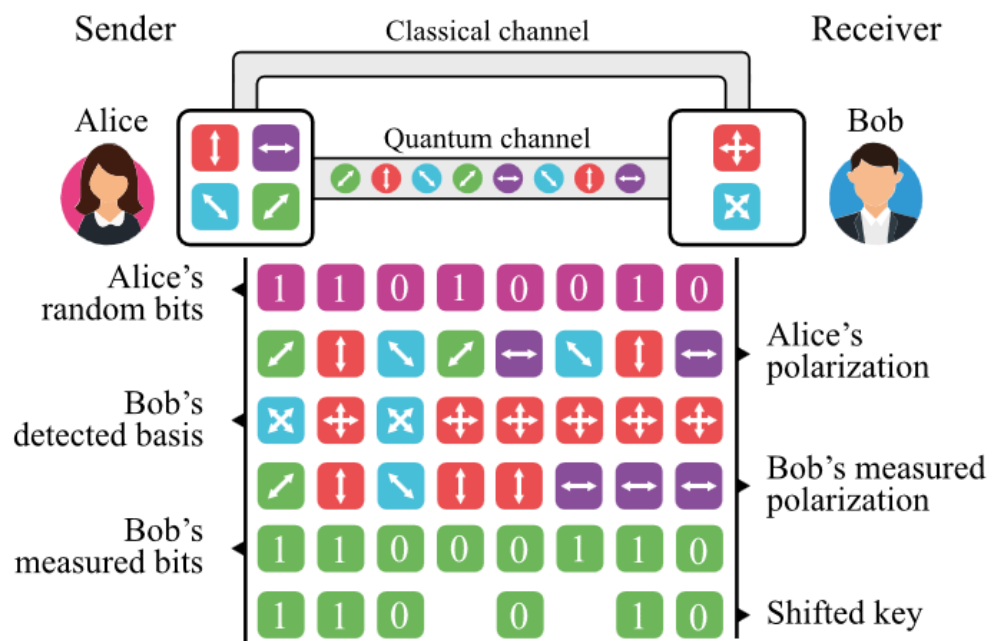
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- Bob generates his basis sequence bits string. As he does not know about Alice's basis sequences, Bob measures each qubit in $|0\rangle$ or $|1\rangle$ and $|+\rangle$ or $|-\rangle$ using his basis sequence bits string.
- Bob and Alice share their basis sequences.

::... Quantum Key Distribution - BB84

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 - for X basis ($+45^\circ/-45^\circ$), a rotate polarization and PBS are used.



- When Bob's estimated basis is different than Alice's one, they both scrap the key bit at that location.
- Alice and Bob both check and compare their key bits. The communication will be aborted if disagreements exceed a given threshold.

::... Quantum Key Distribution - BB84

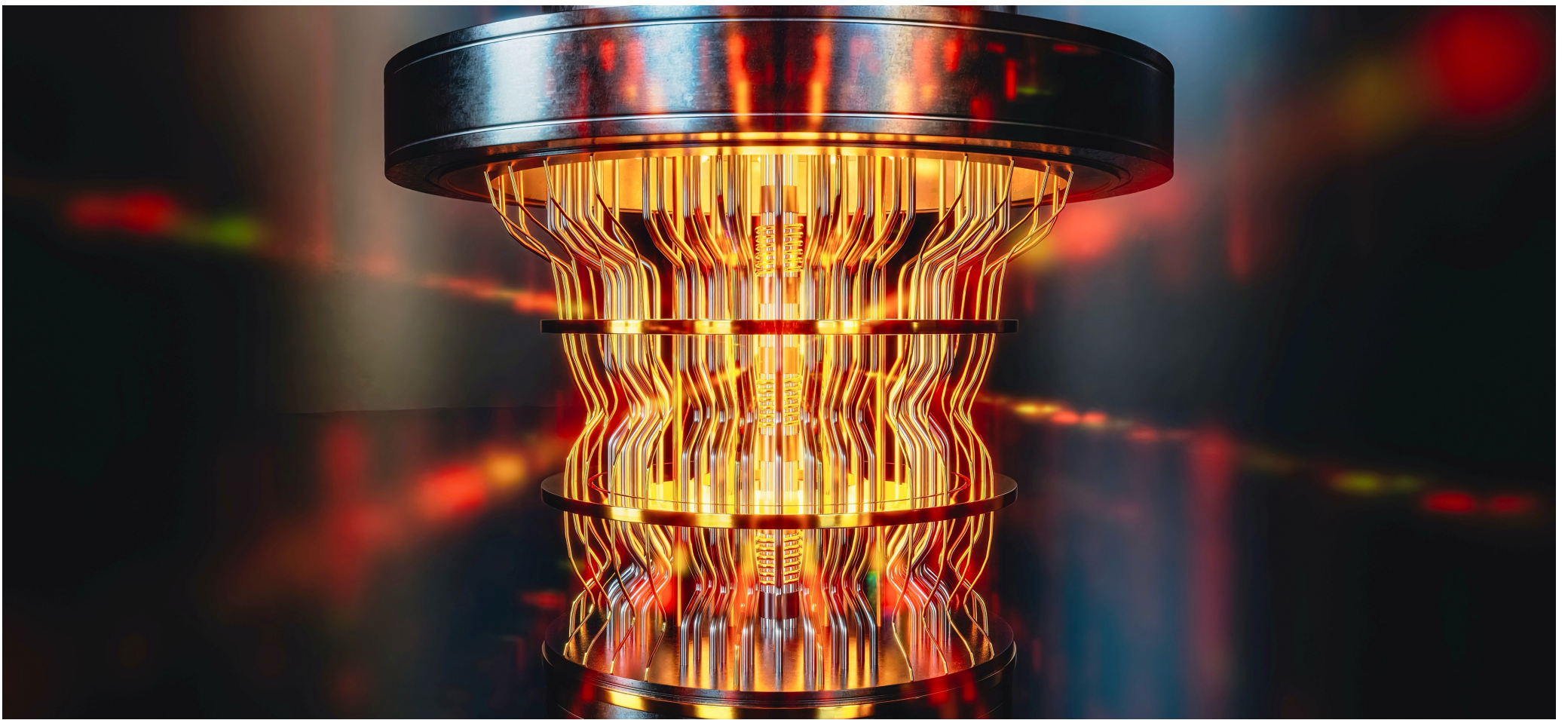
The BB84 protocol, while theoretically secure, faces practical implementation issues:

- **Physical Layer Vulnerabilities:** Real-world implementations use imperfect single-photon sources and detectors. Attackers can exploit these, such as by using blinding attacks on detectors or Trojan horse attacks in which light is injected into the system to analyze its internal settings.
- **Range and Rate Limitations:** Photons are lost over long distances, limiting the practical range of QKD. Moreover, because of the reliance on matching random bases, more than half of the transmitted bits are discarded during the "sifting" phase, reducing the overall secret key rate.
- **Sensitivity to Noise and Environment:** Environmental factors can cause noise in the transmission channel. This noise can be misinterpreted as an eavesdropper, triggering unnecessary security alerts and dropping the key rate.

::... Quantum Key Distribution - BB84

- Imperfect Source Preparation: The protocol assumes perfect polarization states. In practice, imperfect devices may introduce side channels that leak information about the key.
- Need for Authenticated Channel: BB84 requires a pre-existing authenticated classical channel, which can be difficult to implement in some setups.

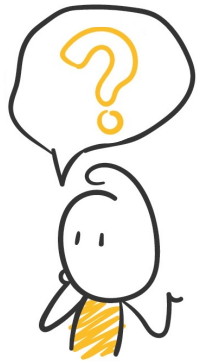
Superior alternatives include entanglement-based protocols such as E91 or BBM92 for improved noise resilience, decoy-state methods to defeat photon-number-splitting attacks, high-dimensional encoding for higher capacity, and quantum memory integration for higher efficiency.



Introduction to Quantum Communications

::... Introduction to Entropy

Alice wants to communicate the results of her random experiments to Bob.



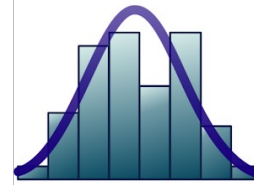
How? What's the information content of these exchanged messages? How can we quantify it?



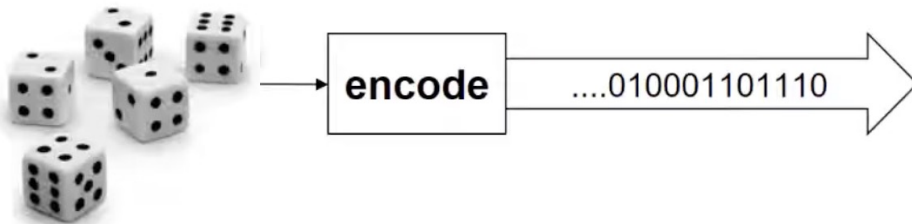
::... Introduction to Entropy

Alice wants to communicate the results of her random experiments to Bob. The communication between Alice and Bob is restricted to only bits being sent over the channel between them. Alice must encode the outcomes of her random experiments in a string of bits.

These outcomes have some probability distribution, which is assumed to be known to both Alice and Bob (who also knows the kind of experiments being done by Alice).



Bob does not know the exact outcome in each repetition of the experiment.

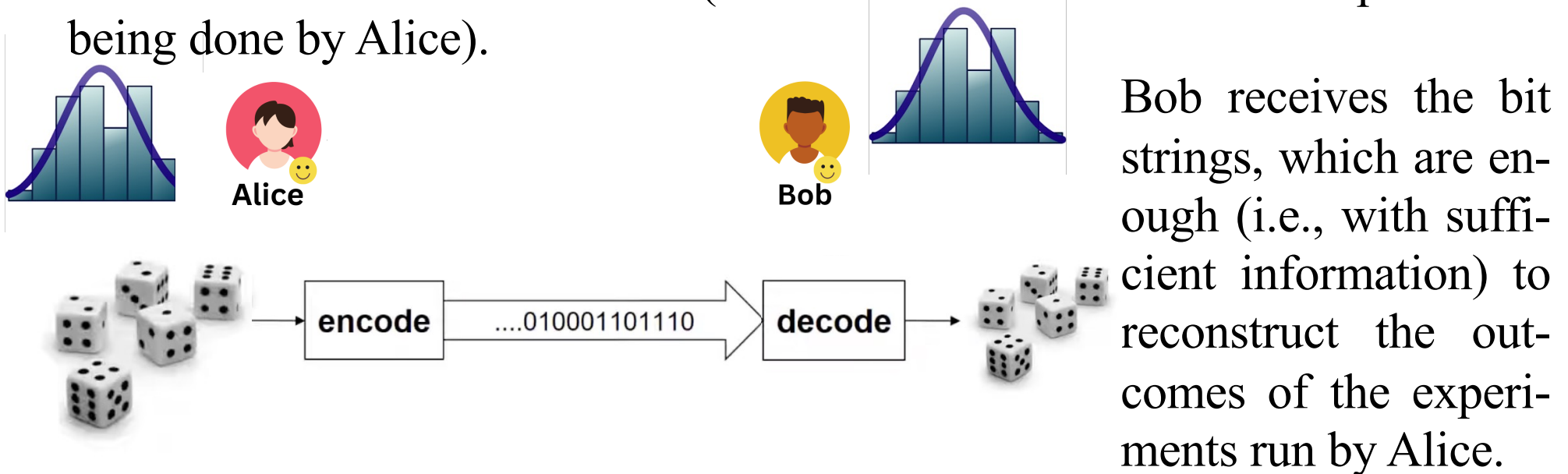


This is the kind of information Alice wants to send to Bob using the bit strings exchanged over the channel.

::... Introduction to Entropy

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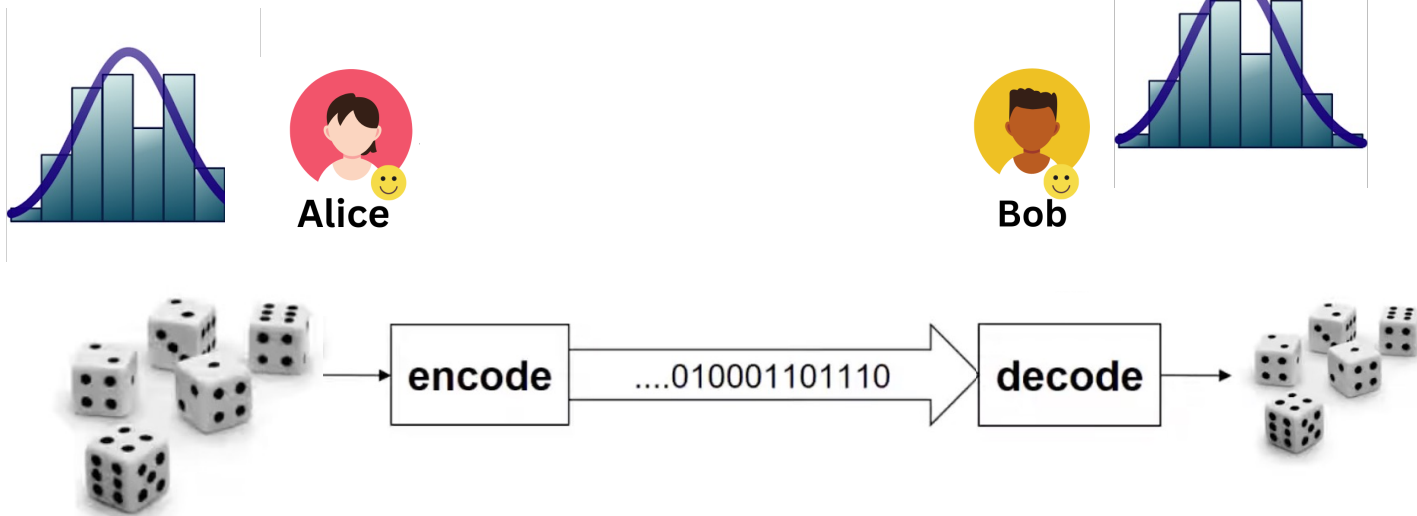


The amount of classical information is defined by means of the **Shannon Entropy**.

::... Introduction to Entropy

Alice is using optimal encoding, by making the bitstrings to be exchanged as short as possible (maximum compression). Alice has performed the experiment n times, and S_n denotes the length of the used bit strings to communicate to Bob the outcomes of these n experiments.

The Shannon entropy is defined as $S := \lim_{n \rightarrow \infty} \frac{S_n}{n}$.



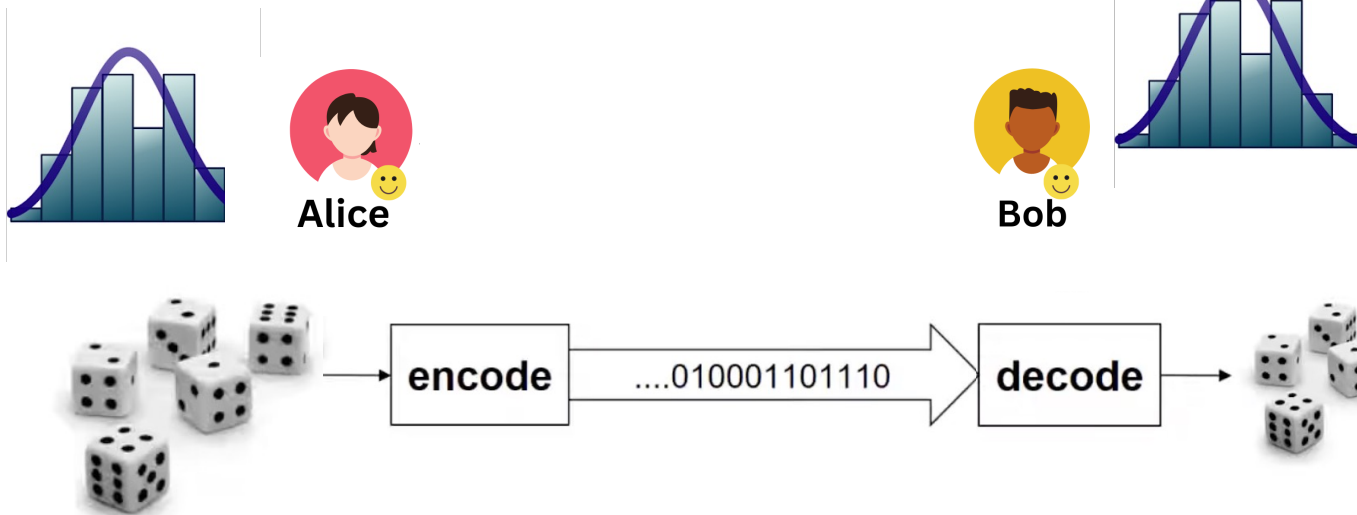
It might not be necessarily optimal if we do not encode each outcome separately with Alice making the experiment, seeing the outcome and encoding it.

It is better to make more experiments in a row, i.e., n times, collect all the outcomes and encode the totality of all collected outcomes to be sent to Bob.

::... Introduction to Entropy

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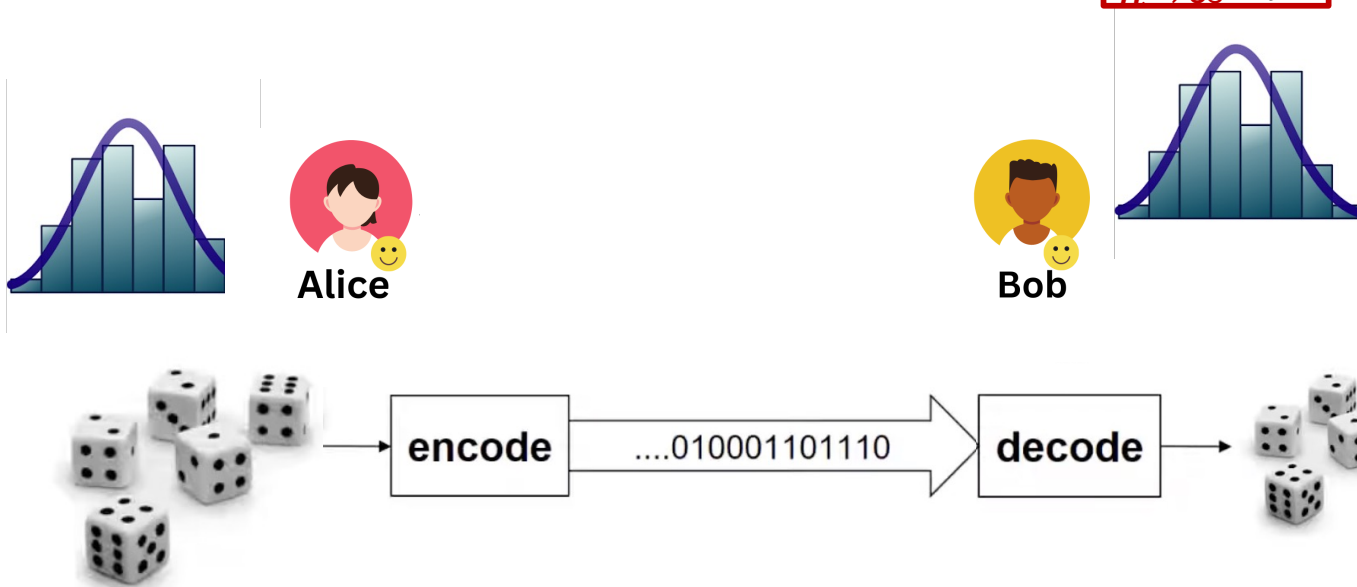
To encode the outcome of rolling a die we need 3 bits to encode the 6 possible outcomes. If Alice rolls the die 3 times, if she encodes separately, she will 9 bits.

An alternative, is to roll the die three times having 6^3 possibilities, which is smaller to 2^8 , so that 8 bits are sufficient to encode the outcomes of three consecutive rolling of a die, having a better compression.

::... Introduction to Entropy

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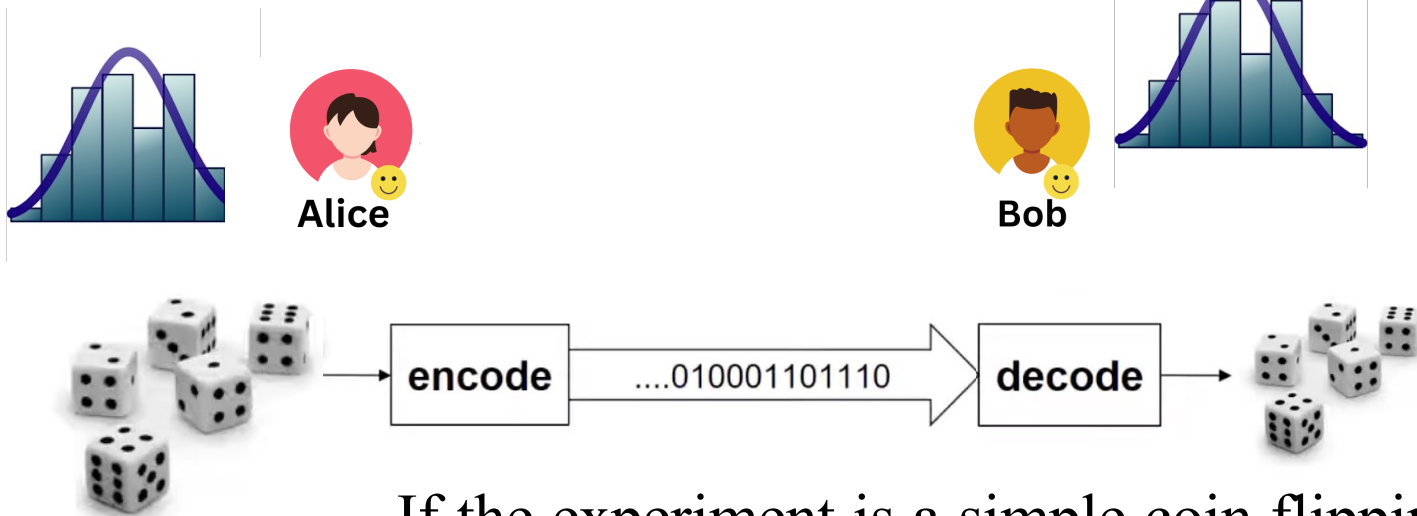
The longer Alice waits before commencing the encoding, the better is the achievable compression.

The limit in the formula represents the optimum one can achieve in an idealised scenario.

::... Introduction to Entropy

Alice is using optimal encoding, by making the bitstrings to be exchanged as short as possible (maximum compression). Alice has performed the experiment n times, and S_n denotes the length of the used bit strings to communicate to Bob the outcomes of these n experiments.

The Shannon entropy is defined as $S := \lim_{n \rightarrow \infty} \frac{S_n}{n} \leq S_1$.



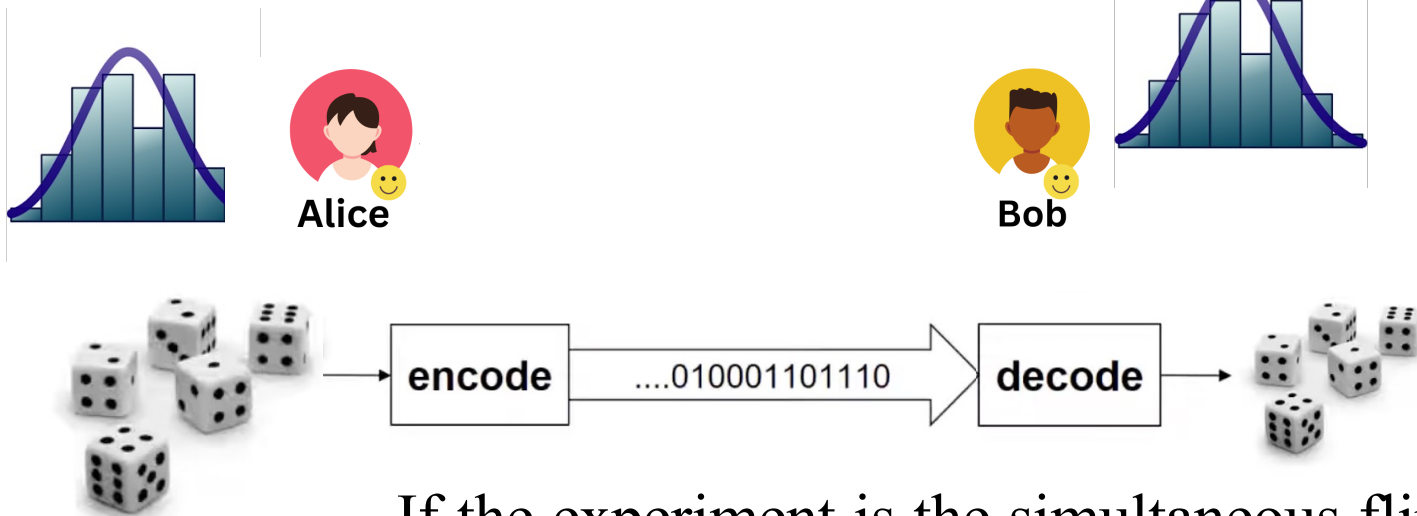
Shannon derived a formula for the entropy concept based on the probability distribution of the stochastic source.

If the experiment is a simple coin flipping (ideal case of 50/50 probabilities) and the encoding of the outcome is one of the possible binary digits. Collecting encoding does not imply an advantage over the separate one, and the entropy is equal to 1.

::... Introduction to Entropy

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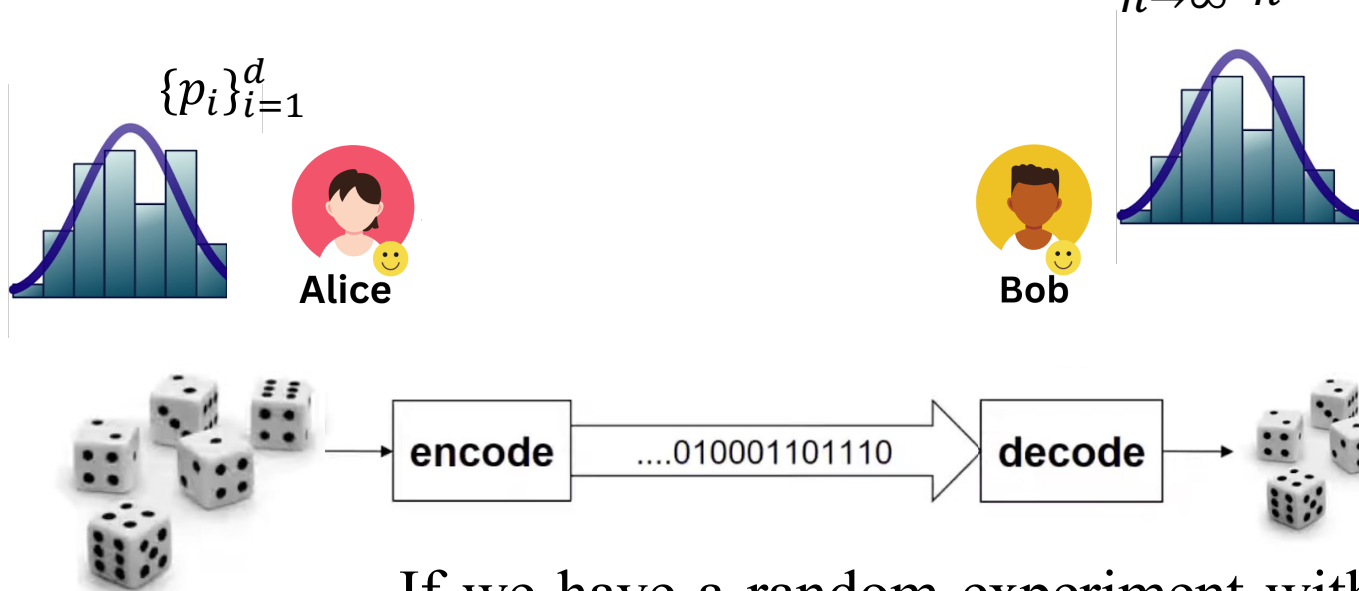
Shannon derived a formula for the entropy concept based on the probability distribution of the stochastic source.

If the experiment is the simultaneous flipping of r coins, there are 2^r possibilities, and Alice needs r bits. The entropy is equal to r , specifically $S = \log_2 2^r = r$, where each outcome has probability $p_i = \frac{1}{2^r}$.

::... Introduction to Entropy

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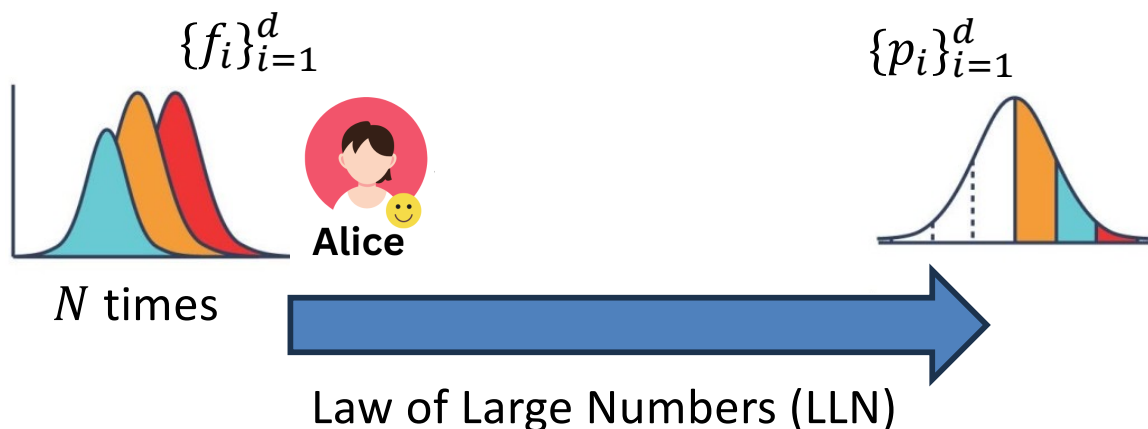
Shannon derived a formula for the entropy concept based on the probability distribution of the stochastic source.

If we have a random experiment with d outcomes all equally possible, so that the probability of a particular outcome is $p_i = \frac{1}{d}$, we have that $S = \log_2 d$

::... Introduction to Entropy

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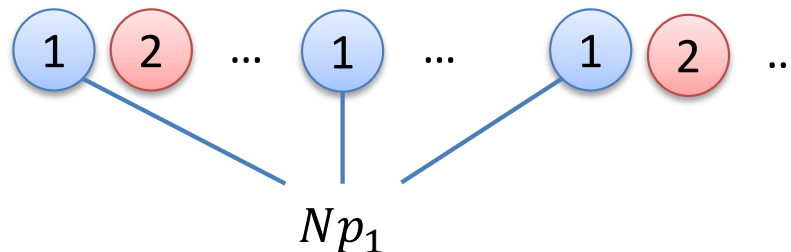
Given the probability distribution and N repetitions of the experiment, if N is large enough, the relative frequencies with which the various outcomes occur coincide with the probabilities.

If we have a sequence of i.i.d. random variables, the sample average converges to the expected value as the number of trials N approaches infinity, or the difference between the Empirical Frequency f_i and Theoretical Probability p_i shrinks toward zero, $f_i - p_i \rightarrow 0$ as $N \rightarrow \infty$.

::... Introduction to Entropy

For the LLN, if N goes to infinity, encountering sequences being typical, where the sample frequencies of the outcomes are very close to their actual probabilities, is highly probable. A typical sequence has the following formulation for the probabilities of its outcomes:

- for outcome 1, its occurrence probability is p_1 while the probability this outcome happens in the sequence is $p_1^{Np_1}$, the LLN tells us that $k_1 \approx Np_1$, where k_1 is the number of occurrences of outcome 1.



- for outcome 2, its occurrence probability is p_2 while the probability this outcome happens in the sequence is $p_2^{Np_2}$.
- this applies for all the outcomes up to the last one d , whose occurrence probability is p_d while the probability this outcome happens in the sequence is $p_d^{Np_d}$.

::... Introduction to Entropy

For large N , the total probability of the typical sequence approaches 1, and given $\mathfrak{D}$ as the number of typical sequences, we have that

$$\mathfrak{D} = \frac{1}{p_1^{Np_1} \cdots p_d^{Np_d}}$$

Even if there are $|X|^N$ total possible sequences, only $\mathfrak{D}$ of them are likely to happen. The ratio of typical sequences to total sequences ($\mathfrak{D} / |X|^N$) usually goes to zero, which is why data compression is possible.

The number of bit necessary for the communication of the collected outcomes is $\log_2 \mathfrak{D}$, so per each trial we have the Shannon entropy

$$\begin{aligned} S(\rho) &= \frac{\log_2 \mathfrak{D}}{N} = -\frac{1}{N} \log_2 p_1^{Np_1} \cdots p_d^{Np_d} = -\frac{1}{N} \sum_i \underbrace{\log_2 p_i^{Np_i}}_{Np_i \log_2 p_i} = \\ &= -\sum_i p_i \log_2 p_i \end{aligned}$$

This is the general formula per for the Shannon entropy, for $\rho = \{p_i\}_{i=1}^d$.

::... Characteristics of Entropy

The Shannon entropy indicates the minimum number of bits needed to inform Bob of the outcomes of the random experiments done by Alice over a very large number of repetitions N , for $\rho = \{p_i\}_{i=1}^d$:

$$S(\rho) = - \sum_{i=1}^d p_i \log_2 p_i .$$

Here are some key features:

- non-negative values, $S(\rho) \geq 0$, and it can be null if there is zero uncertainty or no diversity in the random experiment being communicated or one outcome in the experiment occurs deterministically and has probability equal to 1, $S(\rho) = 0 \iff \rho = \{0, \dots, 1, \dots, 0\}$. This is due to the fact that Bob knows ρ , so Alice has to send nothing to Bob as he knows the outcome occurred from ρ .
- permutation invariant, $S(\pi(\rho)) = S(\rho)$, where $\pi(\cdot)$ is a permutation or reordering of the underlying probability distribution of the outcomes.

::... Characteristics of Entropy

- additive, $S(\rho_{AB}) = S(\rho_A) + S(B|A)$, consider Alice making two kinds of experiments in parallel, e.g., rolling a die and flipping a coin, and sending to Bob information of these combined outcomes.
 - If the experiments are perfectly independent, the number of bits to send is just the sum of those needed for each of them.

$$S(\rho_{AB}) = S(\rho_A) + S(\rho_B)$$

- We cannot add them up if the experiments are correlated. One random experiment can influence the probability distribution of a second random experiment, which uses conditional probabilities. The entropy is still additive but the second contribution $S(B|A)$ is called the **conditional entropy**:

$$S(B|A) = \sum_i \text{prob}(a_i) S(\rho_{B|a_i})$$

It quantifies the amount of bits needed to describe a random experiment B knowing the outcome of another experiment A. This conditional entropy is never greater than the one of B.

::... Characteristics of Entropy

When there is correlation, knowing the outcome of A tells something also about B , so the uncertainty is lower and less bits are needed to encode the results of B : $S(B|A) < S(\rho_B)$.

Therefore, the **joint entropy** satisfies the properties that

$$S(\rho_{AB}) \leq S(\rho_A) + S(\rho_B).$$

These three properties uniquely, in addition to continuity, specify the Shannon entropy, up to a positive multiplicative constant. In fact, the Khinchin-Shannon's Theorem (or the Faddeev-Chintschin axiomatization) states that a specific mathematical function for the entropy is identified by imposing the three properties to it, among any possible information measure one can think of.

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$$I(A : B) = S(\rho_B) - S(B|A) \geq 0$$

The **Mutual Information** of the two experiments A and B quantifies how much the outcome of A tells about the outcome of B.

::... Characteristics of Entropy

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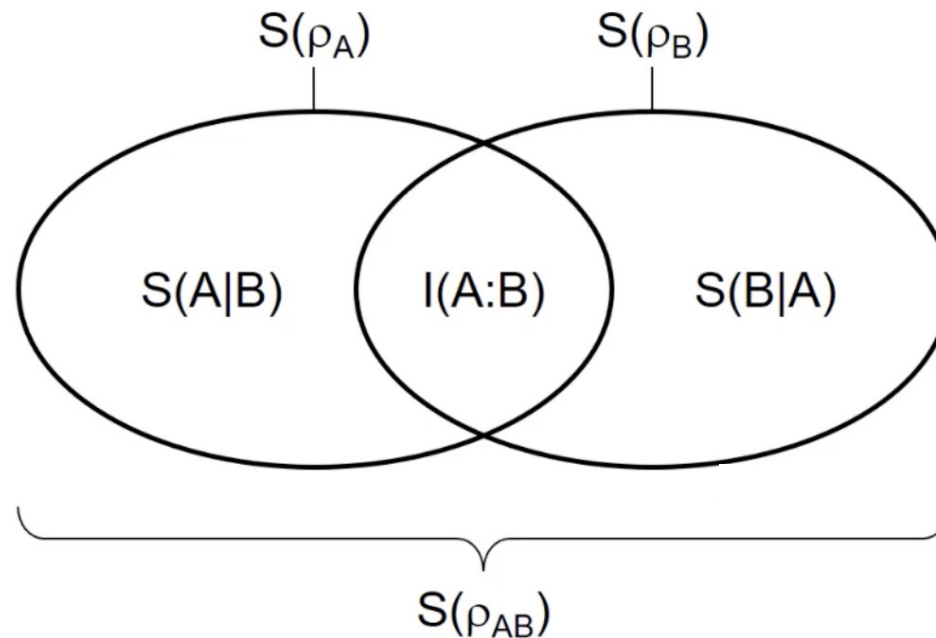
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$$I(A : B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}) \geq 0$$

The **Mutual Information** has a symmetric definition in A and B. It is zero if A and B are uncorrelated and positive otherwise.

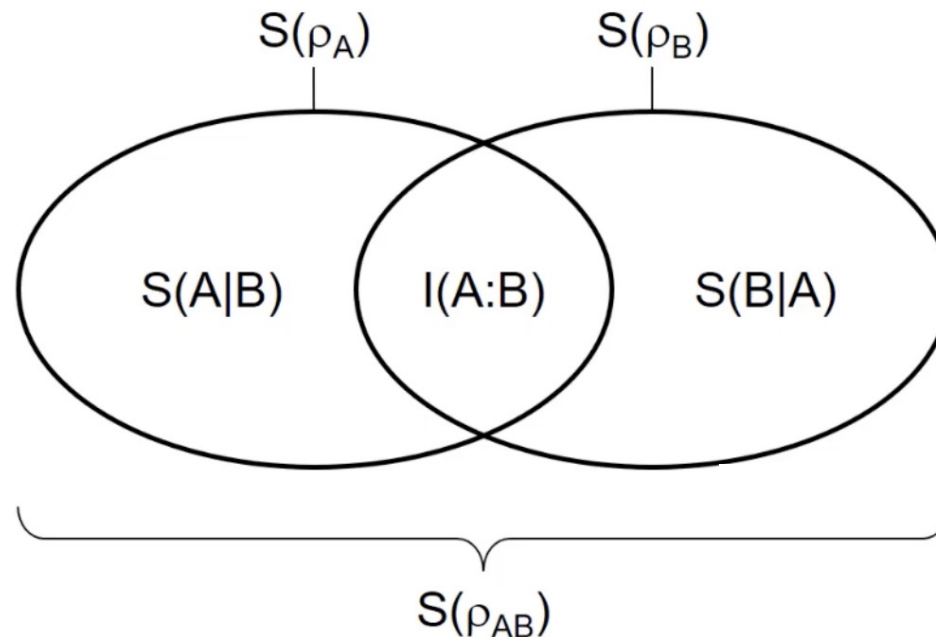
::... Characteristics of Entropy

The relationships among these classical entropy concepts can be illustrated with a Venn diagram as follows:



::... Characteristics of Entropy

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There is another entropy concept: given two probability distributions $\rho = \{p_i\}_{i=1}^d$ and $\sigma = \{q_i\}_{i=1}^d$, their **relative entropy** is

$$S(\rho||\sigma) := \sum_i p_i (\log_2 p_i - \log_2 q_i)$$

and measures the "distance" between the two distributions ρ and σ .

::... Characteristics of Entropy

This concept holds two key interesting properties:

- It is never negative:

$$S(\rho||\sigma) = \begin{cases} = 0 & : \rho = \sigma \\ > 0 & : \rho \neq \sigma \end{cases}$$

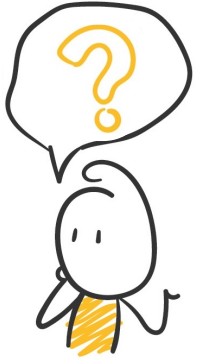
The large the distance between the distributions, the large is the value assumed by the relative entropy, making this a good distance indication and Kullback-Leibler Divergence (D_{KL}). However, it is not a metric (but a divergence) as it is not symmetric, i.e., $S(\rho||\sigma) \neq S(\sigma||\rho)$, but if the two distribution are close each other, the relative entropy is approximately symmetric.

- It can be related to the mutual information:

$$I(A : B) = \sum_i prob(a_i) S(\rho_{B|a_i} || \rho_B)$$

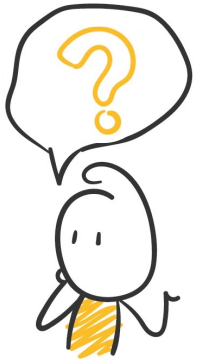
The mutual information is essentially a measure of how much two distribution are "not independent", or measures the relative entropy between those two distributions.

::... Quantum Communication



Whether or not and to what extent quantum systems can be used as carriers of classical information?

::... Quantum Communication



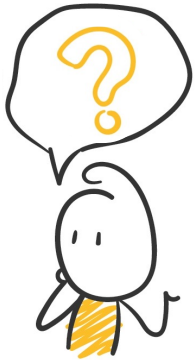
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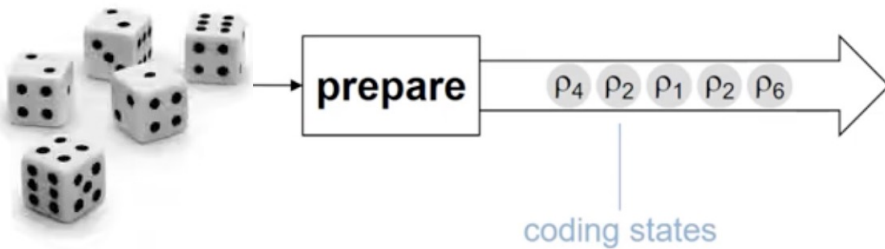
Let us consider Bob and Alice running a classical random experiment, with the outcomes communicated to Bob.



::... Quantum Communication



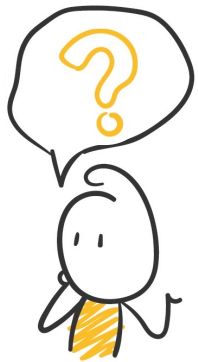
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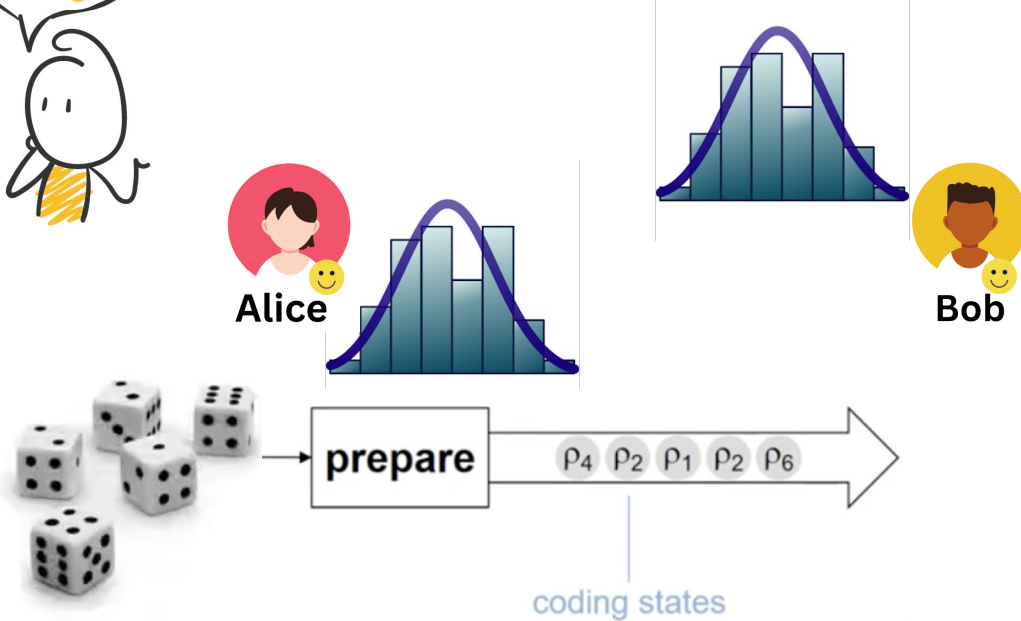
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Now, Alice does not use classical bit strings; instead, she prepares quantum states and sends them to Bob.

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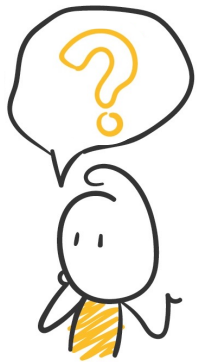


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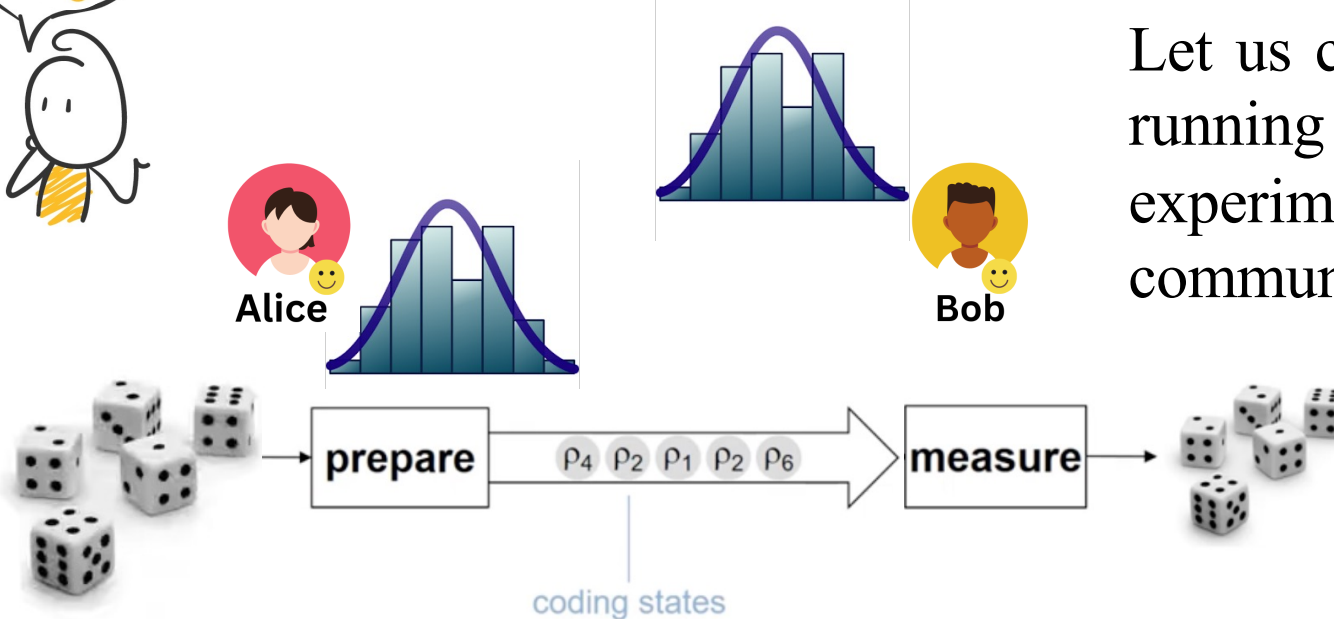
Bob and Alice know the probability distribution of the experiment's outcomes, and Bob wants to know the exact outcomes seen by Alice.

::... Quantum Communication



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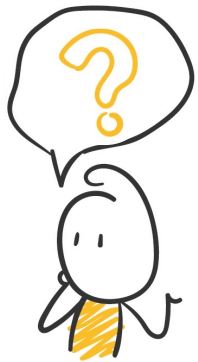


Bob receives these prepared states and performs measurements.

Is even possible for Bob to reconstruct the exact outcomes by looking at the results of the measurements made over the quantum states received from and prepared by Alice?

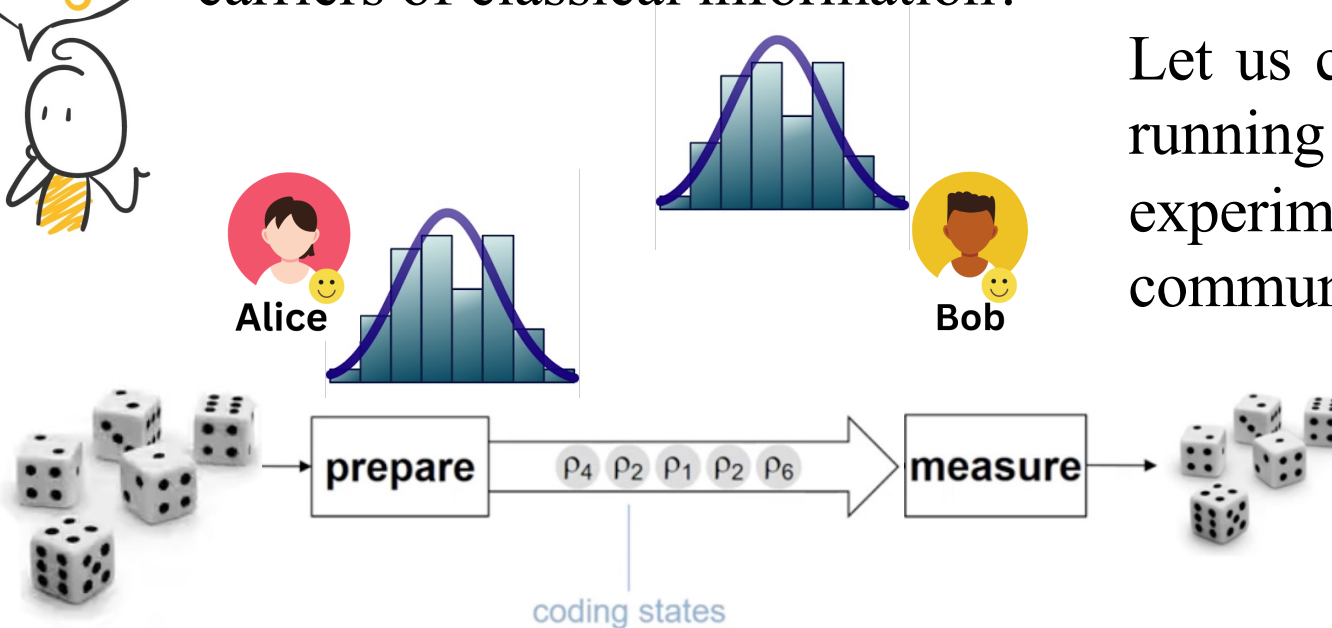


::... Quantum Communication



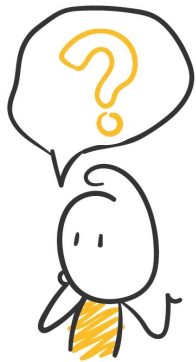
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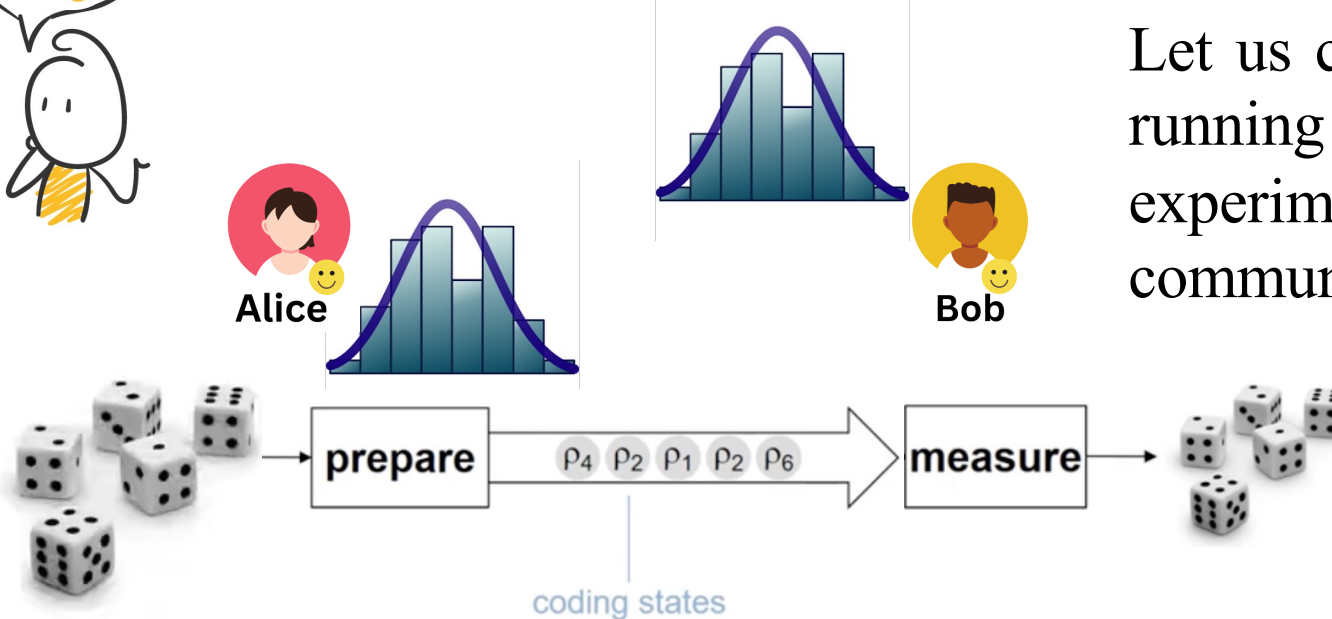
a_i denotes the outcome of the random experiment of Alice at the i -th execution and ρ_A indicates the probability distribution over the possible outcomes. Bob's measurements can be considered another random experiment: in quantum mechanics, the measurement is not certain, but there are various possible outcomes, which occur with certain probabilities. b_j denotes the outcome of Bob's measurements.

::... Quantum Communication



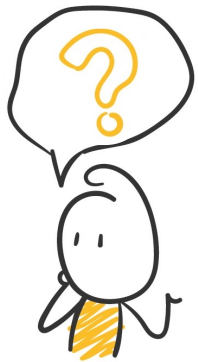
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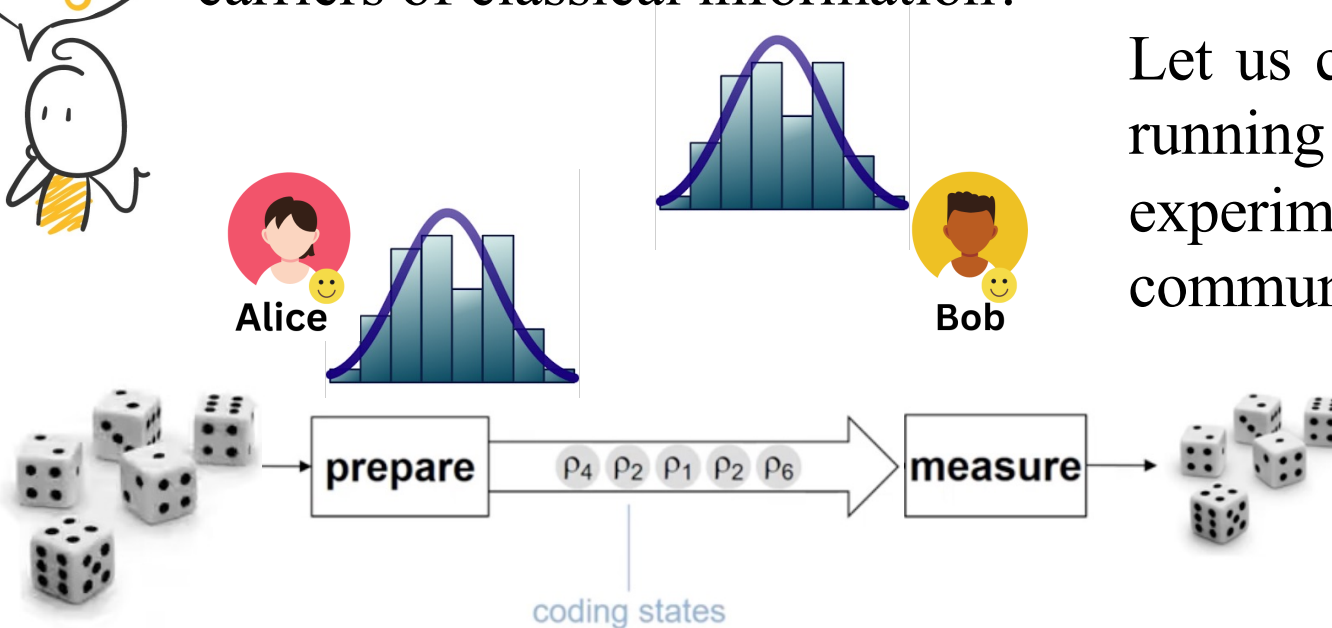
$\hat{\rho}_i$ is a coding state prepared by Alice based on a_i and sent to Bob. The hat indicates a statistical operator and this is according to the Gleason's theorem saying for a Hilbert space $\mathcal{H}$ with a dimension $d \geq 3$, any "measure" μ that assigns a probability to every closed subspace can be represented by a unique density operator ρ . A state is not just a vector $|\psi\rangle$; it is a rule for assigning probabilities to measurement outcomes.

::... Quantum Communication



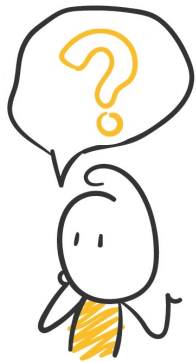
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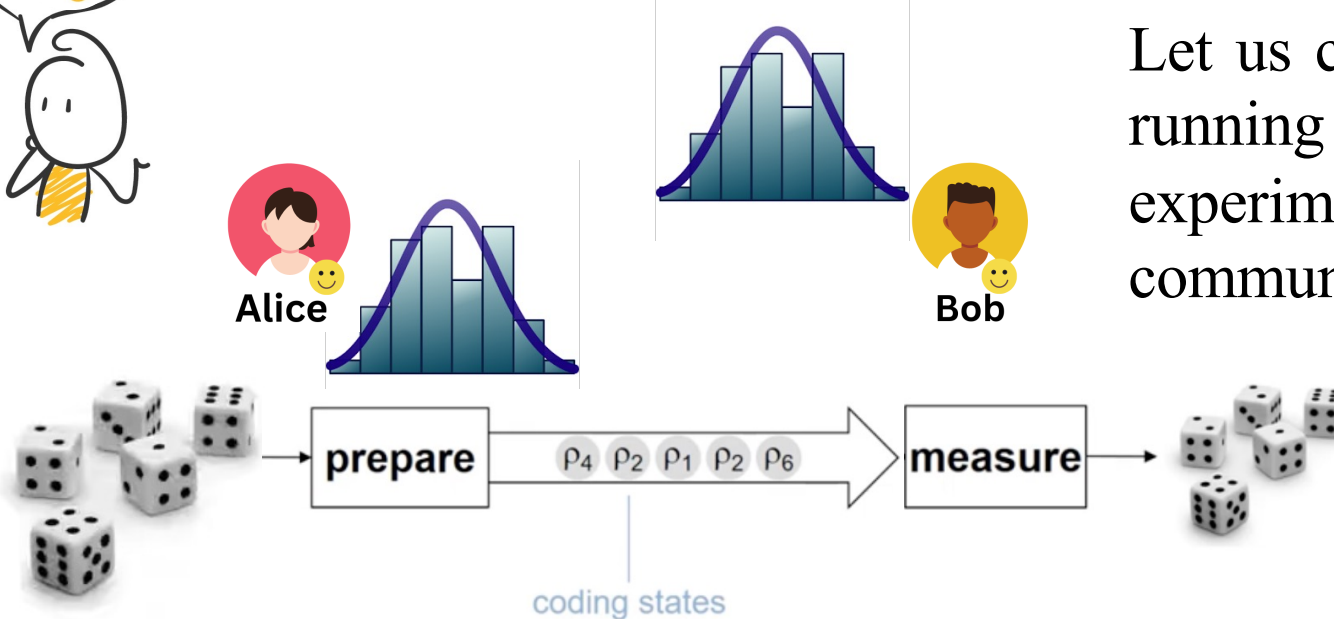
If Bob already knows the outcome of Alice's experiment, or a_i , the quantum system received from Alice must have been prepared in state $\hat{\rho}_i$ and b_j corresponds to the projector (or a closed subspace in $\mathcal{H}$) onto the relevant Eigenspace $\hat{P}_j$. The Gleason's theorem tells us to calculate the probability of b_j known a_i by the trace: $p(b_j|a_i) = \text{tr}(\hat{\rho}_i \hat{P}_j)$.

::... Quantum Communication



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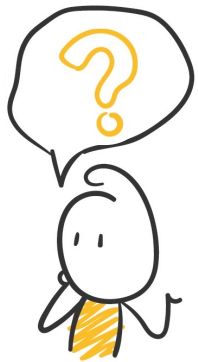
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If Bob already knows the outcome of the quantum system received from Alice, then the probability of b_j corresponding to the projection onto the relevant Eigenspace $\hat{P}_j$. The probability of b_j known a_i by the **trace**: $p(b_j|a_i) = \text{tr}(\hat{\rho}_i \hat{P}_j)$.

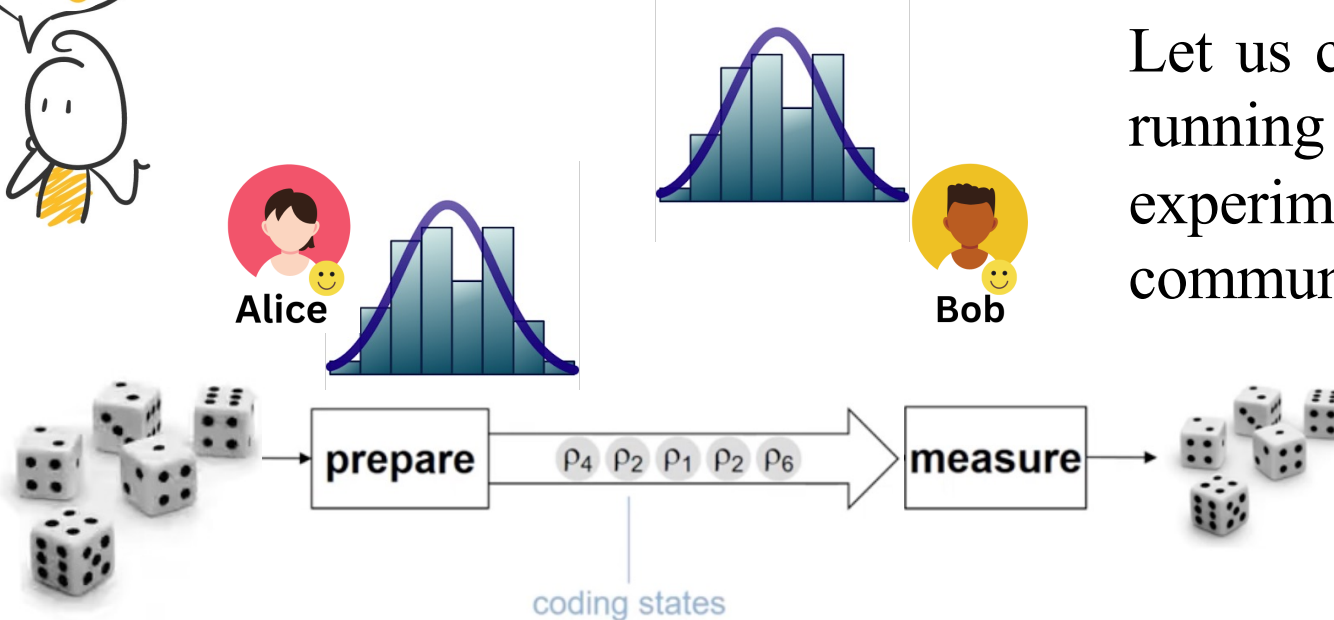
The trace of a square matrix A is the sum of the elements on its main diagonal. $\text{tr}(\rho P)$ is essentially "averaging" the measurement operator P over the state ρ .

::... Quantum Communication



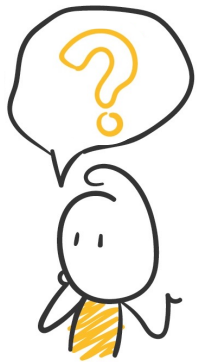
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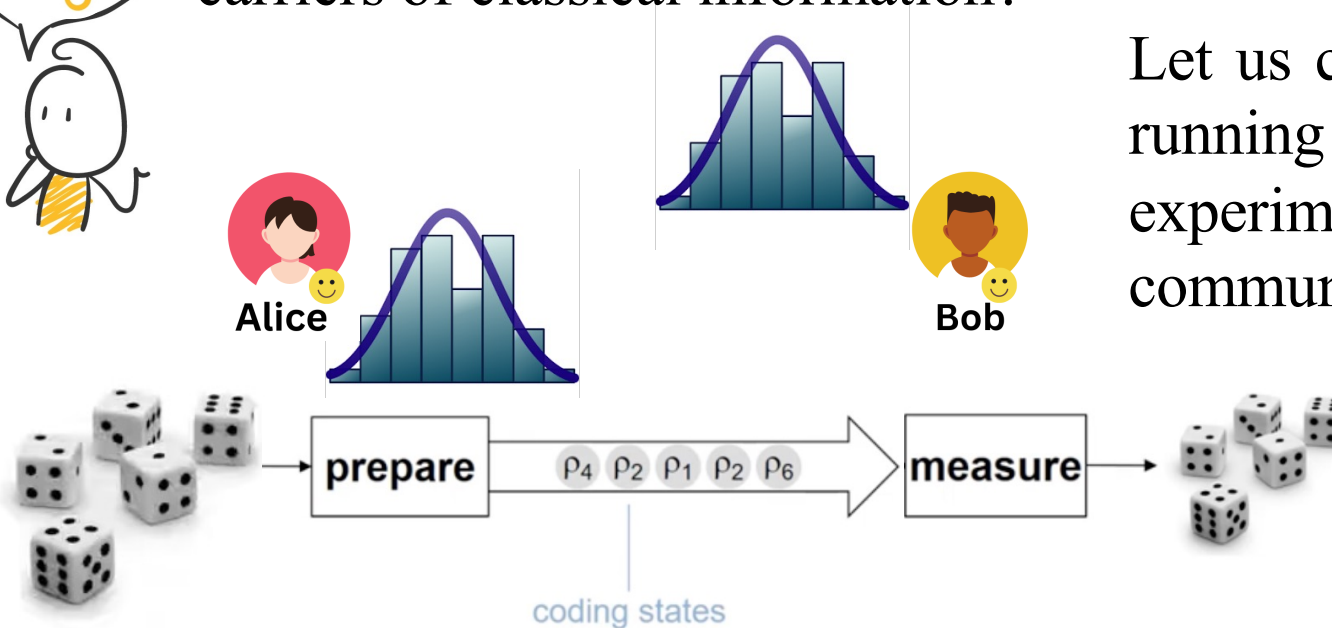
If Bob does not know the outcome of Alice's experiment, or a_i , but he does know $\rho_A = \{p(a_i)\}_{i=1}^d$, then the distributed probability of the measurement done by Bob is the weighted average of probabilities of the previous case weighted with the probabilities of the various outcomes at Alice side: $p(b_j) = \text{tr}(\hat{\rho}\hat{P}_j)$, where $\hat{\rho} := \sum_i p(a_i) \hat{\rho}_i$.

::... Quantum Communication



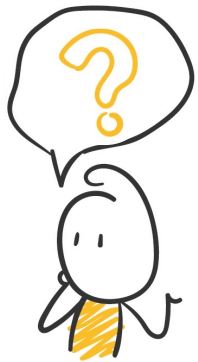
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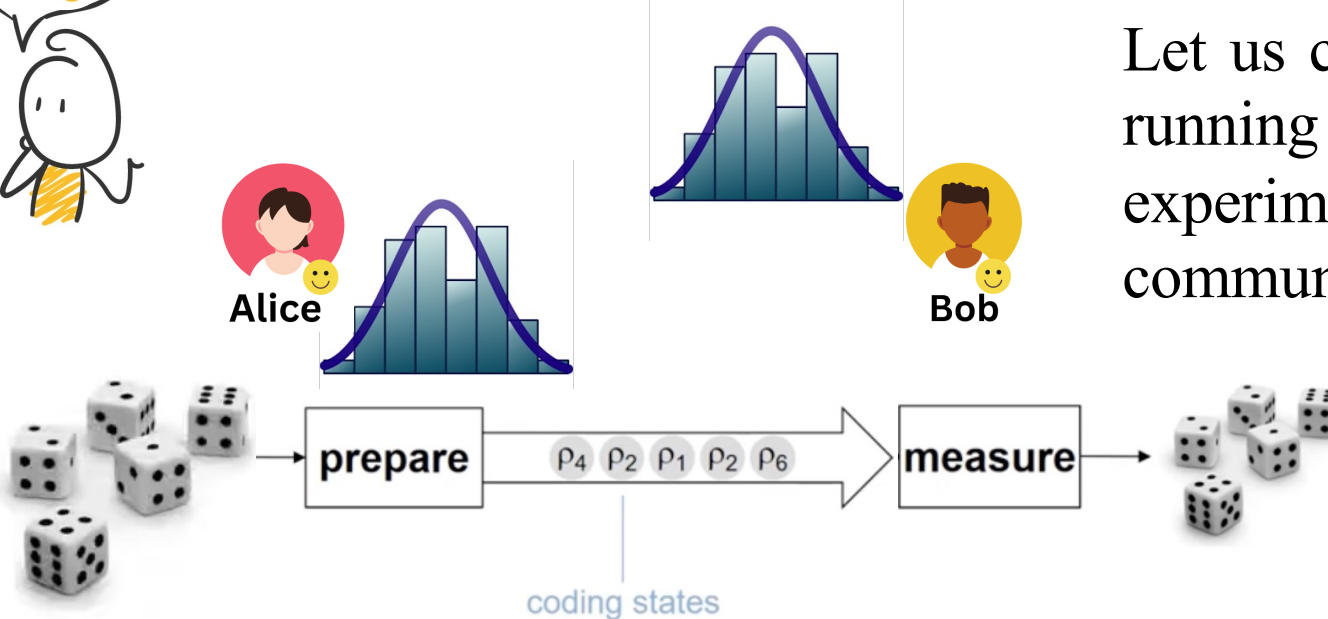
The communication from Alice to Bob is successful if the amount of information that was encoded by Alice equals the information obtained by Bob's measurements revealing about the outcome of Alice's experiment.

::... Quantum Communication



Whether or not and to what extent quantum systems can be used as carriers of classical information?

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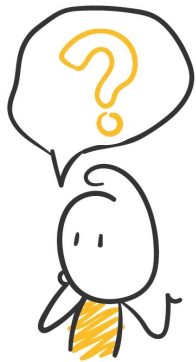


The communication from Alice to Bob is successful if **the amount of information that was encoded by Alice** equals the information obtained by Bob's measurements revealing about the outcome of Alice's experiment.

Shannon entropy associated to Alice's random experiment

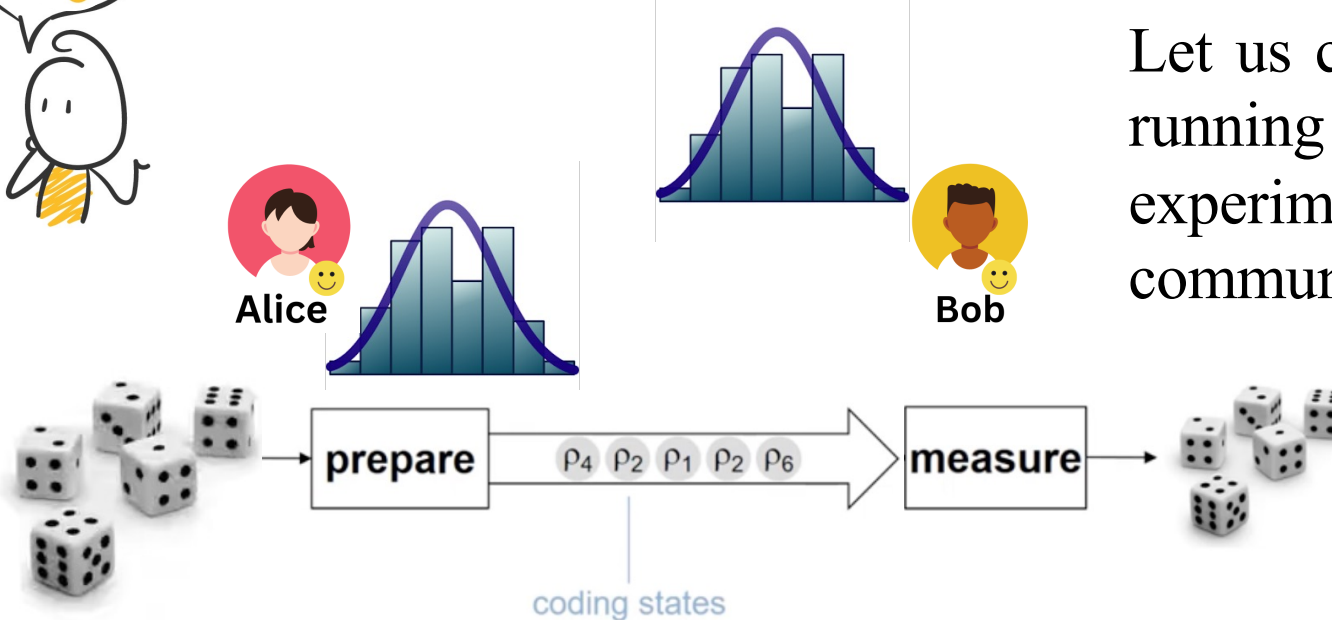
$$S(\rho_A)$$

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The mutual information of A and B

$$S(\rho_A) = I(A : B)$$



::... Quantum Communication

To have a formalization of such equality we need to introduce two key concepts:

::... Quantum Communication

To have a formalization of such equality, we need to introduce two key mathematical objects:

- **Quantum relative entropy** (also known as the Umegaki relative entropy) is essentially a measure of "distance" or distinguishability between two quantum states. While it isn't a distance in the strict mathematical sense (it's not symmetric), it tells you how much one quantum state $\hat{\rho}$ differs from a reference state $\hat{\sigma}$.

$$S(\hat{\rho}||\hat{\sigma}) := \text{tr}(\hat{\rho} \log_2 \hat{\rho} - \hat{\rho} \log_2 \hat{\sigma})$$

This is never negative, and zero if and only if the two quantum states are identical and strictly positive if different, and it gets larger if the two statistical operators differ more strongly than each other. As the classical counterpart, it is not symmetric, but is approximately symmetric when the two states are closer.

::... Quantum Communication

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- **Holevo bound** tells us exactly how much classical information we can reliably extract from a quantum system, and is the upper limit for the Mutual Information between Alice's encoding and Bob's measurement result.

$$H[\{p(a_i), \hat{\rho}_i\}] := \sum_i p(a_i) S(\hat{\rho}_i || \hat{\rho})$$

::... Quantum Communication

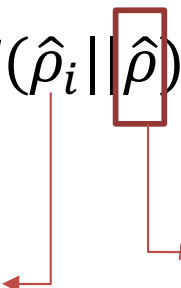
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The encoding state chosen by Alice for outcome a_i  $\hat{\rho} := \sum_i p(a_i) \hat{\rho}_i$

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The von Neumann entropy of a quantum state

$$= S(\hat{\rho}) - \sum_i p(a_i) S(\hat{\rho}_i)$$


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If we want a successful quantum communication, the used coding states should be distinguishable. If each outcome of Alice's experiment were encoded in the same quantum state, its weighted average would be the same quantum state, so relative entropies would be zero and the Holevo function would be zero. In this case we cannot transmit any information, as there is no way to distinguish the outcomes.

::... Quantum Communication

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- **Holevo bound** tells us exactly how much classical information we can reliably extract from a quantum system.

If the coding states are far apart from each other in a mathematical sense, it should be more possible to transmit the classical information of Alice's experiment to Bob. The Holevo function is the mathematical definition of the distinguishability and distance among quantum states.

::... Quantum Communication

The Holevo function satisfies a number of interesting inequalities:

- The amount of information that Bob can extract from his measurements about the outcome of the Alice's experiment is bounded from above by the Holevo function:

$$I(A : B) \leq H[\{p(a_i), \hat{\rho}_i\}]$$

This indicates that the coding states should be clearly distinguishable, and the Holevo function has to take a large value to enable Bob to extract a lot of information from his measurements about the Alice's outcomes. The equality is obtained whenever the coding state commute each other.

In quantum mechanics, two operators (observables) A and B commute if their order of application does not matter, expressed as the commutator $[A, B] = AB - BA = 0$. Commuting operators share a common set of eigenvectors and can be measured simultaneously with arbitrary precision, meaning the measurement of one does not affect the other.

::... Quantum Communication

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The Holevo function, or the amount of information to be encoded in quantum states, making them as distinguishable as possible, is bounded from above by the Shannon entropy of the outcomes of the Alice's experiment. This is intuitive as we cannot encode more information than what we have.

The equality holds whenever the coding quantum states are orthogonal: in the case of pure states their scalar product vanishes.

Two quantum states $|\psi\rangle$ and $|\phi\rangle$ are orthogonal if their inner product is zero: $\langle\psi | \phi\rangle = 0$. In the language of vectors, this means they are at a "right angle" to each other in Hilbert space.

::... Quantum Communication

The transmission is successful if the mutual information of A and B is equal to the amount of information Alice encoded.

$$I(A : B) = S(\rho_A)$$

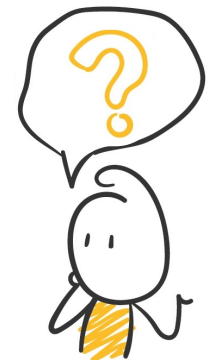
These are the two extremes of the previous inequalities:

$$I(A : B) \leq H[\{p(a_i), \hat{\rho}_i\}] \leq S(\rho_A).$$

Then, we need equalities in both cases, so we have to satisfy both conditions. The condition on the right side is stronger than the one on the left side: if two quantum states are orthogonal, they automatically commute.

The transmission of classical information via quantum states is possible (successful) if and only if the coding states are orthogonal.

If we have a qubit, how much classical information can be encoded in it so that the recipient can read it out?



::... Quantum Communication

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A qubit has a continuum of pure states (i.e., the points on the Bloch sphere), and hypothetically, we can have a lot of classical information and encode them on a specific position of the sphere, but that would be unsefull for the receipient, as he/she is incapable to extract this information via a measurement. As quantum states must be orthogonal, there are only two orthogonal states in the two-dimensional Hilbert space, depending on the measurement basis being used.

::... Quantum Communication

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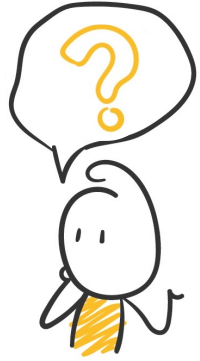
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These two only-possible orthogonal states corresponds to the two possible values of the classical information.

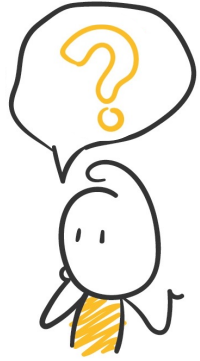
A qubit cannot carry more information than a classical bit.
Quantum communication does not increase the amount of information carries by qubits than what possible in the classical communication.

::... Quantum Information



What is genuine quantum information?

::... Quantum Information

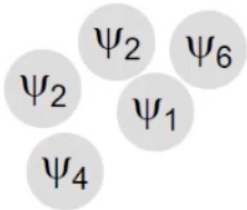


What is genuine quantum information?

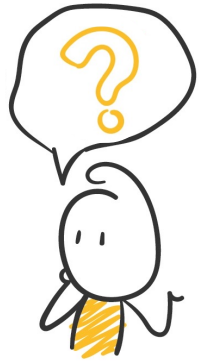
Let's reason around it using the scenario with Alice and Bob.



However, Alice no longer performs the classical random experiment; rather, with a certain probability distribution, she produces various quantum states, and these have to be pure states.

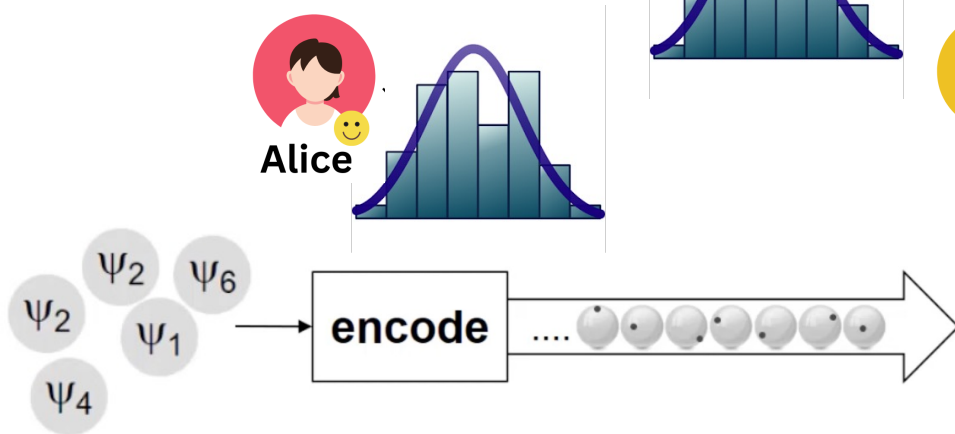


::: Quantum Information



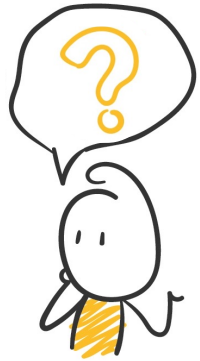
What is genuine quantum information?

Let's reason around it using the scenario with Alice and Bob.



Alice wants to send to Bob these quantum states via a quantum channel using qubits. As in the classical setting, the probability distribution with which the Alice's pure states are produced is known to Alice and Bob.

::... Quantum Information



What is genuine quantum information?

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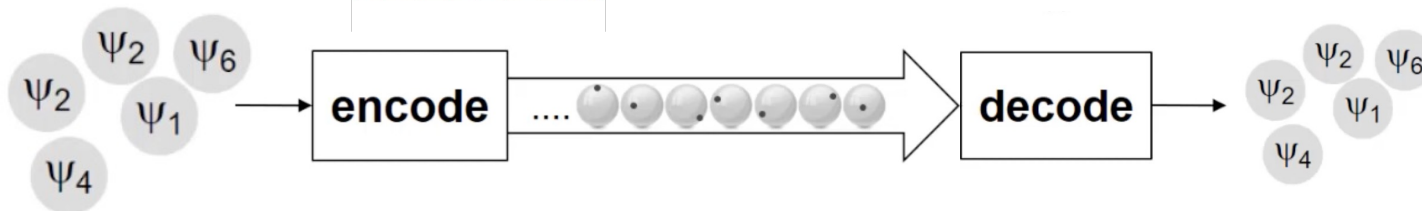


Alice



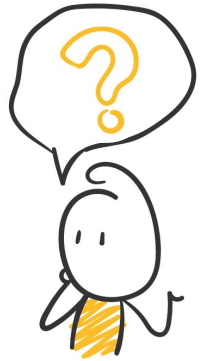
Bob

Alice wants to send to Bob these quantum states via a quantum channel using qubits.



Bob receives the qubits and decodes them to reconstructs the quantum states produces by Alice during her quantum experiment.

::... Quantum Information



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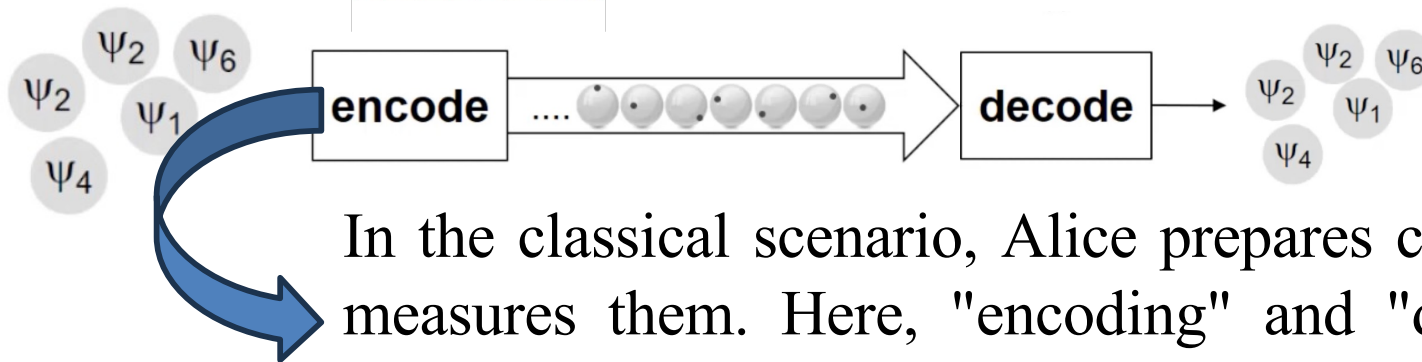


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Bob

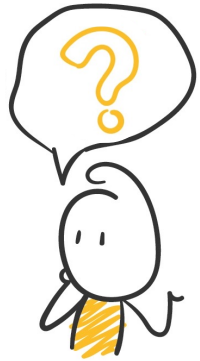
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In the classical scenario, Alice prepares coding states, and Bob measures them. Here, "encoding" and "decoding" are unitary transformations, so there is no preparation or measurement.

Alice has $|\psi_{i_1}\rangle \otimes \dots \otimes |\psi_{i_N}\rangle$ and does $U|\psi_{i_1}\rangle \otimes \dots \otimes |\psi_{i_N}\rangle$, Bob performs a dual unitary transformation to obtain back $|\psi_{i_1}\rangle \otimes \dots \otimes |\psi_{i_N}\rangle$.

::... Quantum Information



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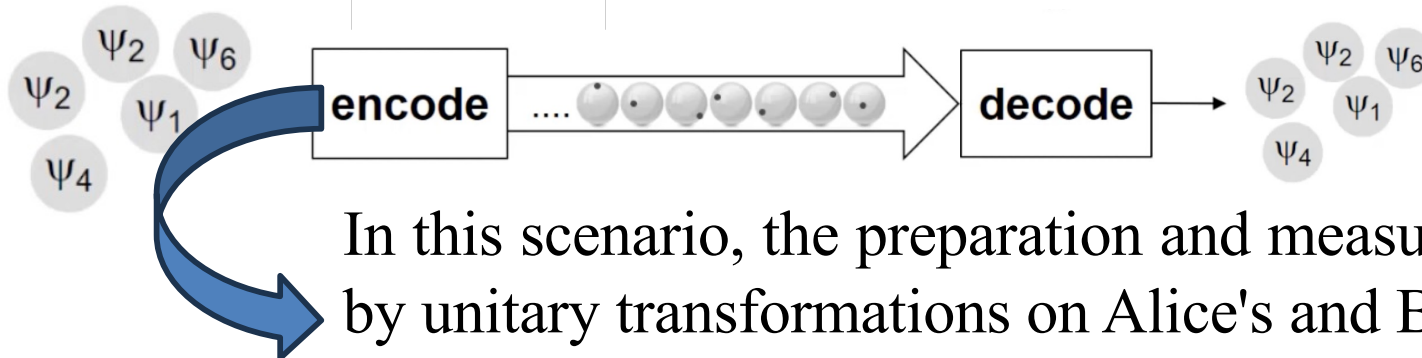


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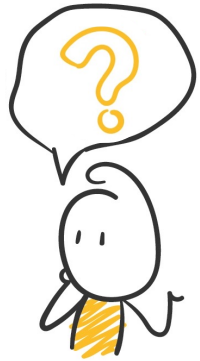
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In this scenario, the preparation and measurement are substituted by unitary transformations on Alice's and Bob's sides.

Alice performs the tensor product of her individual pure states (the order is important, and the outcomes follow a specific probability distribution). This is the tensor product Bob wants to reconstruct.

::... Quantum Information



What is genuine quantum information?

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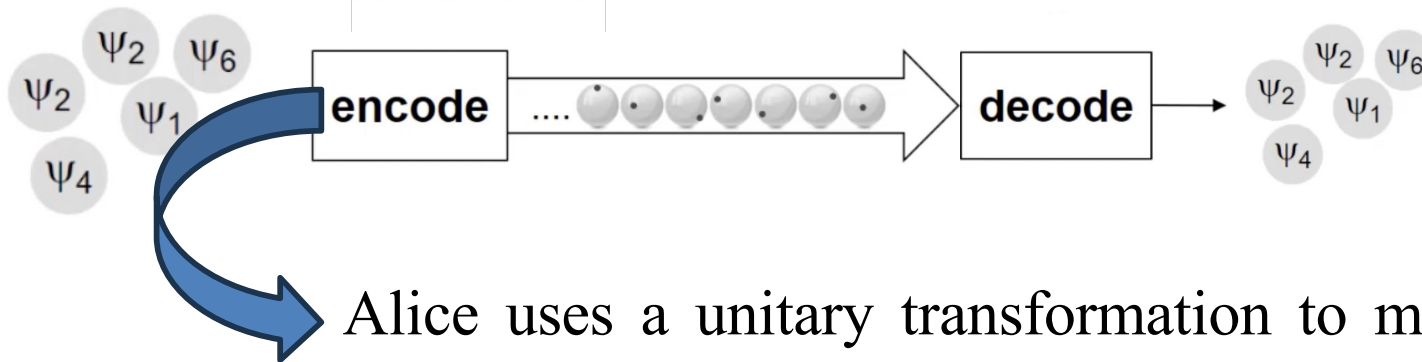


Alice



Bob

Alice wants to send to Bob these quantum states via a quantum channel using qubits.



Alice uses a unitary transformation to map the tensor product onto the state of the quantum system used for the communication. Bob has another unitary transformation that maps the states of the received qubits in his own quantum state.

::... Quantum Entropy

Quantum information is the average number of qubits that Alice has to send for each pure state she produces. The measure of how much quantum information can be compressed, known as the **von Neumann entropy**, is closely related to Shannon's definition of entropy.

$$S := \lim_{N \rightarrow \infty} \frac{S_N}{N} \leq S_1$$

Given a set of pure states $\{|\psi_i\rangle\}$ and their respective probabilities p_i , we can define a mathematical operator called the density operator:

$$\hat{\rho} = \sum_{i=0}^n p_i |\psi_i\rangle \langle \psi_i|$$

$\hat{\rho}$ must satisfy three primary mathematical conditions:

- Hermiticity: $\hat{\rho} = \hat{\rho}^\dagger$, the eigenvalues (probabilities) are real.
- Unit Trace: $\text{tr}(\hat{\rho}) = 1$, the total probability of all possible outcomes is 1.
- Positivity: $\hat{\rho}$ is a positive semi-definite operator. For any state $|\phi\rangle$, $\langle \phi | \hat{\rho} | \phi \rangle \geq 0$, so that no "negative" probabilities are possible.

::... Quantum Entropy

The use of the projection operator $|\psi_i\rangle\langle\psi_i|$ in the definition of the density operator $\hat{\rho}$ isn't just a mathematical convenience, but it is the bridge between the abstract wave function and the physical reality of measurement.

For a 2-level system (like a qubit), every pure state lives on the surface of the Bloch Sphere. These are represented by the projection $|\psi_i\rangle\langle\psi_i|$. In case of a mixed state $\hat{\rho}$, a weighted average of those points on the surface are taken. Because the sphere is convex, this average (the density matrix) ends up inside the sphere.

- Pure State: A single projection $|\psi_i\rangle\langle\psi_i|$ is a point on the surface (distance $r = 1$ from origin).
- Mixed State: A sum of projections $|\psi_i\rangle\langle\psi_i|$ is a point inside the sphere $r < 1$).
- Maximally Mixed: The "center" of the sphere, where we have zero knowledge of the state.

::... Quantum Entropy

The phenomenon entropy is a function of only the density operator, and the quantum information does not depend on the choice of the quantum states produced by Alice. Using this operator, we can rewrite the definition of the von Neumann entropy as:

$$S(\hat{\rho}) := -\text{tr}(\hat{\rho} \log \hat{\rho})$$

and this is again very similar to the classical case. However, the physical meaning is very different. This entropy has three properties that parallel the ones of the Shannon entropy:

1. Non-negative $S(\hat{\rho}) \geq 0$ - It is equal to 0 if and only if the states are pure. entropy is a measure of our ignorance or "missing information" about which state a system is in. For a pure state, there is no missing information; we know exactly which state the system occupies.

If a state is pure, the density operator is a single projection: $\hat{\rho} = |\psi\rangle\langle\psi|$. When we look at the eigenvalues of this matrix, one is 1 (corresponding to the state $|\psi\rangle$, and all the others are 0. Plugging these into the entropy formula $S(\hat{\rho}) = -(1 \ln 1 + 0 \ln 0 + 0 \ln 0 + \dots) = 0$.

::... Quantum Entropy

2. Invariant on the unitary transformations - $S(\hat{U}\hat{\rho}\hat{U}^\dagger) = S(\hat{\rho})$: when we change basis, a unitary transformation is applied, and this does not affect the von Neumann entropy. This is a fundamental property because unitary transformations represent "reversible" processes (like time evolution under the Schrödinger equation), and reversible processes do not create or destroy information. The Von Neumann entropy is determined solely by the eigenvalues of the density matrix. Unitary transformations change the eigenvectors (the orientation), but leave the eigenvalues (the probabilities p_i) completely untouched.

There is no equivalent quantum analog to non-negative conditional entropy. In classical information theory, Shannon's conditional entropy $H(X|Y)$ is always non-negative. It essentially represents how many additional bits you need to describe X if you already know Y . In the quantum realm, however, the conditional von Neumann entropy $S(A|B)$ can indeed be negative.

::... Quantum Entropy

The "culprit" is quantum entanglement.

- In a classical system, the whole cannot be more "certain" than its parts.
- But in quantum mechanics, a composite system can be in a pure state (total certainty, $S(A, B) = 0$) while its individual subsystems are mixed (uncertainty, $S(B) > 0$).

If the value is negative, Alice doesn't need to send Bob any information to communicate her state; in fact, they gain the potential to perform future quantum communication (like teleportation) for free. In this sense, negative conditional entropy represents a credit of future communication capacity.

3. Additivity: the entropy associated with a combined experiment is given by the entropy of one of them plus the conditional entropy of the outcome from the first experiment given the outcome from the second. This only holds true if the systems are completely uncorrelated (product states). Otherwise, we move into the realm of Subadditivity.

::... Quantum Entropy

3. Subadditivity $S(\hat{\rho}_{AB}) \leq S(\hat{\rho}_A) + S(\hat{\rho}_B)$: the joint entropy of composite states is at most the sum of the individual entropies. The condition for equality is the uncorrelation among the two systems $\hat{\rho}_{AB} = \hat{\rho}_A \otimes \hat{\rho}_B$.

Subadditivity is a main deviation from the classical entropy but we have another one:

- Classically, the fact that conditional entropy was never negative mean that the entropy of the joint experiments was never smaller than the one associated to each individual experiment in isolation.

$$S_{class}(\rho_{AB}) \geq S_{class}(\rho_A)$$

- In the quantum case, this inequality is no longer guaranteed due to entanglement. There is only a weaker lower bound to the joint entropy, called Araki-Lieb Inequality:

$$S(\hat{\rho}_{AB}) \geq |S(\hat{\rho}_A) - S(\hat{\rho}_B)|$$

::... Quantum Entropy

While subadditivity says the joint entropy cannot be too large, Araki-Lieb says it cannot be too small relative to the difference of its parts.

$$S(\hat{\rho}_{AB}) \geq |S(\hat{\rho}_A) - S(\hat{\rho}_B)|$$

In the case of entanglement, each entropy is equal to 1, so their difference is 0. The entangled qubits are in pure states, so their entropy is 0, and the Araki-Lieb Inequality is satisfied.

If we have two different systems entangled with each other, the Araki-Lieb Inequality tells us that the left and right sides needs to be 0, which derives that the entropy of the first system must be equal to the one of the second.

It shows that you cannot have a system where one part is highly disordered (high $S(A)$) and another part is highly ordered (low $S(B)$) if the total system is pure. They must "share" the disorder equally.

::... Quantum Entropy

The **Monogamy of Entanglement** is one of the most fundamental differences between classical and quantum correlations. In simple terms, it states that if two qubits are maximally entangled with each other, they cannot be entangled with anything else.

While classical correlations can be shared infinitely (everyone in a room can know the same secret), quantum entanglement is a "limited resource".

Consider a tripartite system A, B and C, and consider the composite state of all of them described by $\hat{\rho}_{ABC}$ and its entropy $S(\hat{\rho}_{ABC})$. There are two inequalities that we can write:

- Sub-additivity tells that this joint entropy should be lower than the sum of the entropy of two parts of the system (A and B) with the entropy of the remaining one (C).

$$S(\hat{\rho}_{ABC}) \leq S(\hat{\rho}_{AB}) + S(\hat{\rho}_C)$$

::... Quantum Entropy

- Araki-Lieb Inequality tells that this joint entropy should be greater than or equal to the absolute value of the difference of the entropy of two parts of the system (A and B) with the entropy of the remaining one (C).

$$S(\hat{\rho}_{ABC}) \geq |S(\hat{\rho}_{AB}) - S(\hat{\rho}_C)|$$

Let us consider the special case where A and B are in pure entangled states, so $S(\hat{\rho}_{AB}) = 0$.


$$|S(\hat{\rho}_{AB}) - S(\hat{\rho}_C)| \leq S(\hat{\rho}_{ABC}) \leq S(\hat{\rho}_{AB}) + S(\hat{\rho}_C)$$
$$S(\hat{\rho}_C) \leq S(\hat{\rho}_{ABC}) \leq S(\hat{\rho}_C)$$

The entropy of the tripartite system is equal to the one of the C part.

$$S(\hat{\rho}_{ABC}) = S(\hat{\rho}_C) \implies S(\hat{\rho}_{AB}) + S(\hat{\rho}_C)$$

The state of ABC is the tensor product of the state of AB with the state of C, and tensor product implies the systems are statistically uncorrelated.

::... Quantum Entropy

$$S(\hat{\rho}_{AB}) = 0 \implies \hat{\rho}_{AB} \otimes \hat{\rho}_C$$

When we have a tripartite system, whenever two of them are entangled then the third one C cannot be correlated in anyway with A and B, and the monogamy comes from this.

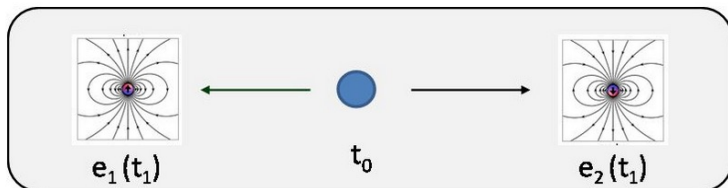
Such a phenomenon can be exploited in cryptography: unlike classical encryption, which relies on the mathematical difficulty of factoring large numbers, quantum security relies on the fact that quantum correlations cannot be shared.

- Imagine that A and B are the legitimate communication partners and they use for the communication entangled qubit pairs in the EPR protocol (no BB84 one). An evedropped C would have to couple to these in order to extract information, become correlated to them. However, this is impossible, but this attempt will destroy the entanglement and can be detected.

::... Entanglement

Entanglement is when two or more particles are so deeply linked that the state of one cannot be described independently of the others, regardless of their distance. A measurement performed on one of two entangled particles instantly determines the possible measurement outcomes of the other.

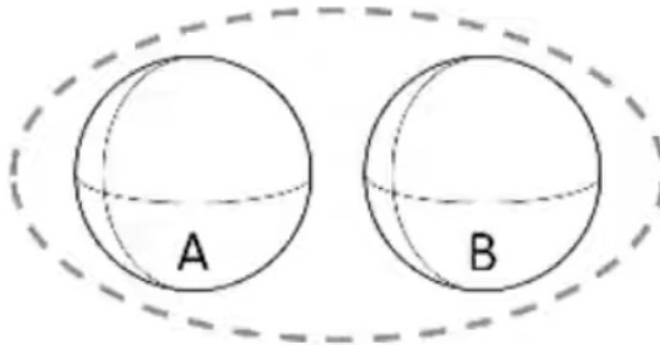
- Classical correlation: Imagine a pair of shoes: the left is in a box and the right in another. If you send one box to the moon and open it to find the "Left", you know instantly that the other contains the "Right". There is no mystery; the shoes were always "Left" and "Right".
- Quantum: Imagine two "entangled" coins. Until you look at them, they aren't "Heads" or "Tails", but they exist in a superposition of both. But the moment you flip one, and it lands on Heads, the other coin instantly becomes Tails, even though it was undecided before.



In t_0 , two paired particles e_1 and e_2 are produced, which drift apart. If particle e_1 is measured in t_1 with spin-up, instantaneously, the spin of e_2 will be spin-down, and vice versa.

::... Entanglement

Let us consider two qubits A and B; if they are individually in pure states, their composite state is a tensor product of them, and it is a pure state.



We can construct an arbitrary superposition of this product space, and a particularly important family of these superposition states is the **Bell State**:

$$|\beta_{jk}\rangle := \frac{|0, k\rangle + (-1)^j |1, 1 - k\rangle}{\sqrt{2}}, \quad j, k \in \{0, 1\}$$

If both j and k are 0, $|\beta_{00}\rangle := \frac{|0,0\rangle + |1,1\rangle}{\sqrt{2}}$ is the superposition of the basis states for the two qubits, and is a pure state, but cannot be written as tensor product of two pure states.

::... Entanglement

Consider $\hat{A}$ observable for the first qubit and we want to have the expected value of the Bell state with respect to the first qubit:

$$\langle \beta_{00} | \hat{A} \otimes \hat{I} | \beta_{00} \rangle$$

the tensor product is needed to make the measurement only on the first qubit when the Bell state is defined on the combined Hilbert space of the two qubits.

::... Entanglement

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$$\langle \beta_{00} | \hat{A} \otimes \hat{I} | \beta_{00} \rangle$$

the tensor product is needed to make the measurement only on the first qubit when the Bell state is defined on the combined Hilbert space of the two qubits.


$$\begin{aligned} & \frac{1}{2} (\langle 00 | + \langle 11 |) (\hat{A} \otimes \hat{I}) (|00\rangle + |11\rangle) = \\ &= \frac{1}{2} \langle 00 | (\hat{A} \otimes \hat{I}) | 00 \rangle + \frac{1}{2} \langle 00 | (\hat{A} \otimes \hat{I}) | 11 \rangle + \frac{1}{2} \langle 11 | (\hat{A} \otimes \hat{I}) | 00 \rangle \\ &+ \frac{1}{2} \langle 11 | (\hat{A} \otimes \hat{I}) | 11 \rangle \end{aligned}$$

::... Entanglement

Consider $\hat{A}$ observable for the first qubit and we want to have the expected value of the Bell state with respect to the first qubit:

$$\langle \beta_{00} | \hat{A} \otimes \hat{I} | \beta_{00} \rangle$$

the tensor product is needed to make the measurement only on the first qubit when the Bell state is defined on the combined Hilbert space of the two qubits.



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::... Entanglement

Consider $\hat{A}$ observable for the first qubit and we want to have the expected value of the Bell state with respect to the first qubit:

$$\langle \beta_{00} | \hat{A} \otimes \hat{I} | \beta_{00} \rangle$$

the tensor product is needed to make the measurement only on the first qubit when the Bell state is defined on the combined Hilbert space of the two qubits.


$$\begin{aligned} & \frac{1}{2} (\langle 00 | + \langle 11 |) (\hat{A} \otimes \hat{I}) (|00\rangle + |11\rangle) = \\ &= \frac{1}{2} \langle 00 | (\hat{A} \otimes \hat{I}) | 00 \rangle + \cancel{\frac{1}{2} \langle 00 | (\hat{A} \otimes \hat{I}) | 11 \rangle} + \cancel{\frac{1}{2} \langle 11 | (\hat{A} \otimes \hat{I}) | 00 \rangle} \\ &+ \frac{1}{2} \langle 11 | (\hat{A} \otimes \hat{I}) | 11 \rangle \end{aligned}$$

::... Entanglement

Consider $\hat{A}$ observable for the first qubit and we want to have the expected value of the Bell state with respect to the first qubit:

$$\langle \beta_{00} | \hat{A} \otimes \hat{I} | \beta_{00} \rangle$$

the tensor product is needed to make the measurement only on the first qubit when the Bell state is defined on the combined Hilbert space of the two qubits.

$$\frac{1}{2} (\langle 00 | + \langle 11 |) (\hat{A} \otimes \hat{I}) (|00\rangle + |11\rangle) =$$

$$\frac{1}{2} \langle 00 | (\hat{A} \otimes \hat{I}) | 00 \rangle + \cancel{\frac{1}{2} \langle 00 | (\hat{A} \otimes \hat{I}) | 11 \rangle} + \cancel{\frac{1}{2} \langle 11 | (\hat{A} \otimes \hat{I}) | 00 \rangle}$$

$$+ \frac{1}{2} \langle 11 | (\hat{A} \otimes \hat{I}) | 11 \rangle$$

$$\langle 0 | \hat{A} | 0 \rangle \cdot \langle 0 | \hat{I} | 0 \rangle = \langle 0 | \hat{A} | 0 \rangle \cdot 1$$

$$\langle 1 | \hat{A} | 1 \rangle \cdot \langle 1 | \hat{I} | 1 \rangle = \langle 1 | \hat{A} | 1 \rangle \cdot 1$$

::... Entanglement

Consider $\hat{A}$ observable for the first qubit and we want to have the expected value of the Bell state with respect to the first qubit:

$$\langle \beta_{00} | \hat{A} \otimes \hat{I} | \beta_{00} \rangle$$

the tensor product is needed to make the measurement only on the first qubit when the Bell state is defined on the combined Hilbert space of the two qubits.

$$\begin{aligned} & \frac{1}{2} (\langle 00 | + \langle 11 |) (\hat{A} \otimes \hat{I}) (|00\rangle + |11\rangle) = \\ &= \frac{1}{2} \langle 00 | (\hat{A} \otimes \hat{I}) | 00 \rangle + \frac{1}{2} \langle 00 | (\hat{A} \otimes \hat{I}) | 11 \rangle + \frac{1}{2} \langle 11 | (\hat{A} \otimes \hat{I}) | 00 \rangle \\ &+ \frac{1}{2} \langle 11 | (\hat{A} \otimes \hat{I}) | 11 \rangle = \\ &= \frac{1}{2} \langle 0 | \hat{A} | 0 \rangle + \frac{1}{2} \langle 1 | \hat{A} | 1 \rangle = \frac{1}{2} \text{tr}_A(\hat{A}) = \text{tr}_A \left(\frac{1}{2} \hat{I} \hat{A} \right) = \text{tr}_A(\hat{\rho}_A \hat{A}) \end{aligned}$$

The state of **Maximum Entropy** has been reached.
This is not a pure state, but a reduced state of A.

::... Entanglement

The **Bell State**:

$$|\beta_{jk}\rangle := \frac{|0, k\rangle + (-1)^j |1, 1 - k\rangle}{\sqrt{2}}, \quad j, k \in \{0, 1\}$$

but the individual qubits are described by mixed states.

As we discussed with the density operator $\hat{\rho}$, most systems can be broken down into their parts. An entangled state cannot.

The notation $|\beta_{xy}\rangle$ is a standard way to index the four Bell states using two bits:

- x (The Phase Bit): Determines if the sign is positive or negative.
- y (The Parity Bit): Determines if the qubits are the same (00, 11) or different (01, 10).

The indexing of the Bell states as $|\beta_{xy}\rangle$ is not arbitrary. It is designed to create a direct mathematical link between the two binary bits (x and y) and the operations required to transform the "standard" state into the others.

::... Entanglement

You can generate all four Bell states by taking the "first" state $|\beta_{00}\rangle$ and applying Pauli matrices (X and Z gates) to just one of the qubits. The rule is:

$$|\beta_{xy}\rangle = (Z^x X^y \otimes I) |\beta_{00}\rangle$$

An alternative way of representing the Bell states is using Greek letters or Parity (Same or Different?) and signs or Phase (Constructive or Destructive?):

- Φ : Used when the two qubits are in the same state (both 0 or both 1). This corresponds to $y = 0$.
- Ψ : Used when the two qubits are in different states (one is 0, the other is 1). This corresponds to $y = 1$.
- $+$: Corresponds to $x = 0$. No Z operator is applied, so the terms are added.
- $-$: Corresponds to $x = 1$. A Z operator is applied, which introduces a relative phase of π (a minus sign).

::... Entanglement

You can generate all four Bell states by taking the "first" state $|\beta_{00}\rangle$ and applying Pauli matrices (X and Z gates) to just one of the qubits. The rule is:

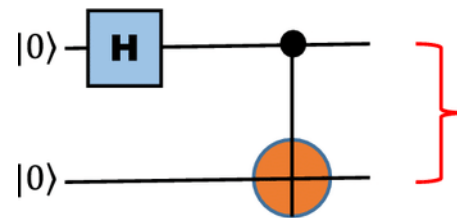
$$|\beta_{xy}\rangle = (Z^x X^y \otimes I) |\beta_{00}\rangle$$

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

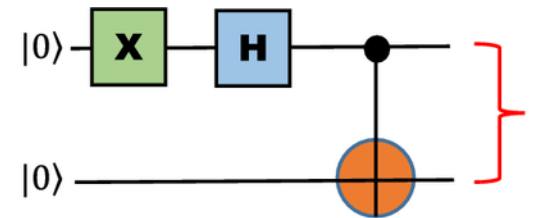
$$|\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$

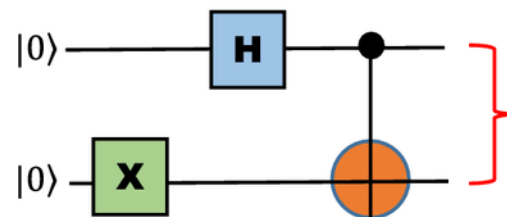
$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$



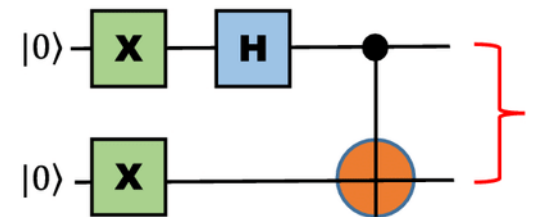
$$|\psi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$



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$$|\phi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$

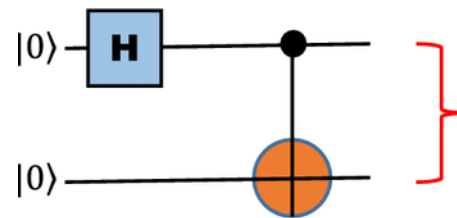


$$|\phi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

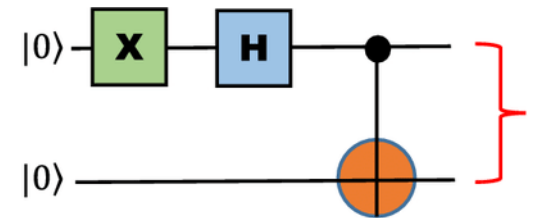
::... Entanglement

You can generate all four Bell states by taking the "first" state $|\beta_{00}\rangle$ and applying Pauli matrices (X and Z gates) to just one of the qubits. The rule is:

$$|\beta_{xy}\rangle = (Z^x X^y \otimes I) |\beta_{00}\rangle$$



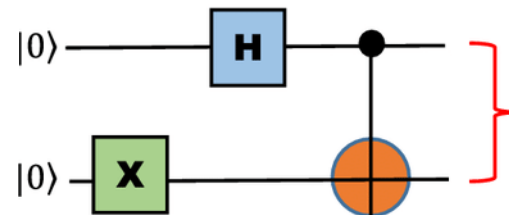
$$|\psi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$



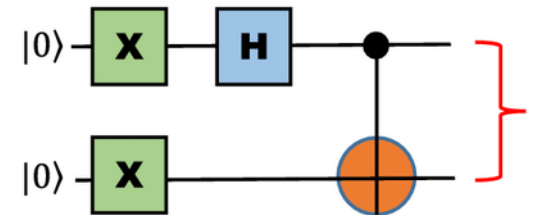
$$|\psi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

$$|00\rangle \xrightarrow{H \otimes I} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

$$\xrightarrow{CNOT} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle$$



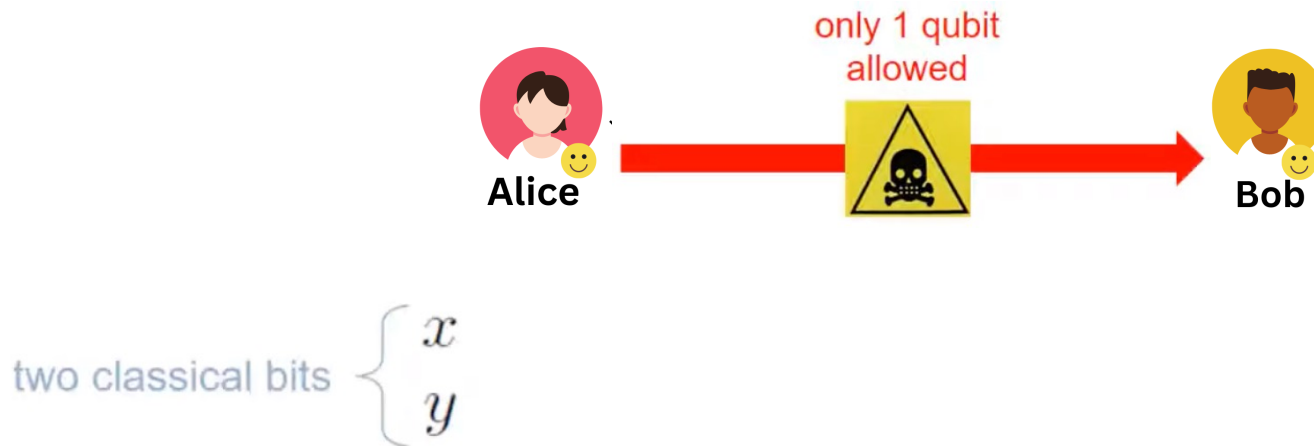
$$|\phi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$



$$|\phi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

::... Superdense Coding

Imagine Alice and Bob, with Alice holding two classical bits, x and y , that she would like to send to Bob. Unfortunately, she can only use a quantum channel to send a single qubit.



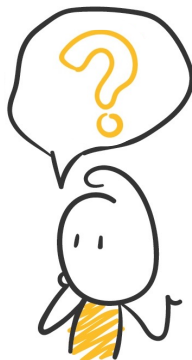
::... Superdense Coding

Imagine Alice and Bob, with Alice holding two classical bits, x and y , that she would like to send to Bob. Unfortunately, she can only use a quantum channel to send a single qubit.



This looks impossible as the Holevo bounds tell us that a qubit can carry at most one classical bit.

two classical bits $\left\{ \begin{array}{l} x \\ y \end{array} \right.$



Is there a trick to resolve such an impossible situation and work around the limitation we learned on the Holevo bounds?

::... Superdense Coding

Imagine Alice and Bob, with Alice holding two classical bits, x and y , that she would like to send to Bob. Unfortunately, she can only use a quantum channel to send a single qubit.



This looks impossible as the Holevo bounds tell us that a qubit can carry at most one classical bit.

two classical bits $\begin{cases} x \\ y \end{cases}$

previously shared Bell pair $|\beta_{00}\rangle$



There is a solution!

The trick consists of using a Bell pair that Alice and Bob previously shared and stored. Each of them has a member of the Bell pair.

Let's see this protocol step by step.

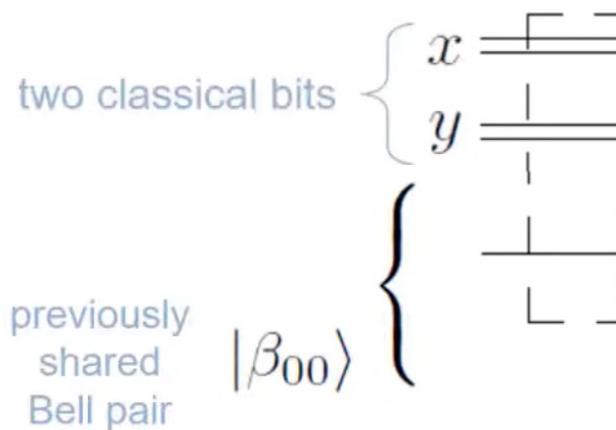
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Imagine Alice and Bob, with Alice holding two classical bits, x and y , that she would like to send to Bob. Unfortunately, she can only use a quantum channel to send a single qubit.



This looks impossible as the Holevo bounds tell us that a qubit can carry at most one classical bit.

The starting point is represented by the two classical bits and the Bell pair in the state $|\beta_{00}\rangle$ with one member qubit in possession of Alice, while the other member qubit is already available at Bob's side.



$\uparrow$
 $|\psi_0\rangle$

$$|\psi_0\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

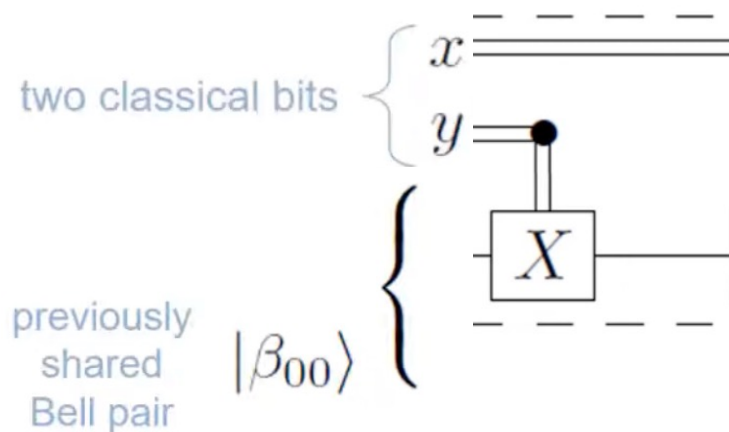
::... Superdense Coding

Imagine Alice and Bob, with Alice holding two classical bits, x and y , that she would like to send to Bob. Unfortunately, she can only use a quantum channel to send a single qubit.



This looks impossible as the Holevo bounds tell us that a qubit can carry at most one classical bit.

Alice uses the second of her classical bits y to control a Pauli-X operator on her member qubit of $|\beta_{00}\rangle$ obtaining state $|\psi_1\rangle$:



$$|\psi_1\rangle = \frac{|y, 0\rangle + |1 \oplus y, 1\rangle}{\sqrt{2}}$$

↑
 $|\psi_1\rangle$

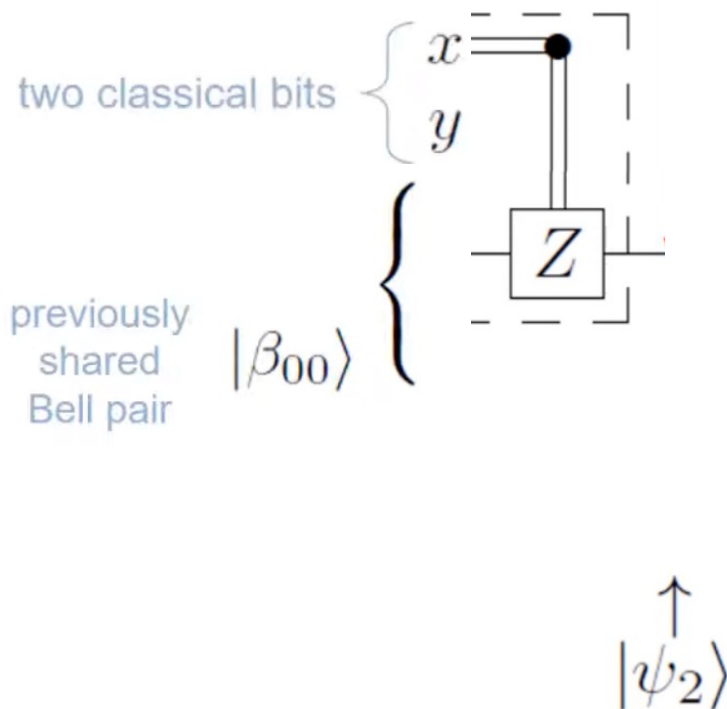
::... Superdense Coding

Imagine Alice and Bob, with Alice holding two classical bits, x and y , that she would like to send to Bob. Unfortunately, she can only use a quantum channel to send a single qubit.



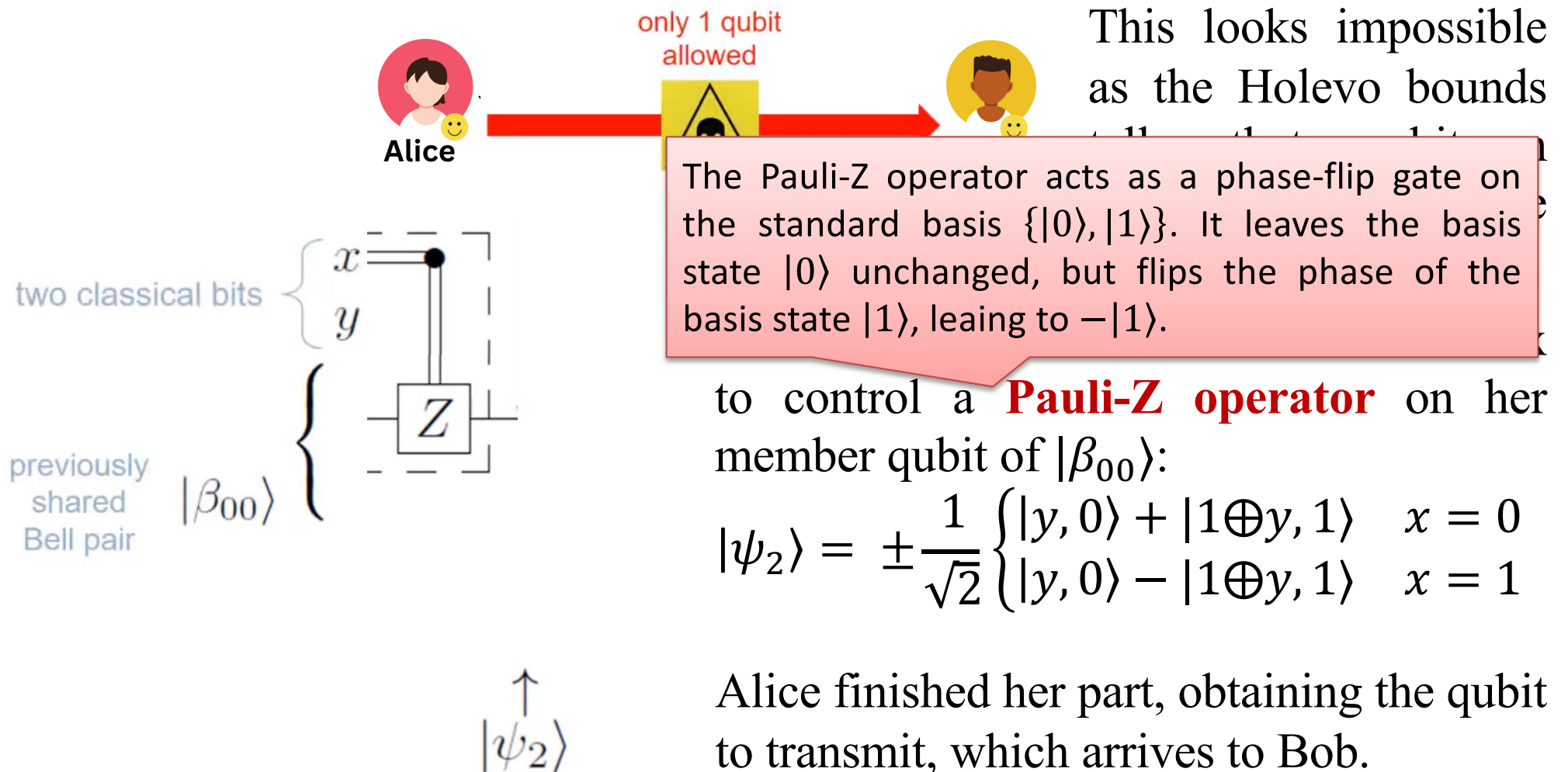
This looks impossible as the Holevo bounds tell us that a qubit can carry at most one classical bit.

Alice uses the first of her classical bits x to control a Pauli-Z operator on her member qubit of $|\beta_{00}\rangle$.



::... Superdense Coding

Imagine Alice and Bob, with Alice holding two classical bits, x and y , that she would like to send to Bob. Unfortunately, she can only use a quantum channel to send a single qubit.



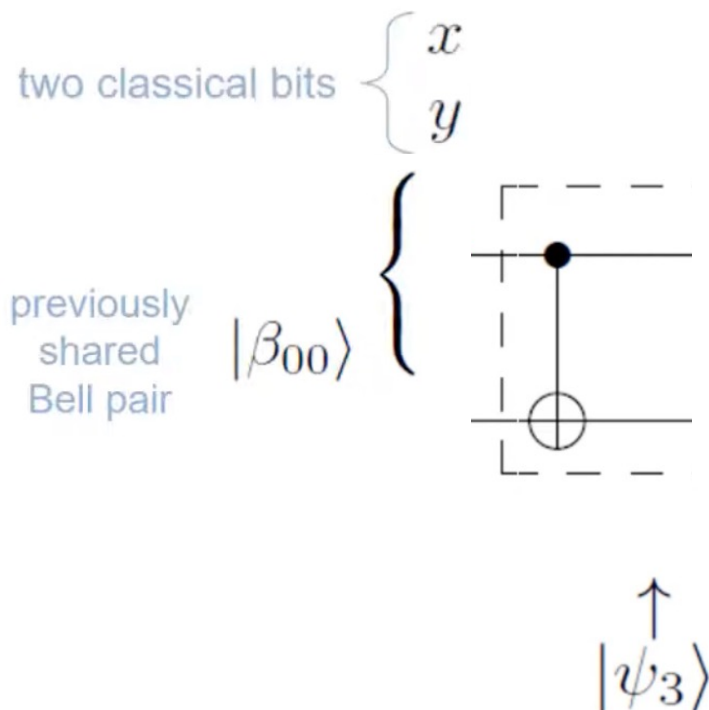
::... Superdense Coding

Imagine Alice and Bob, with Alice holding two classical bits, x and y , that she would like to send to Bob. Unfortunately, she can only use a quantum channel to send a single qubit.



This looks impossible as the Holevo bounds tell us that a qubit can carry at most one classical bit.

Bob has his member of the Bell state, and now in addition he has the qubit from Alice. He performs the C-Not on these two qubits, where the qubit from Alice is the controller. Let's ignore all the signs:



$$|\psi_3\rangle \propto \begin{cases} |y, y\rangle + |1 \oplus y, y\rangle & x = 0 \\ |y, y\rangle - |1 \oplus y, y\rangle & x = 1 \end{cases}$$

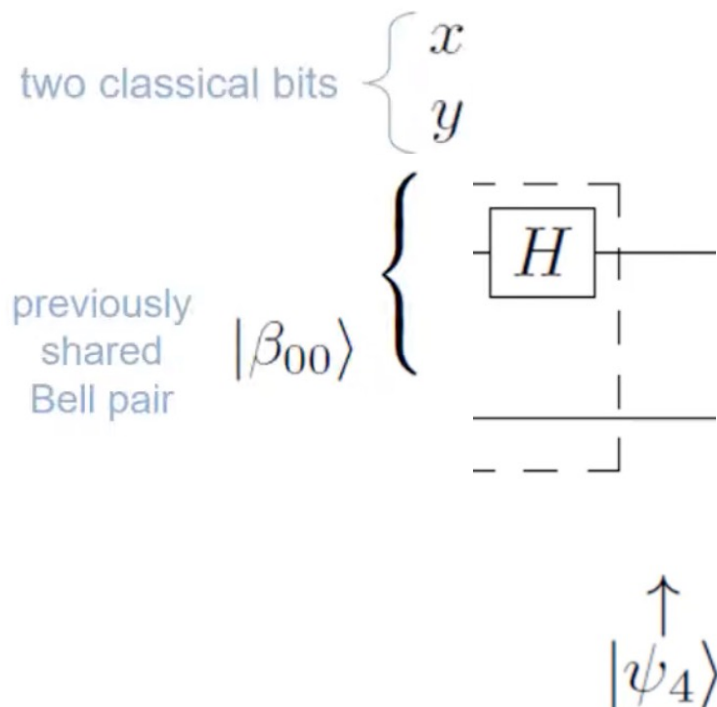
::... Superdense Coding

Imagine Alice and Bob, with Alice holding two classical bits, x and y , that she would like to send to Bob. Unfortunately, she can only use a quantum channel to send a single qubit.



This looks impossible as the Holevo bounds tell us that a qubit can carry at most one classical bit.

Bob applies to the first qubit the Hadamard gate:



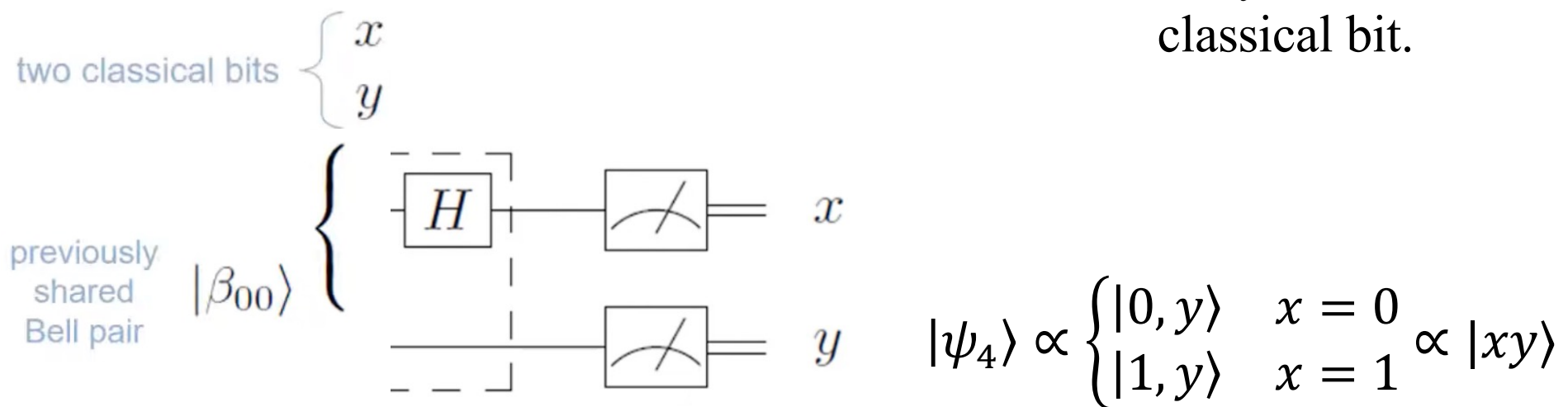
$$|\psi_4\rangle \propto \begin{cases} |0, y\rangle & x = 0 \\ |1, y\rangle & x = 1 \end{cases} \propto |xy\rangle$$

::... Superdense Coding

Imagine Alice and Bob, with Alice holding two classical bits, x and y , that she would like to send to Bob. Unfortunately, she can only use a quantum channel to send a single qubit.



This looks impossible as the Holevo bounds tell us that a qubit can carry at most one classical bit.

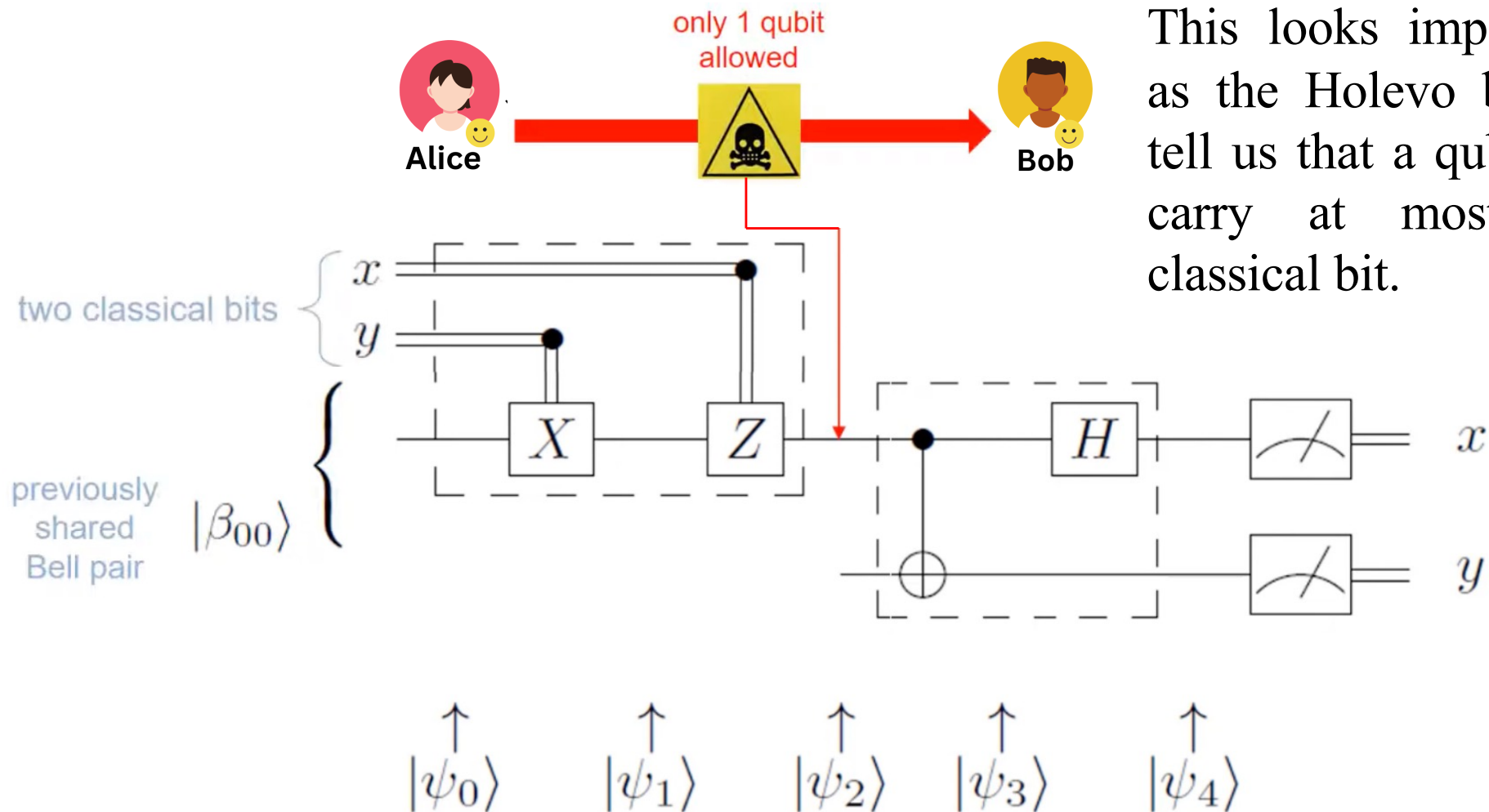


$\uparrow$
 $|\psi_4\rangle$

If now Bob measures in the standard basis the two qubits, he will obtain the classical bits.

::... Superdense Coding

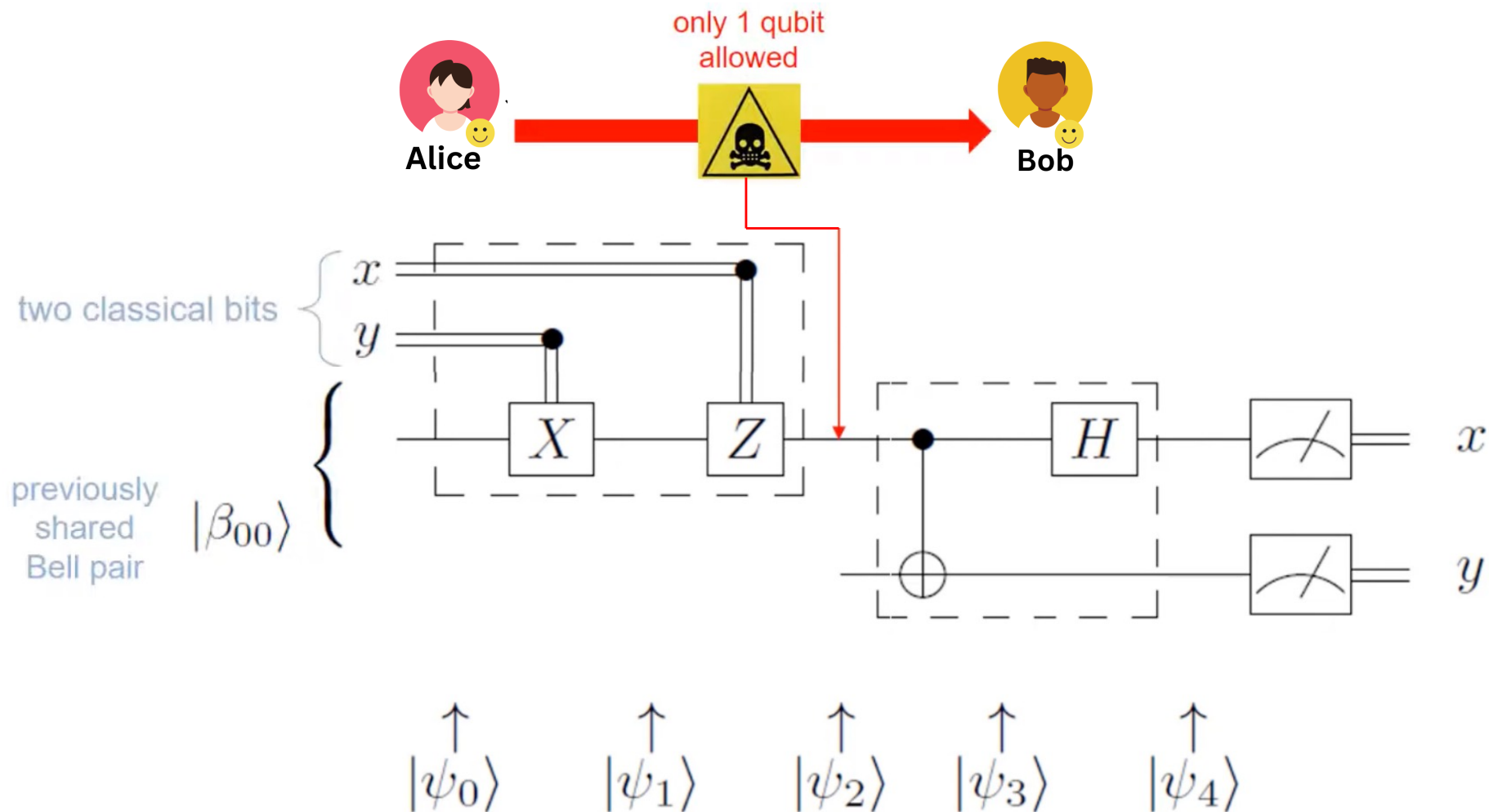
Imagine Alice and Bob, with Alice holding two classical bits, x and y , that she would like to send to Bob. Unfortunately, she can only use a quantum channel to send a single qubit.



This looks impossible as the Holevo bounds tell us that a qubit can carry at most one classical bit.

::... Superdense Coding

Superdense coding is a quantum communication protocol that transmits some classical bits by sending only a smaller number of qubits, assuming that the sender and receiver have pre-shared an entangled resource.



::... Teleportation

Let us now consider a different scenario where Alice holds a qubit in a given state $|\psi\rangle$, which is any arbitrary possible state, and he wants to transmit it to Bob, but the available channel conveys only a finite number of classical bits.



arbitrary
qubit state

$|\psi\rangle$

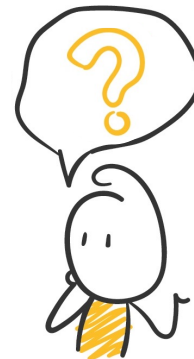
::... Teleportation

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arbitrary
qubit state

$|\psi\rangle$



Is such a task even possible? $|\psi\rangle$ is a point on the Bloch sphere and specified by given coordinates there.

If you want infinite precision in specifying the coordinates of a pure state, then you need an infinite number of digits. However, Alice can only send a finite number of classical bits.

::... Teleportation

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arbitrary
qubit state $|\psi\rangle$

previously
shared
Bell pair $|\beta_{00}\rangle$ {

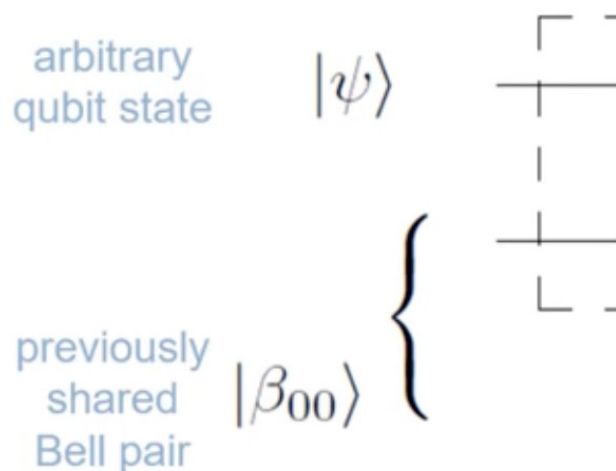


Teleportation is a quantum communication protocol that allows sending a qubit by only using two classical bits, if the sender and the receiver have pre-shared an entangled resource.

If you want infinite precision in specifying the coordinates of a pure state, then you need an infinite number of digits. However, Alice can only send a finite number of classical bits.

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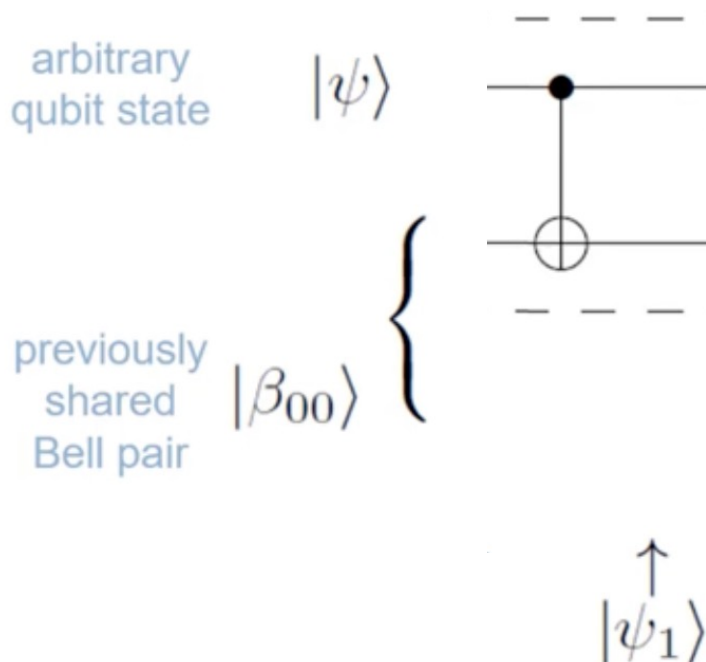


At the start of the protocol, we have two qubits at Alice's side: the one to be sent to Bob and her member of the Bell state. The other member of the Bell state is already at Bob's side.

$$\begin{aligned}
 |\psi_0\rangle &= |\psi\rangle \otimes |\beta_{00}\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes \frac{|00\rangle + |11\rangle}{\sqrt{2}} \\
 &\propto \alpha|0\rangle \otimes (|00\rangle + |11\rangle) + \beta|1\rangle \otimes (|00\rangle + |11\rangle)
 \end{aligned}$$

::... Teleportation

Let us now consider a different scenario where Alice holds a qubit in a given state $|\psi\rangle$, which is any arbitrary possible state, and he wants to transmit it to Bob, but the available channel conveys only a finite number of classical bits.

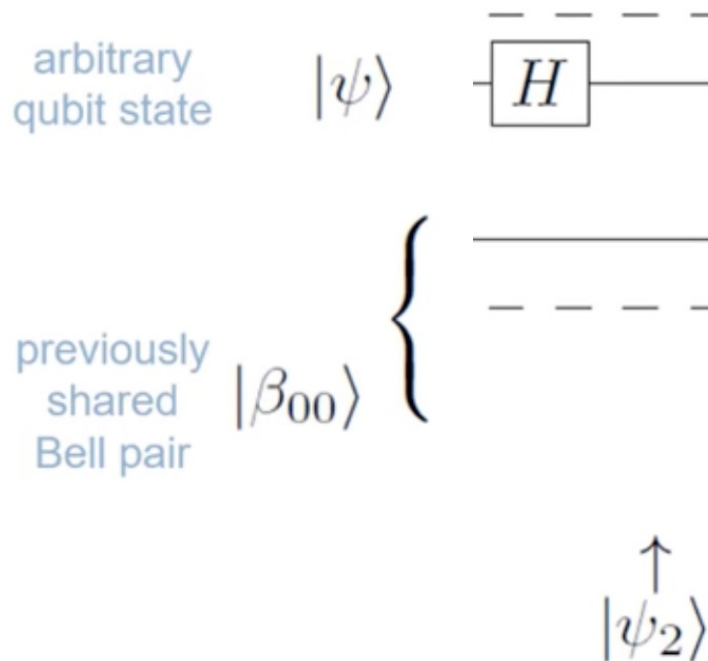


Now Alice performs a C-NOT controlled by the qubit Alice wants to send to Bob.

$$|\psi_1\rangle \propto \alpha|0\rangle \otimes (|00\rangle + |11\rangle) + \beta|1\rangle \otimes (|10\rangle + |01\rangle)$$

::... Teleportation

Let us now consider a different scenario where Alice holds a qubit in a given state $|\psi\rangle$, which is any arbitrary possible state, and he wants to transmit it to Bob, but the available channel conveys only a finite number of classical bits.

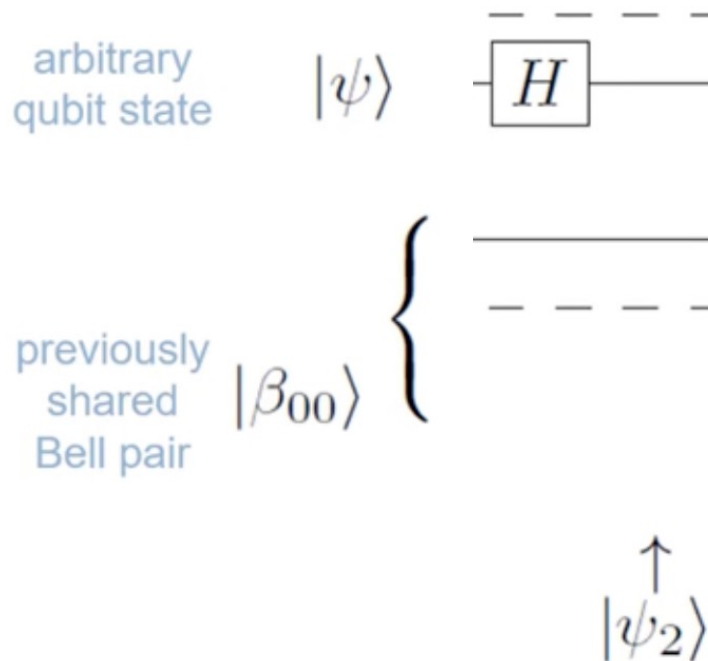


Then, Alice applies the Hadamard gate to the first qubit.

$$|\psi_2\rangle \propto \alpha(|0\rangle + |1\rangle) \otimes (|00\rangle + |11\rangle) + \beta(|0\rangle - |1\rangle) \otimes (|10\rangle + |01\rangle)$$

::... Teleportation

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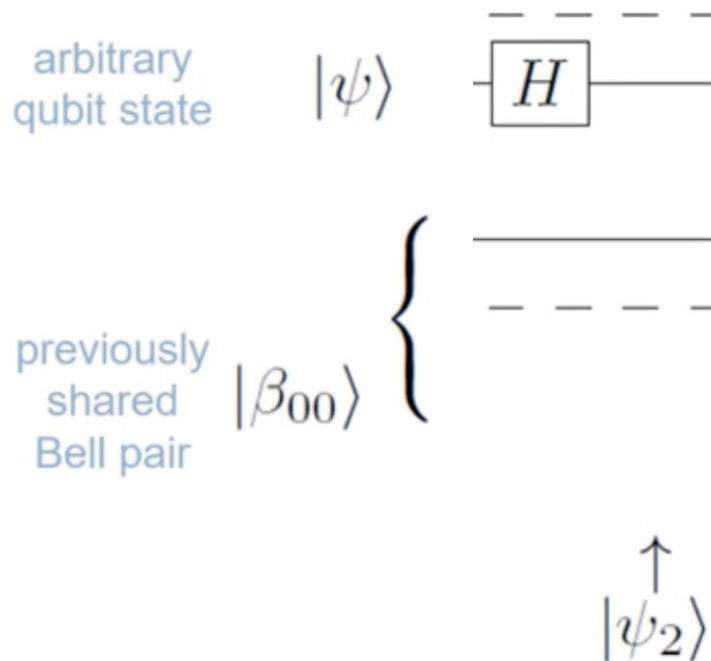


Then, Alice applies the Hadamard gate to the first qubit.

$$\begin{aligned}
 |\psi_2\rangle &\propto \alpha(|0\rangle + |1\rangle) \otimes (|00\rangle + |11\rangle) + \\
 &\quad + \beta(|0\rangle - |1\rangle) \otimes (|10\rangle + |01\rangle) \propto \\
 &\propto |00\rangle \otimes (\alpha|0\rangle + \beta|1\rangle)
 \end{aligned}$$

::... Teleportation

Let us now consider a different scenario where Alice holds a qubit in a given state $|\psi\rangle$, which is any arbitrary possible state, and he wants to transmit it to Bob, but the available channel conveys only a finite number of classical bits.

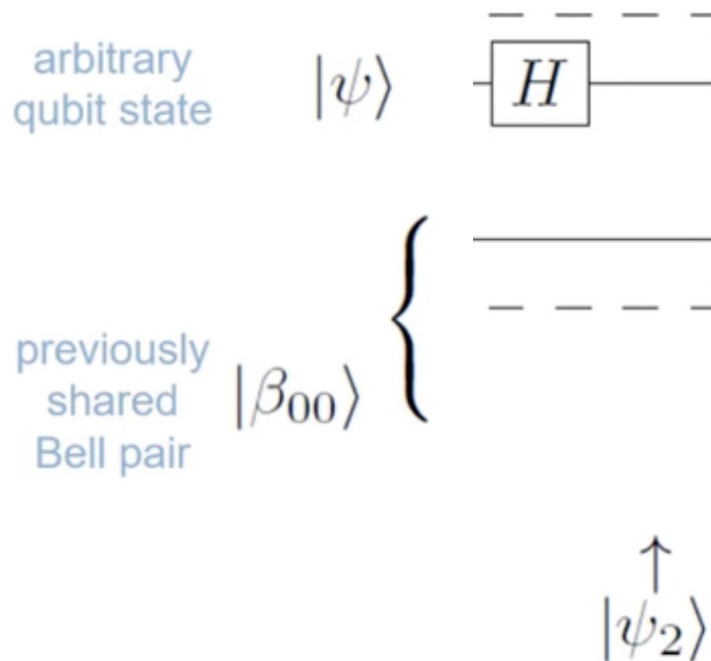


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 &\quad + \beta(|0\rangle - |1\rangle) \otimes (|10\rangle + |01\rangle) \propto \\
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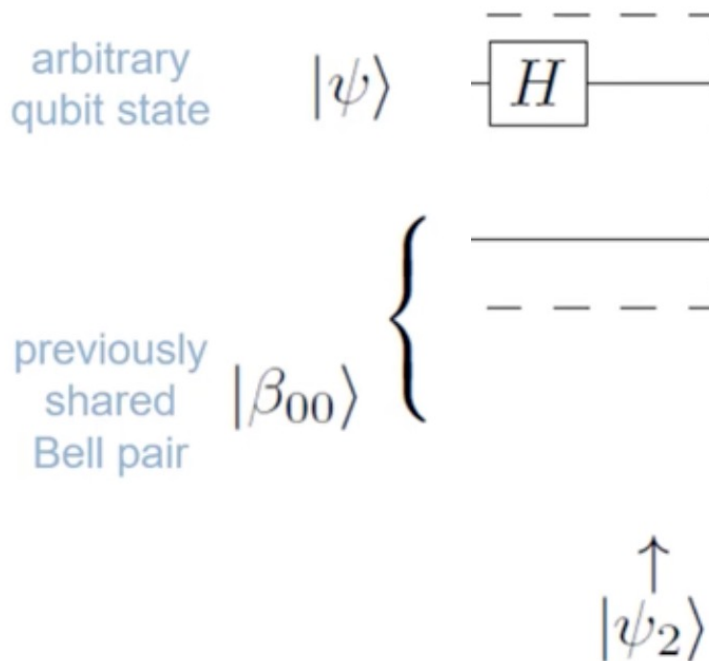


Then, Alice applies the Hadamard gate to the first qubit.

$$\begin{aligned}
 |\psi_2\rangle &\propto \alpha(|0\rangle + |1\rangle) \otimes (|00\rangle + |11\rangle) + \\
 &\quad + \beta(|0\rangle - |1\rangle) \otimes (|10\rangle + |01\rangle) \propto \\
 &\propto |00\rangle \otimes (\alpha|0\rangle + \beta|1\rangle) + |01\rangle \otimes (\alpha|1\rangle \\
 &\quad + \beta|0\rangle) + |10\rangle \otimes (\alpha|0\rangle - \beta|1\rangle)
 \end{aligned}$$

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Let us now consider a different scenario where Alice holds a qubit in a given state $|\psi\rangle$, which is any arbitrary possible state, and he wants to transmit it to Bob, but the available channel conveys only a finite number of classical bits.

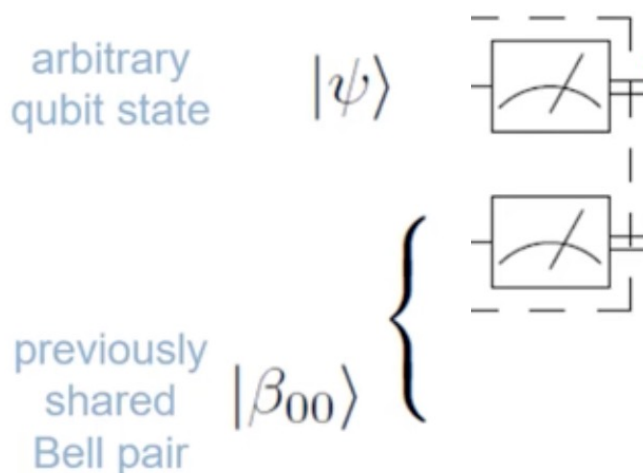


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$$\begin{aligned}
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 &\quad + \beta(|0\rangle - |1\rangle) \otimes (|10\rangle + |01\rangle) \propto \\
 &\propto |00\rangle \otimes (\alpha|0\rangle + \beta|1\rangle) + |01\rangle \otimes (\alpha|1\rangle \\
 &\quad + \beta|0\rangle) + |10\rangle \otimes (\alpha|0\rangle - \beta|1\rangle) \\
 &\quad + |11\rangle \otimes (\alpha|1\rangle - \beta|0\rangle)
 \end{aligned}$$

::... Teleportation

Let us now consider a different scenario where Alice holds a qubit in a given state $|\psi\rangle$, which is any arbitrary possible state, and he wants to transmit it to Bob, but the available channel conveys only a finite number of classical bits.



Last, Alice performs measurements in the standard basis on her qubits, obtaining two classical bits she sends to Bob.

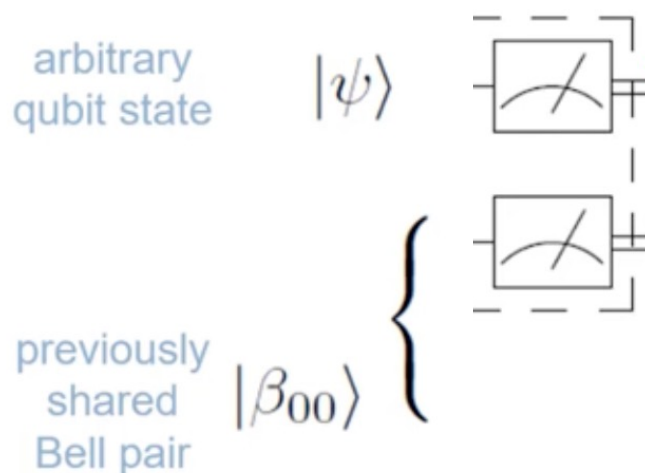
Looking at $|\psi_2\rangle$, there are four cases:

- if she obtained 00, $|\psi_2\rangle$ is projected onto the associated eigenspace, and there is only one term of $|\psi_2\rangle$ in that eigenspace, so we know that Bob's qubit must be in superposition state $(\alpha|0\rangle + \beta|1\rangle)$;
- likewise all the other cases.

$$|\psi_2\rangle \propto |00\rangle \otimes (\alpha|0\rangle + \beta|1\rangle) + |01\rangle \otimes (\alpha|1\rangle + \beta|0\rangle) + |10\rangle \otimes (\alpha|0\rangle - \beta|1\rangle) + |11\rangle \otimes (\alpha|1\rangle - \beta|0\rangle)$$

::... Teleportation

Let us now consider a different scenario where Alice holds a qubit in a given state $|\psi\rangle$, which is any arbitrary possible state, and he wants to transmit it to Bob, but the available channel conveys only a finite number of classical bits.



- if Alice measures 00, the Bob's qubit is already in the desired state $|\psi\rangle$, and nothing has to be done;
- if Alice measures 01, the Bob's qubit is not in the desired state $|\psi\rangle$, and he has to apply a Pauli-X to flip all the basis;
- if Alice measures 10, the Bob's qubit is not in the desired state $|\psi\rangle$, and he has to apply a Pauli-Z to flip the sign of $|1\rangle$;

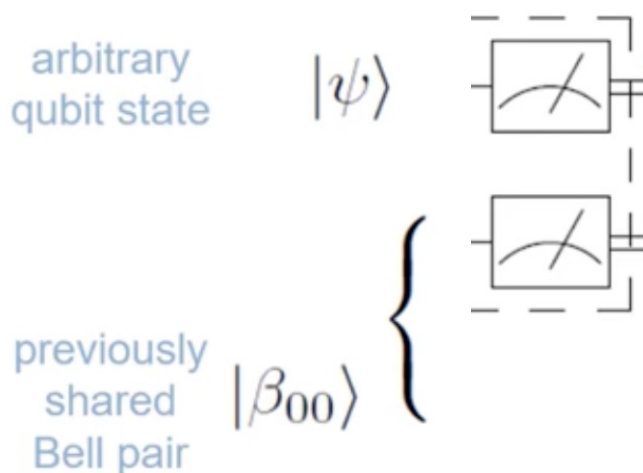
$$|\psi_2\rangle \propto |00\rangle \otimes (\alpha|0\rangle + \beta|1\rangle) + |01\rangle \otimes (\alpha|1\rangle + \beta|0\rangle) + |10\rangle \otimes (\alpha|0\rangle - \beta|1\rangle) + |11\rangle \otimes (\alpha|1\rangle - \beta|0\rangle)$$

::... Teleportation

Let us now consider a different scenario where Alice holds a qubit in a given state $|\psi\rangle$, which is any arbitrary possible state, and he wants to transmit it to Bob, but the available channel conveys only a finite number of classical bits.



- if Alice measures 11, the Bob's qubit is not in the desired state $|\psi\rangle$, and he has to apply a Pauli-X and Pauli Z to get the desired state.



$$|\psi_2\rangle \propto |00\rangle \otimes (\alpha|0\rangle + \beta|1\rangle) + \\ + |01\rangle \otimes (\alpha|1\rangle + \beta|0\rangle) + |10\rangle \otimes (\alpha|0\rangle \\ - \beta|1\rangle) + |11\rangle \otimes (\alpha|1\rangle - \beta|0\rangle)$$

::... Teleportation

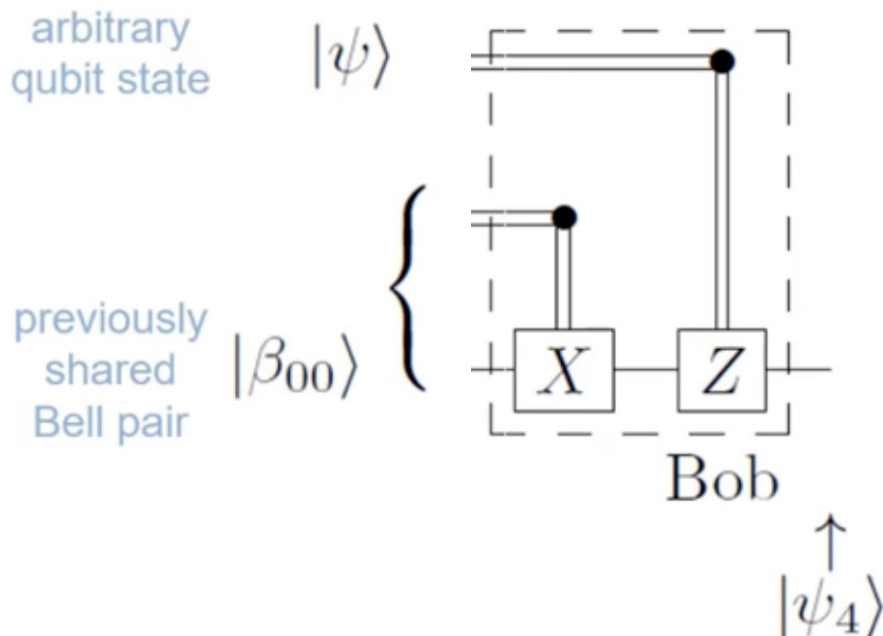
Let us now consider a different scenario where Alice holds a qubit in a given state $|\psi\rangle$, which is any arbitrary possible state, and he wants to transmit it to Bob, but the available channel conveys only a finite number of classical bits.



- if Alice measures 11, the Bob's qubit is not in the desired state $|\psi\rangle$, and he has to apply a Pauli-X and Pauli Z to get the desired state.

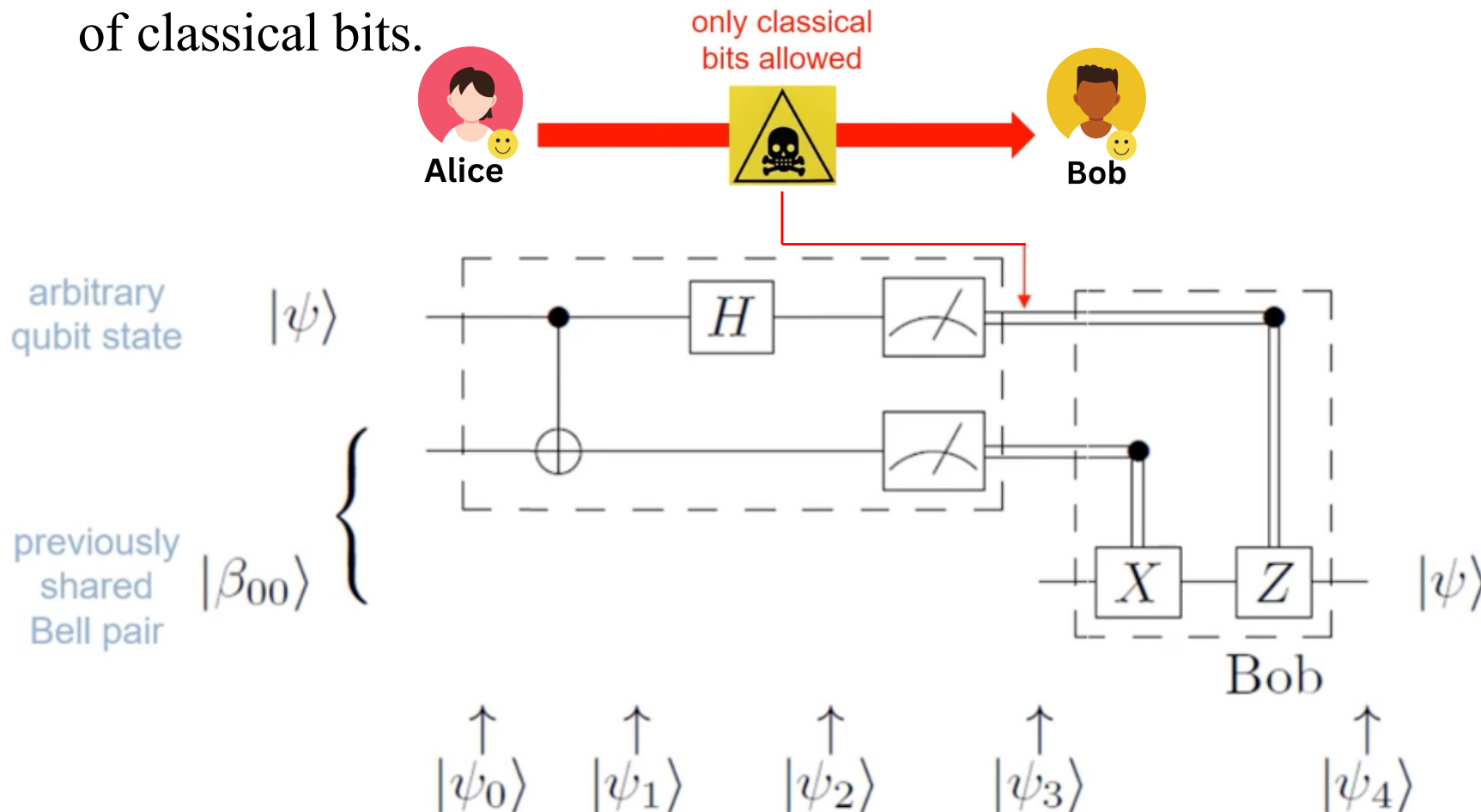
Bob uses the second classical bit from Alice to control a Pauli-X gate and the first one to control a Pauli-Z gate.

These two gates corrects precisely the deviation of the Bob's qubits from state $|\psi\rangle$.



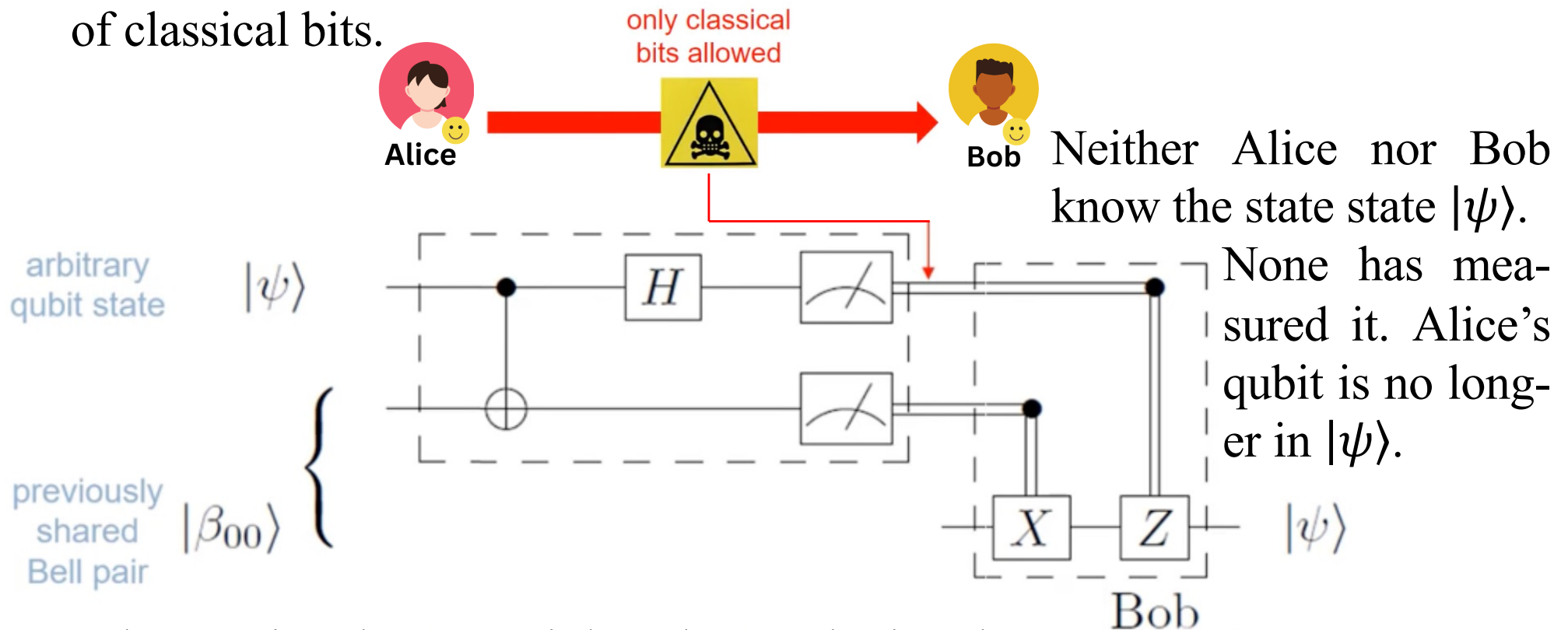
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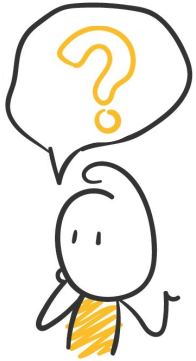
::... Teleportation

Let us now consider a different scenario where Alice holds a qubit in a given state $|\psi\rangle$, which is any arbitrary possible state, and he wants to transmit it to Bob, but the available channel conveys only a finite number of classical bits.



Teleportation does not violate the no-cloning theorem. Also, the Bell state is destroyed, and the two members are resources for the teleportation, but cannot be reused.

::... Superdense Coding vs Teleportation



Are these two protocols optimal?

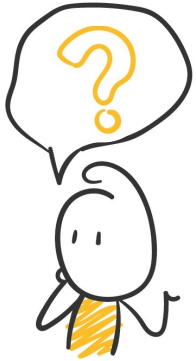


Optimum:

- There exists a protocol that allows to pack s classical bits in 1 qubit, more than the 2 classical bits of the superdense coding;
- There exists a protocol that allows sending a 1-qubit state with t classical bits, with t less than the 2 needed by teleportation.

There are alternatives better than the two protocols we seen.

::... Superdense Coding vs Teleportation



Are these two protocols optimal?



Optimum:

- There exists a protocol that allows to pack s classical bits in 1 qubit, more the 2 classical bits of the superdense coding;
- There exists a protocol that allows sending a 1-qubit state with t classical bits, with t less than the 2 needed by teleportation.

There are alternative better than the two protocols we seen.

The existence of superdense coding and teleportation poses bounds on the values of s and t : $s \geq 2$ and $t \leq 2$.

Imagine we can combine these alternatives if exists: we can use a qubit to send t classical bits, and use s classical bits to send this qubit. So, at the end, we have conveyed s classical bits with t classical bits.

::... Superdense Coding vs Teleportation

If we imagine that s is strictly larger than t ,

- a super-teleportation protocol allows sending 1 qubit with 1 bit, and a super-superdense coding that allows sending 4 bits with 1 qubit, so effectively we send 4 classical bits with only 1. If we iterate, we can send infinite classical information with just 1 bit, burning an infinite number of Bell pairs, and this is absurd.

S must be less than or equal to t , no "ratcheting up": $s \leq t$.

If we combine these three inequalities, the optimal values for s and t are 2. The two protocols are, in fact, already the best possible.

... Peres-Mermin Square

A fundamental theorem in quantum theory is the so-called **Kochen-Specker (KS) theorem**, which proves that quantum mechanics is incompatible with "non-contextual" hidden-variable theories.

The theorem states that it is impossible to consistently assign definite values (± 1) to all physical observables of a quantum system simultaneously. The value of a physical property is inherent to the particle and remains the same regardless of what other compatible measurements are performed along with it.

$$\begin{pmatrix} 0 & \hat{I} \\ \hat{I} & 0 \end{pmatrix} \quad \begin{pmatrix} \hat{X} & 0 \\ 0 & \hat{X} \end{pmatrix} \quad \begin{pmatrix} 0 & \hat{X} \\ \hat{X} & 0 \end{pmatrix}$$

$$\begin{pmatrix} \hat{Y} & 0 \\ 0 & \hat{Y} \end{pmatrix} \quad \begin{pmatrix} 0 & -i\hat{I} \\ i\hat{I} & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & -i\hat{Y} \\ i\hat{Y} & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & \hat{Y} \\ \hat{Y} & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & -i\hat{X} \\ i\hat{X} & 0 \end{pmatrix} \quad \begin{pmatrix} \hat{Z} & 0 \\ 0 & -\hat{Z} \end{pmatrix}$$

While Kochen-Specker provides the general theorem, the **Peres-Mermin square** is a specific, elegant, and "economic" example that proves it using nine observables. It is three by a three-by-three scheme of observables in a four-dimensional Hilbert space, made of two qubits.

::... Peres-Mermin Square

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$\begin{pmatrix} 0 & \hat{I} \\ \hat{I} & 0 \end{pmatrix} \quad \begin{pmatrix} \hat{X} & 0 \\ 0 & \hat{X} \end{pmatrix} \quad \begin{pmatrix} 0 & \hat{X} \\ \hat{X} & 0 \end{pmatrix}$	Each observable is described by a two-by-two matrix, which is also a Hermitian (self-adjoint) matrix so it can represent an observable.
$\begin{pmatrix} \hat{Y} & 0 \\ 0 & \hat{Y} \end{pmatrix} \quad \begin{pmatrix} 0 & -i\hat{I} \\ i\hat{I} & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & -i\hat{Y} \\ i\hat{Y} & 0 \end{pmatrix}$	The way these matrices are arranged gives some interesting properties: all those in the same row
$\begin{pmatrix} 0 & \hat{Y} \\ \hat{Y} & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & -i\hat{X} \\ i\hat{X} & 0 \end{pmatrix} \quad \begin{pmatrix} \hat{Z} & 0 \\ 0 & -\hat{Z} \end{pmatrix}$	commute with each other, as well as those in the same column, but not all commute with each other.

::... Peres-Mermin Square

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This implies that we can measure all the observables in a given row, simultaneously, or those in a given column.

... Peres-Mermin Square

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$$\begin{pmatrix} 0 & \hat{I} \\ \hat{I} & 0 \end{pmatrix} \quad \begin{pmatrix} \hat{X} & 0 \\ 0 & \hat{X} \end{pmatrix} \quad \begin{pmatrix} 0 & \hat{X} \\ \hat{X} & 0 \end{pmatrix} \rightarrow 1$$

$$\begin{pmatrix} \hat{Y} & 0 \\ 0 & \hat{Y} \end{pmatrix} \quad \begin{pmatrix} 0 & -i\hat{I} \\ i\hat{I} & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & -i\hat{Y} \\ i\hat{Y} & 0 \end{pmatrix} \rightarrow 1$$

$$\begin{pmatrix} 0 & \hat{Y} \\ \hat{Y} & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & -i\hat{X} \\ i\hat{X} & 0 \end{pmatrix} \quad \begin{pmatrix} \hat{Z} & 0 \\ 0 & -\hat{Z} \end{pmatrix} \rightarrow 1$$

$$\downarrow \\ 1$$

$$\downarrow \\ 1$$

$$\downarrow \\ -1$$

If we take observables in a given row and multiply them, we obtain the unity operator. Something very similar happens over the columns, with the only exception in the third column, where the result is minus the unity operator.

Such an exception is pivotal in the argumentation we want to do.

::... Peres-Mermin Square

Quantum theory states that we cannot measure two observables with arbitrary accuracy, and measuring one might preclude measuring the other. After measurement, the state changes, and the probabilities of future outcomes change, so knowing both with perfect precision simultaneously. There is quite a debate around this "Measurement Problem", where the clean, deterministic math of the Schrödinger equation meets the messy, unpredictable reality of a laboratory. To be precise, this is rooted in the Heisenberg Uncertainty Principle and the Collapse of the Wavefunction.

- The Uncertainty Principle - particles don't actually possess a definite position and momentum simultaneously. The more you "pin down" where a particle is, the more its velocity becomes a blurry smear of possibilities.
- State Collapse and Back-Action - When measuring a system, the used tool interacts with it, causing the wavefunction—a mathematical cloud of probabilities—to "collapse" into a single definite state. This is why future measurements change.

::... Peres-Mermin Square

Another side, however, assumes that observables do not, in general, have very well-defined values and that measurement tools do not alter them. The KS theorem, in general, and the Peres-Mermin scheme, in a particular case, can give a glimpse into this issue.

Let us imagine that all of these nine observables have well-defined values, and all of them have eigenvalues ± 1 , and they are the only ones, so if, on some fundamental level, there are some definite values assigned to each of

these observables, and they are ± 1 .

$$\begin{pmatrix} 0 & \hat{I} \\ \hat{I} & 0 \end{pmatrix} \quad \begin{pmatrix} \hat{X} & 0 \\ 0 & \hat{X} \end{pmatrix} \quad \begin{pmatrix} 0 & \hat{X} \\ \hat{X} & 0 \end{pmatrix} \rightarrow 1$$

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$$\begin{pmatrix} 0 & \hat{Y} \\ \hat{Y} & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & -i\hat{X} \\ i\hat{X} & 0 \end{pmatrix} \quad \begin{pmatrix} \hat{Z} & 0 \\ 0 & -\hat{Z} \end{pmatrix} \rightarrow 1$$

$$\downarrow \\ 1$$

$$\downarrow \\ 1$$

$$\downarrow \\ -1$$

The relationships for these nine observables (all those over a row/column when multiplied return 1) imply a proper assignment scheme of these values such that the product of the assignments of the assigned values in a row/column gives 1.

::... Peres-Mermin Square

Let us assume that such an assignment exists, and all observables can be measured simultaneously. The value assigned to an observable does not depend on whatever it has been measured jointly with another observable in the same row/column, and this is called **non-contextuality**.

Let us consider the product of all nine observables, row by row, and then multiply the results; the result is 1. If we look at the columns and perform the same operations, the result is -1. This is a logical contradiction.

$$\begin{pmatrix} 0 & \hat{I} \\ \hat{I} & 0 \end{pmatrix} \quad \begin{pmatrix} \hat{X} & 0 \\ 0 & \hat{X} \end{pmatrix} \quad \begin{pmatrix} 0 & \hat{X} \\ \hat{X} & 0 \end{pmatrix} \rightarrow 1$$

$$\begin{pmatrix} \hat{Y} & 0 \\ 0 & \hat{Y} \end{pmatrix} \quad \begin{pmatrix} 0 & -i\hat{I} \\ i\hat{I} & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & -i\hat{Y} \\ i\hat{Y} & 0 \end{pmatrix} \rightarrow 1$$

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$$\downarrow \\ 1$$

$$\downarrow \\ 1$$

$$\downarrow \\ -1$$

One of the two initial assumptions must be false: either assigning a pre-existing, well-defined value to each observable or assuming non-contextuality in the joint measures.

... Peres-Mermin Square

Let us assume that such an assignment exists, and all observables can be measured simultaneously. The value assigned to an observable does not depend on whatever it has been measured jointly with another observable in the same row/column, and this is called **non-contextuality**.

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$\begin{pmatrix} 0 & \hat{I} \\ \hat{I} & 0 \end{pmatrix} \quad \begin{pmatrix} \hat{X} & 0 \\ 0 & \hat{X} \end{pmatrix} \quad \begin{pmatrix} 0 & \hat{X} \\ \hat{X} & 0 \end{pmatrix} \rightarrow 1$	The Conflict: Realism vs. Quantum Mechanics. In the classical world, a ball has a color and a size. It doesn't matter if you measure "color and weight" or "color and volume", the color is just red. In the quantum world, the KS Theorem proves that values cannot be assigned to all observables in accordance to the functional relationships between them.
$\begin{pmatrix} \hat{Y} & 0 \\ 0 & \hat{Y} \end{pmatrix} \quad \begin{pmatrix} 0 & -i\hat{I} \\ i\hat{I} & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & -i\hat{Y} \\ i\hat{Y} & 0 \end{pmatrix} \rightarrow 1$	
$\begin{pmatrix} 0 & \hat{Y} \\ \hat{Y} & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & -i\hat{X} \\ i\hat{X} & 0 \end{pmatrix} \quad \begin{pmatrix} \hat{Z} & 0 \\ 0 & -\hat{Z} \end{pmatrix} \rightarrow 1$	
$\downarrow \qquad \qquad \downarrow \qquad \qquad \downarrow$	
$1 \qquad \qquad 1 \qquad \qquad -1$	

::... Peres-Mermin Square

The relationship between the KS theorem and protocols like superdense coding and teleportation is deep but subtle, as they are essentially two sides of the same coin: Quantum Contextuality and Quantum Non-locality.

1. Contextuality vs. Entanglement - Contextuality is the "magic" that makes entanglement useful. For superdense coding to work, Alice must be able to change the joint state of the two particles by only touching one. If the particles had pre-existing, non-contextual classical values, Alice's local action couldn't "unlock" the extra information Bob eventually receives.
2. The "Quantum Resource" View - contextuality (from KS) and non-locality (from Bell/Teleportation) are only different manifestations.
 - In Teleportation: entanglement is used to move a state. The KS theorem implies that the state we are moving to is not a list of fixed values, but a set of contextual potentialities. If the state were just a classical list of "red, heavy, fast", we wouldn't need a complex quantum protocol to move it; we'd just send the list.

::... Peres-Mermin Square

- In Superdense Coding: Alice sends 2 bits of information using only 1 qubit. This "extra" capacity comes from the fact that the quantum state space is contextual. The KS theorem proves that you cannot simulate this "informational density" with a classical model where properties have fixed, context-independent values.
3. Pseudo-Telepathy Games - these are games (like the Mermin-Peres Magic Square) where two players can win a game without communicating—as if they were telepathic—provided they share an entangled state. Every "Pseudo-Telepathy" game is essentially a physical realization of a Kochen-Specker contradiction. The same math that proves you can't assign fixed values to a single system (KS) is what allows two people to win these "telepathic" games using entanglement (the basis for teleportation protocols).

::... Bell Experiment

If the KS theorem tells us the universe is "contextual" (the answer depends on the question), **Bell's Theorem** tells us the universe is "non-local" (the answer over here depends on the question asked way over there).

Both theorems serve as "no-go" proofs that dismantle the classical hope for Local Realism.

- Before Bell, Einstein (and others) hoped that quantum mechanics was just an incomplete description of a classical-like world:
 - Realism: Objects have definite properties (like spin) before we measure them.
 - Locality: Information cannot travel faster than light; an action in one place cannot instantly affect an object far away.

Bell's Theorem proved that you cannot have both. If you want to keep the results of quantum experiments, you have to give up at least one of these "common sense" pillars.

::... Bell Experiment

- While they both target the idea of "hidden variables" (pre-existing values), they do so from different angles:
 - Bell's Theorem is about distance: if particles have hidden values, those must be able to change instantly across the universe.
 - KS Theorem is about internal logic: even for a particle, it cannot have assigned "answers" making sense across all possible contexts.

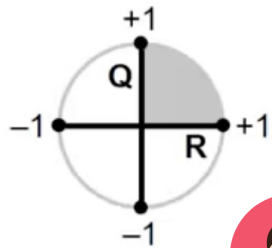
The reason there is still a debate is that we have to choose which "sacrifice" to make:

- The Copenhagen Interpretation says: "Give up Realism. Properties don't exist until measured".
- Bohmian Mechanics says: "Keep Realism, but give up Locality. Particles have positions, but they are guided by a non-local force".
- Many-Worlds says: "Give up the idea of a single outcome. Everything that can happen, does happen".

Both Bell and KS show that the "problem" of measurement isn't a failure of our technology, but a fundamental feature of how the universe is wired.

::... Bell Experiment

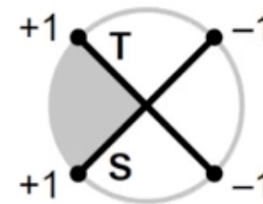
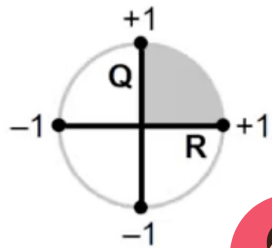
In a Bell experiment, Alice and Bob are two experimenters, each manipulating qubits. Alice performs one of the two measurements in a two-dimensional section of the Bloch sphere.



An observable is an axis through the Bloch sphere. The two observables that Alice can perform are represented by two axes. The observables are called Q and R, and correspond to Pauli-X and Pauli-Z (x and z axes) with corresponding measures ± 1 per each.

::... Bell Experiment

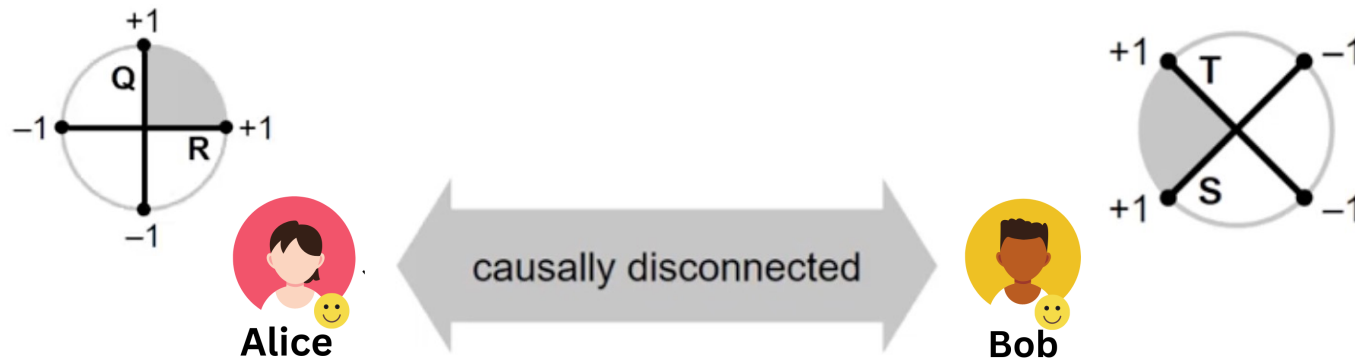
In a Bell experiment, Alice and Bob are two experimenters, each manipulating qubits. Alice performs one of the two measurements in a two-dimensional section of the Bloch sphere.



Bob performs one of two measurements on the qubits with axes rotated against the ones of Alice. Bob's observables are S and T, being the two diagonals of the Bloch sphere, with values ± 1 per each.

... Bell Experiment

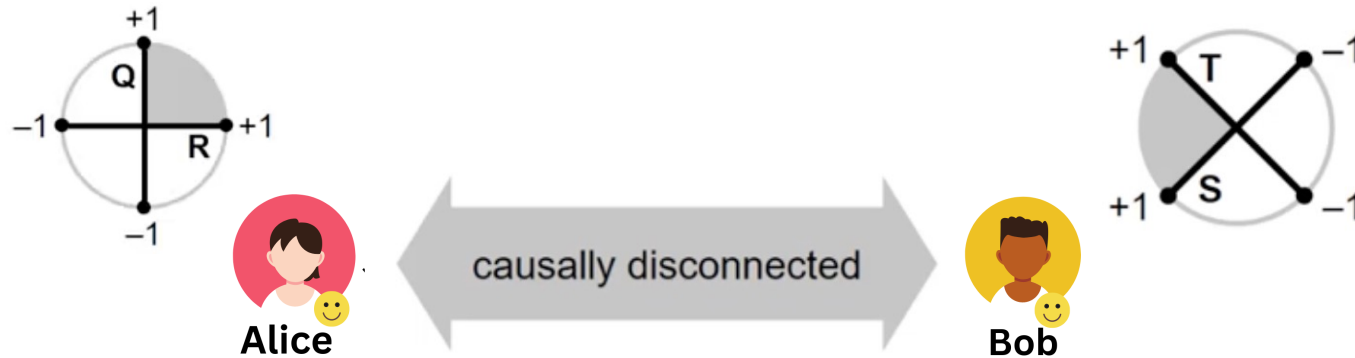
In a Bell experiment, Alice and Bob are two experimenters, each manipulating qubits. Alice performs one of the two measurements in a two-dimensional section of the Bloch sphere.



Alice and Bob are placed far away from each other, and are "causally disconnected", meaning that they are so far apart—and their measurements happen so quickly—that a signal traveling at the speed of light (c) could not possibly pass between them to "tell" Bob's particle what Alice chose to measure. Whatever Alice does on her side, does not influence what in Bob side, and vice versa

::... Bell Experiment

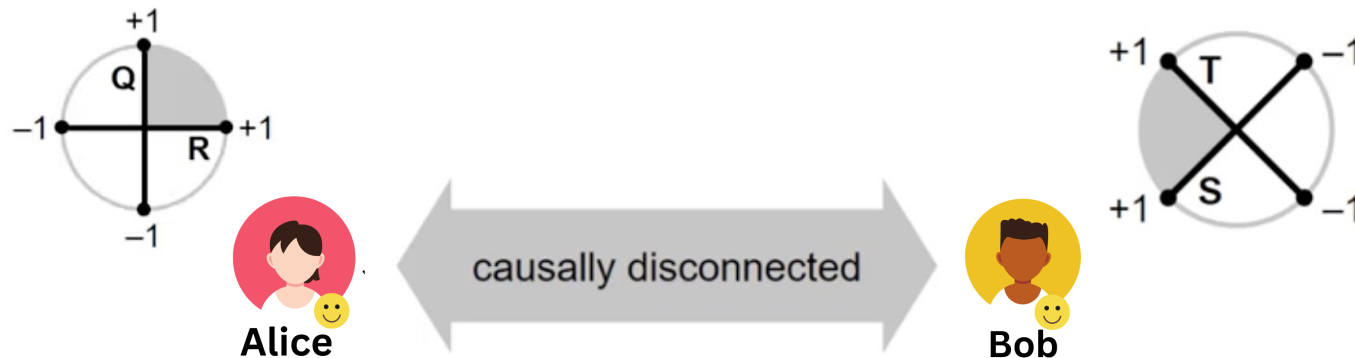
In a Bell experiment, Alice and Bob are two experimenters, each manipulating qubits. Alice performs one of the two measurements in a two-dimensional section of the Bloch sphere.



From Einstein's theory of relativity, which states that signals or any kind of information can travel at most at the speed of light, if we separate Alice and Bob far enough, at the speed of light, we need some minimum time to transfer any kind of information from one side to the other side to affect any kind of disturbance to the other side.

... Bell Experiment

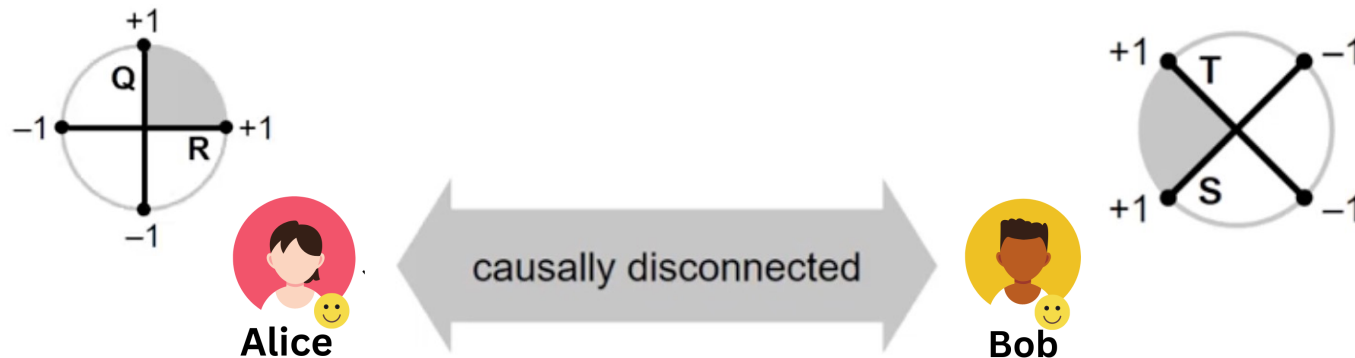
In a Bell experiment, Alice and Bob are two experimenters, each manipulating qubits. Alice performs one of the two measurements in a two-dimensional section of the Bloch sphere.



Quantum entanglement is the phenomenon in which measuring a particle at Alice's location seems to "instantly" determine its state at Bob's location. While this may seem to break Einstein's rule, it actually doesn't. This is known as the **No-Communication Theorem**. While the states are correlated instantaneously, Alice cannot use that correlation to send a specific message (like "Hello") to Bob faster than light. She still needs a classical channel (radio, fiber optics, etc.) to tell Bob what she measured.

::... Bell Experiment

In a Bell experiment, Alice and Bob are two experimenters, each manipulating qubits. Alice performs one of the two measurements in a two-dimensional section of the Bloch sphere.

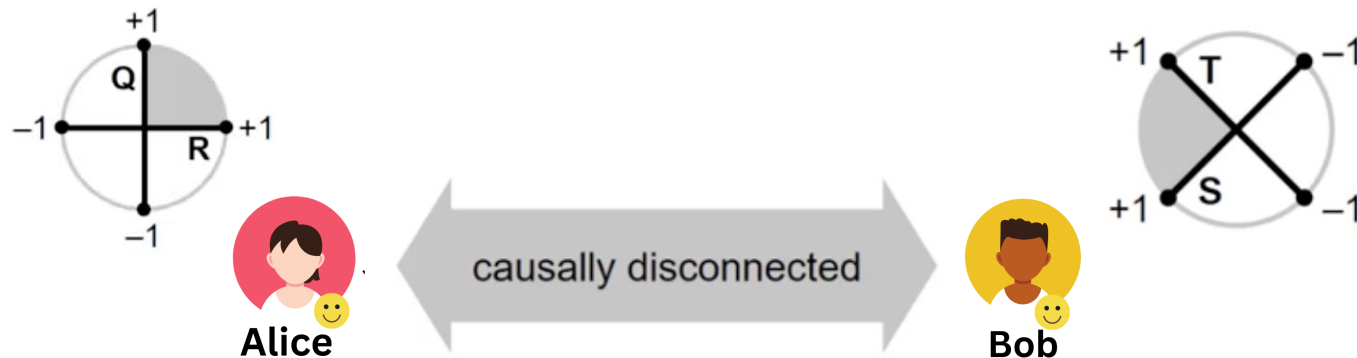


$$A := Q \otimes S + R \otimes S + R \otimes T - Q \otimes T$$

Let us consider a particular two-qubit observable A that is composed of specific one-qubit observables that Alice and Bob measure (the first is an observable at Alice's side, tensor product of the observable at Bob's side).

::... Bell Experiment

In a Bell experiment, Alice and Bob are two experimenters, each manipulating qubits. Alice performs one of the two measurements in a two-dimensional section of the Bloch sphere.



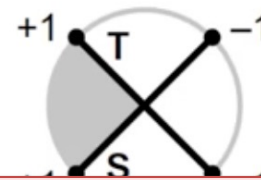
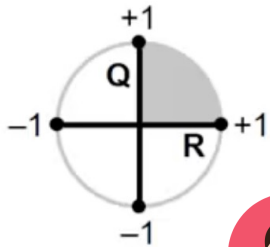
$$A := Q \otimes S + R \otimes S + R \otimes T - Q \otimes T = (Q + R) \otimes S + (R - Q) \otimes T$$

Let us consider a particular two-qubit observable A that is composed of specific one-qubit observables that Alice and Bob measure (the first is an observable at Alice's side, tensor product of the observable at Bob's side).

This particular two-qubit observable can be written in a different form. We cannot measure the one-qubit observables jointly, as they don't commute.

... Bell Experiment

In a Bell experiment, Alice and Bob are two experimenters, each manipulating qubits. Alice performs one of the two measurements in a two-dimensional section of the Bloch sphere.



Alice

$$A := Q \otimes S + R \otimes T$$

Let us choose specific observables (on each side).

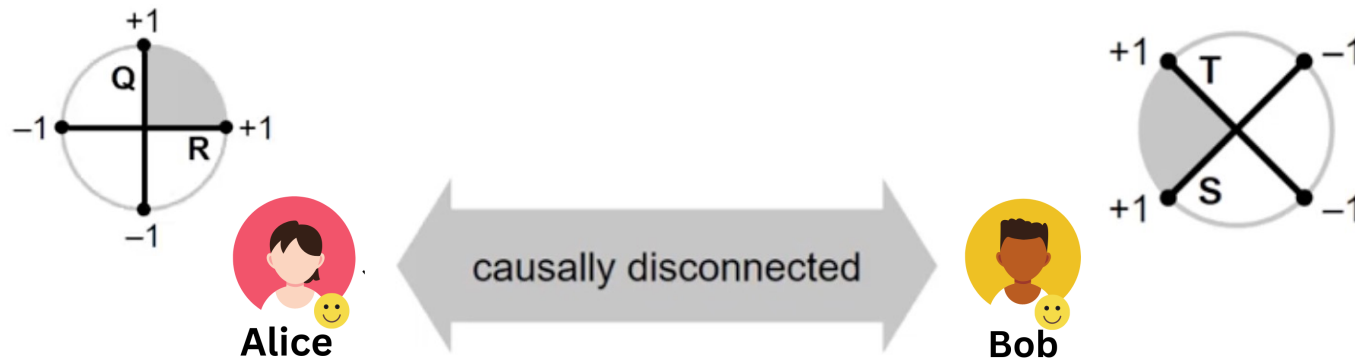
Two observables, let's call them $\hat{A}$ and $\hat{B}$, commute if performing measurement A, then measurement B gives the same result as performing B, then A. Mathematically, we express this using a commutator, denoted by square brackets: $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$

- If $[\hat{A}, \hat{B}] = 0$: They commute. You can measure both simultaneously with infinite precision. They share a "common language" (eigenstates).
- If $[\hat{A}, \hat{B}] \neq 0$: They do not commute. The act of measuring one physically alters the system in a way that "blurs" the result of the other.

This particular two-qubit observable can be written in a different form. We cannot measure the one-qubit observables jointly, as **they don't commute**.

::... Bell Experiment

In a Bell experiment, Alice and Bob are two experimenters, each manipulating qubits. Alice performs one of the two measurements in a two-dimensional section of the Bloch sphere.



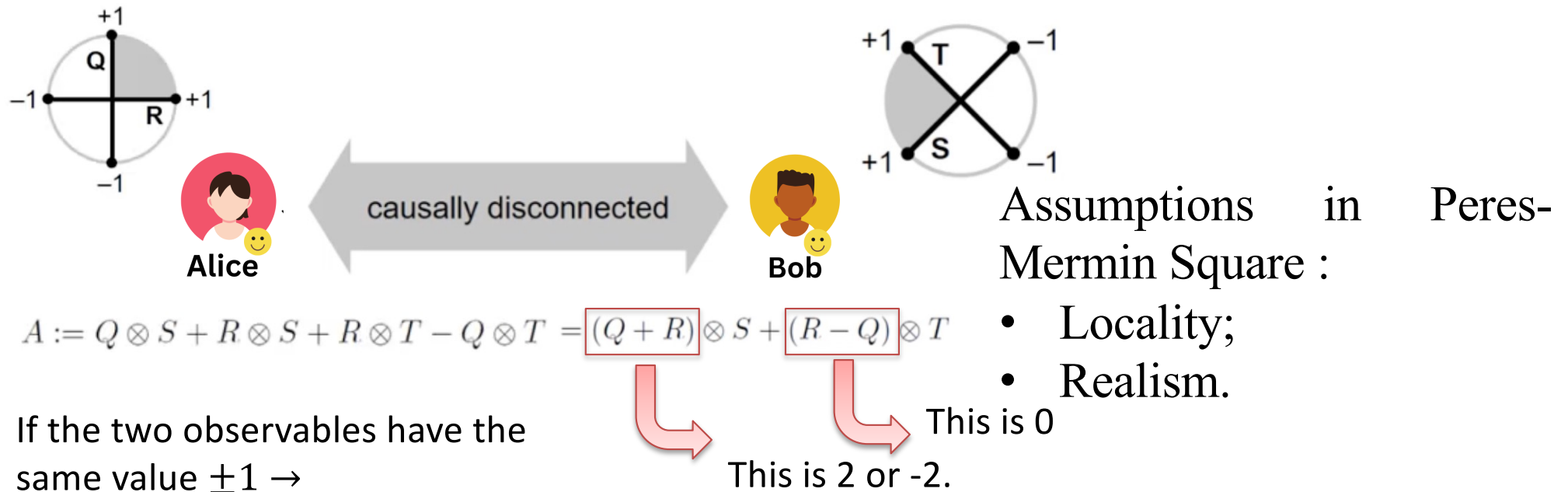
$$A := Q \otimes S + R \otimes S + R \otimes T - Q \otimes T = (Q + R) \otimes S + (R - Q) \otimes T$$

We can compute the expectation value of A by preparing the qubit in this state and measuring it many times. For some sets of measurements, the first product is determined; for others, the second product; and so on. The four contributions to the sum are measured separately and then aggregated to obtain the expected value of the two-qubit observable.

In theory, we can extract constraints on such an expected value.

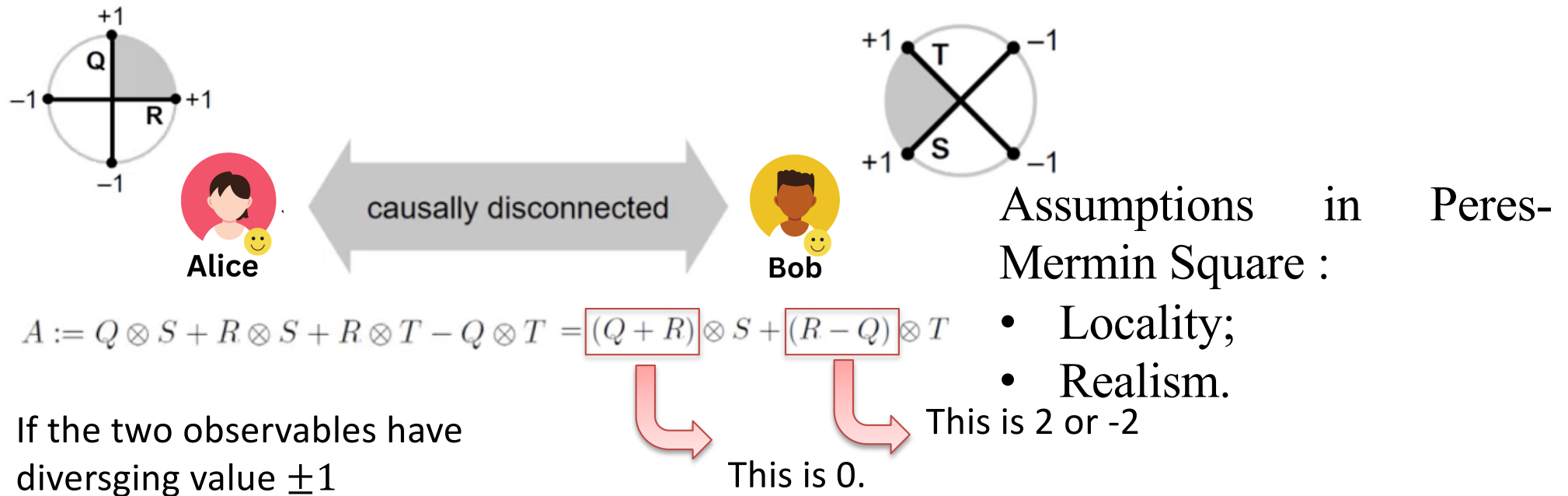
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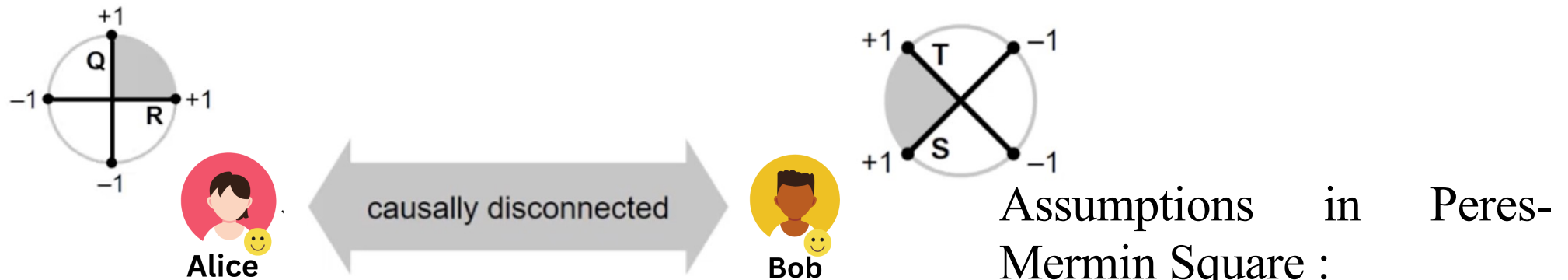
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CHSH:

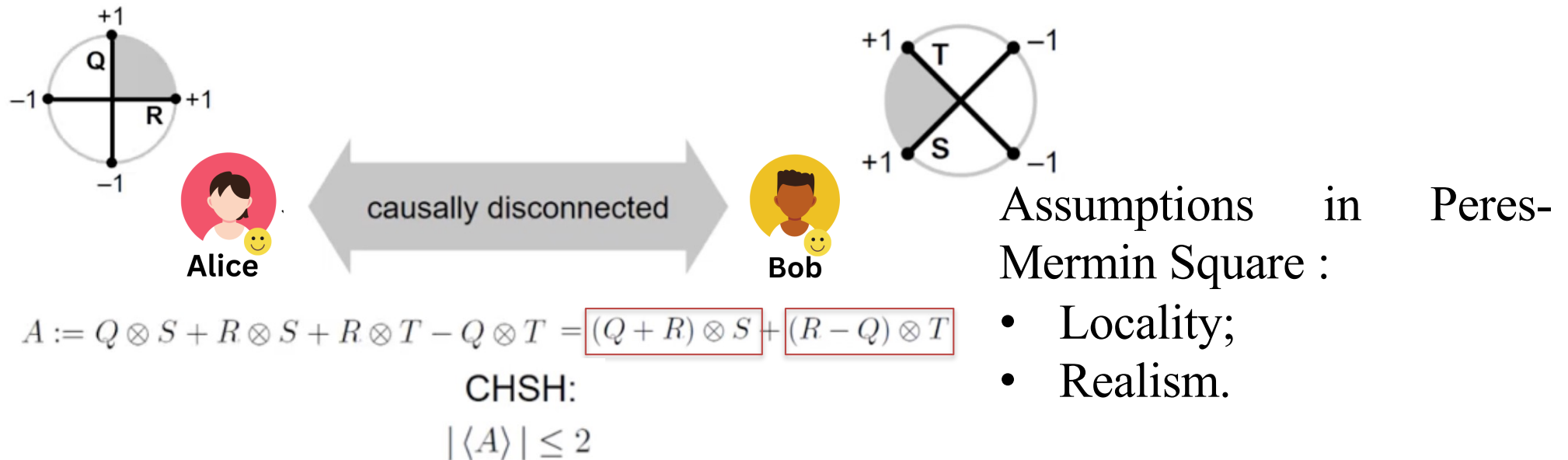
$$|\langle A \rangle| \leq 2$$

- Locality;
- Realism.

Whatever the case we have, one of the two contributions vanishes, and the other is tensor producted with the relative observable of Bob, assuming a value ± 1 . So, the Observable A can have value -2 or 2. This is in theory, but the measures are more probabilistic in nature, so the expected value of the observable has 2 as higher bound.

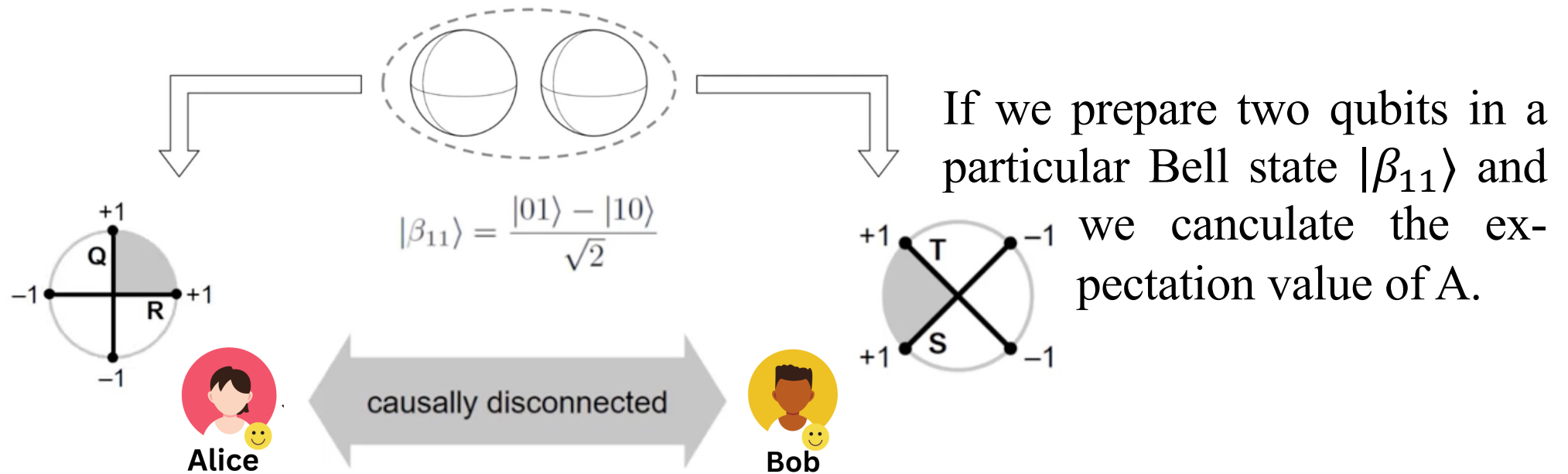
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In a Bell experiment, Alice and Bob are two experimenters, each manipulating qubits. Alice performs one of the two measurements in a two-dimensional section of the Bloch sphere.



This result is called the **CHSH inequality** (named after Clauser, Horne, Shimony, and Holt). This is the theory, and let's see what the Bell experiment actually shows.

... Bell Experiment

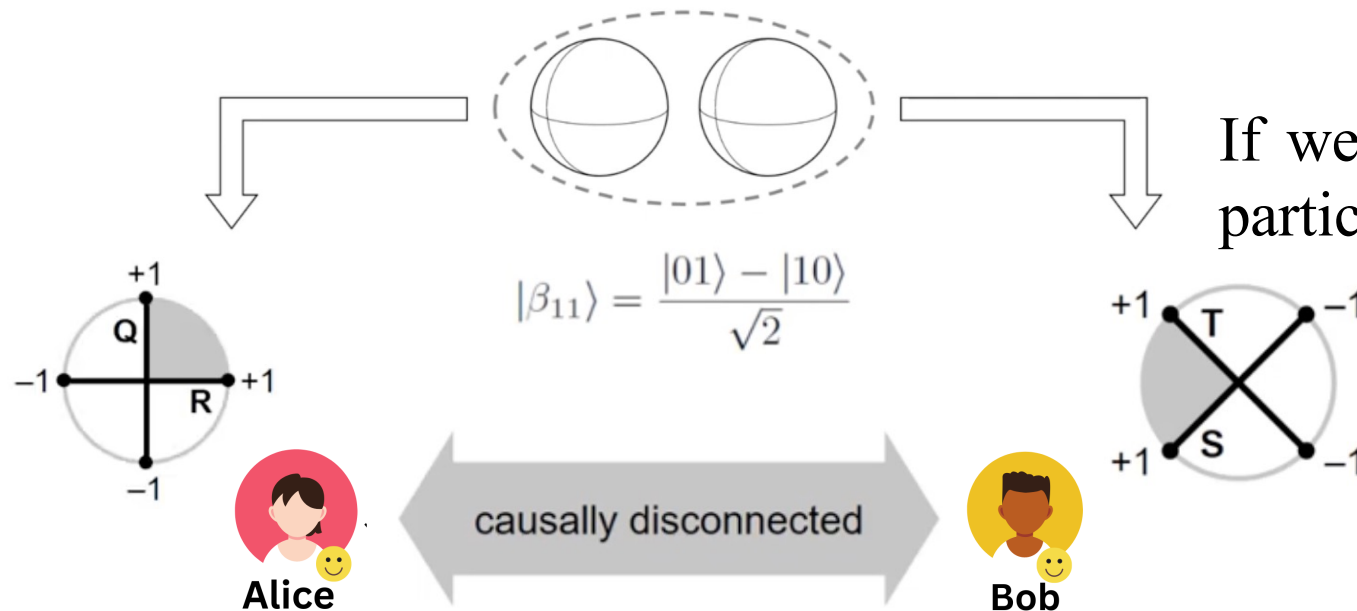


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... Bell Experiment



If we prepare two qubits in a particular Bell state $|\beta_{11}\rangle$ and we calculate the expectation value of A . We find that such a value is $2\sqrt{2}$, as a theoretical result, which has also been found experimentally.

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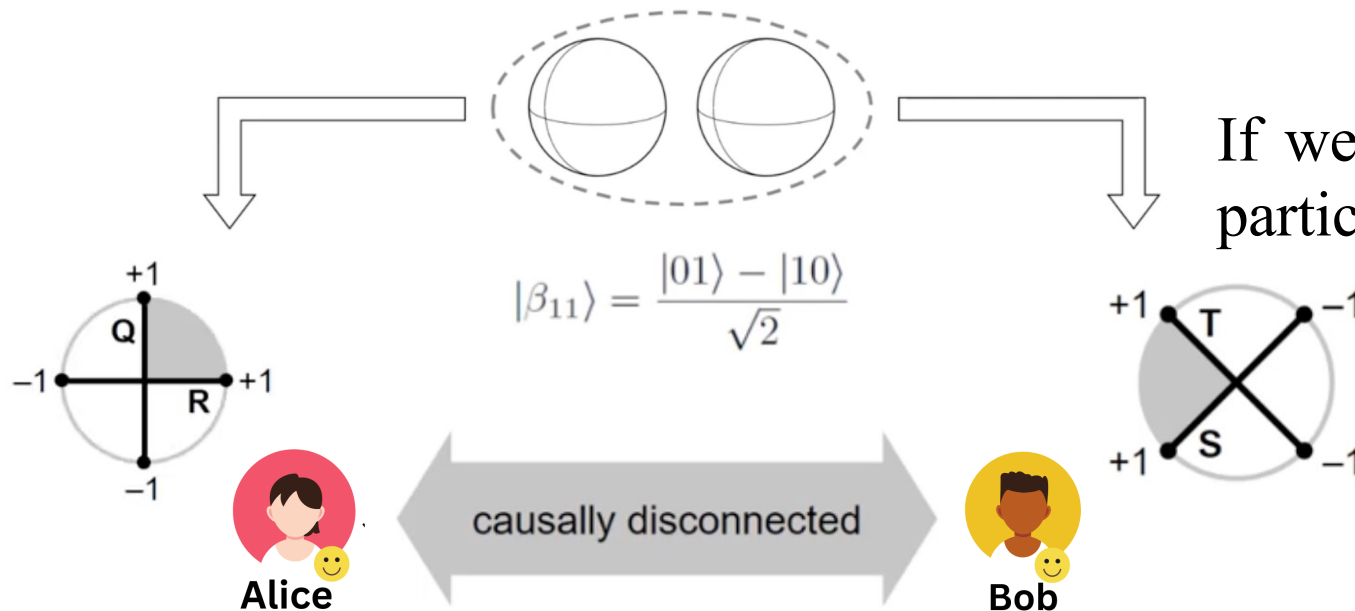
Tsirelson bound:

$$\langle \beta_{11} | \hat{A} | \beta_{11} \rangle = 2\sqrt{2}$$

This Tsirelson's bound violates the CHSH inequality. Even with entanglement, Alice and Bob cannot achieve infinite correlation.

Nature restricts quantum nonlocality to this specific mathematical value.

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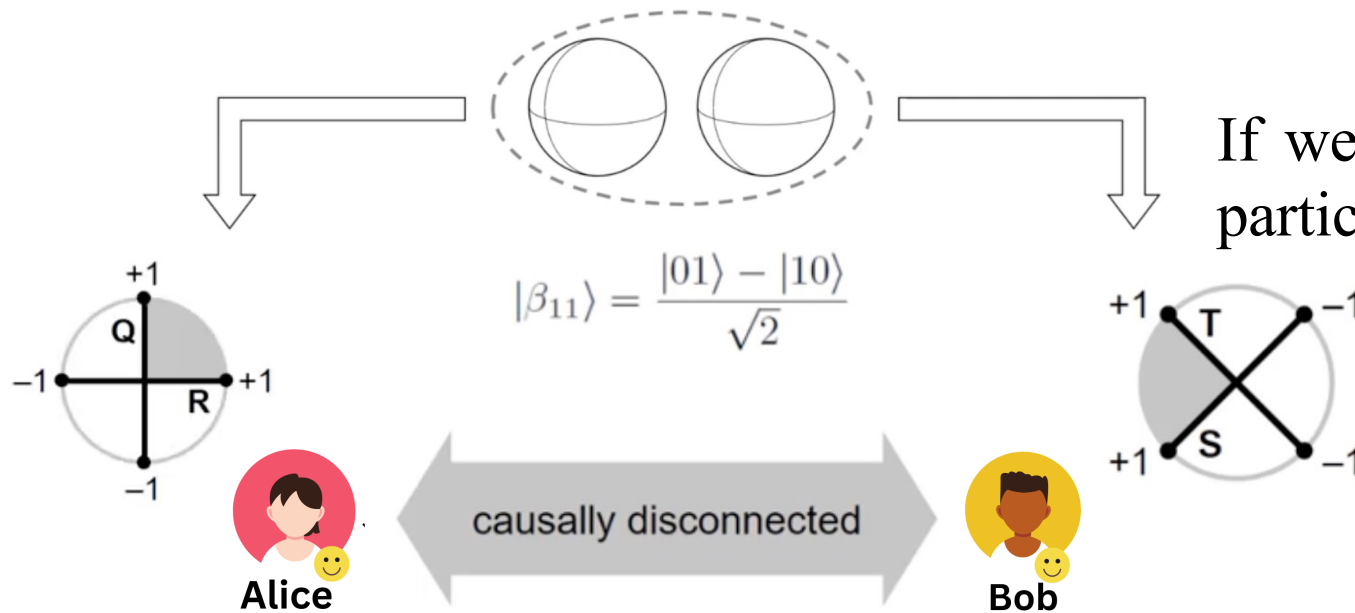
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Why is the quantum bound exactly $2\sqrt{2}$ and not higher? This limit deeply depends on the Information Causality:

if the bound were any higher, Alice could communicate an impossible amount of information to Bob using only a small amount of data, effectively violating the "no-signaling" principle.

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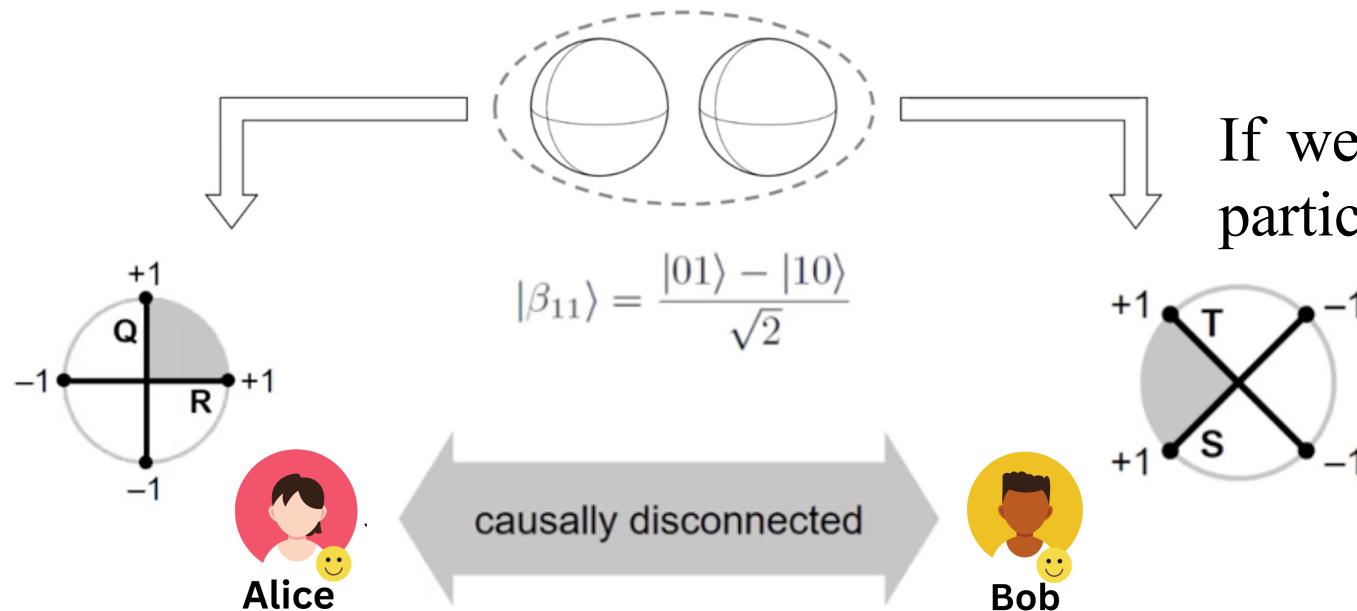
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This contradiction implies that one of the two assumptions does not hold.

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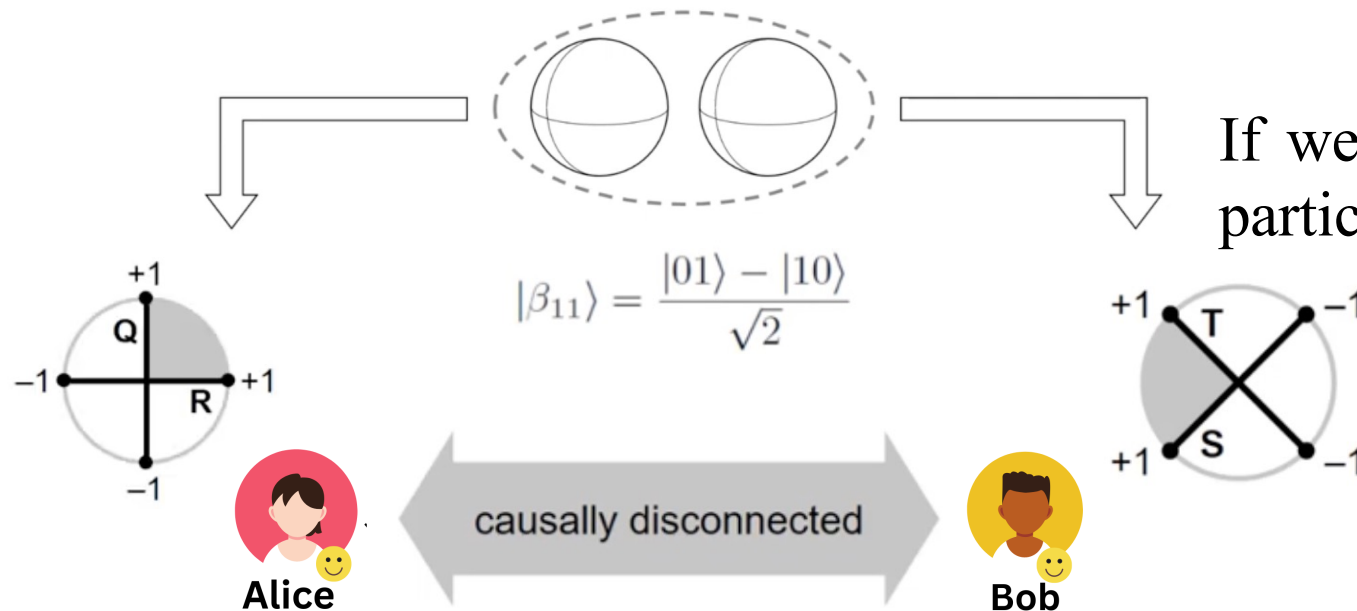
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An interesting aspect is that the observable $\hat{A}$ has eigenvalues $\pm 2\sqrt{2}$ and 0, and the value $2\sqrt{2}$ is the unique maximum, confirming Tsirelson's Bound.

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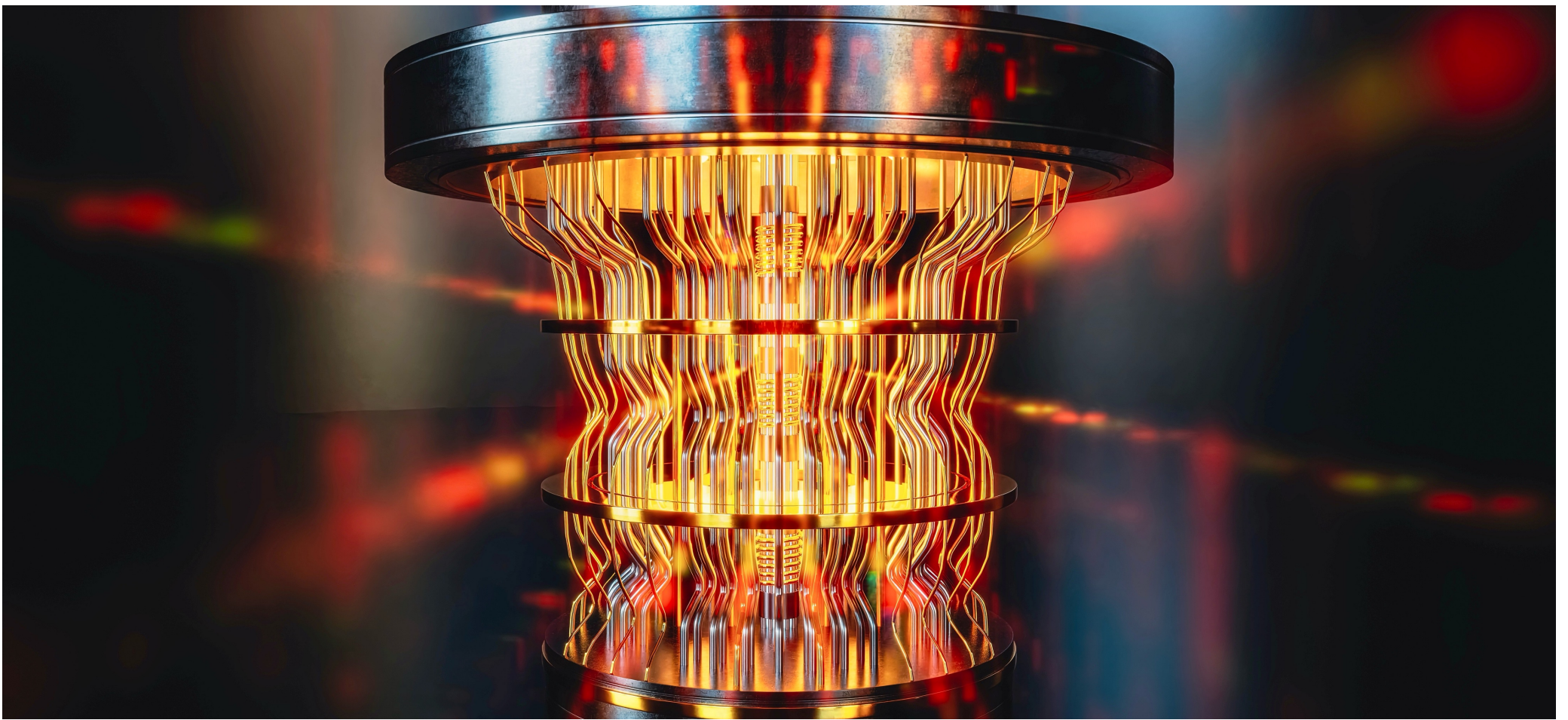
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Tsirelson bound:

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The observable $\hat{A}$ has $|\beta_{11}\rangle$ as its the eigenvector corresponding to the maximum eigenvalue $2\sqrt{2}$.

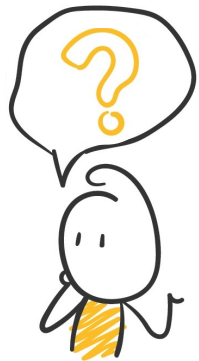
So, given an arbitrary state $|\psi\rangle$ for a two-qubit system if the expected value of the $\hat{A}$ is $2\sqrt{2}$, we can be sure that $|\psi\rangle = |\beta_{11}\rangle$.



Advanced Protocols for Key Distribution

::... EPR Protocol

What we learn from the Bell's experiment can be put to concrete use for cryptography.



How is this possible?

::... EPR Protocol

The **EPR protocol** (named after Einstein-Podolsky-Rosen) uses the outcome of the Bell's experiment as its foundational basis. The Bell's states or pairs are also indicated as EPR states or pairs.

- In 1935, Albert Einstein, Boris Podolsky, and Nathan Rosen published a famous paper titled "Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?", with the example of two particles that had interacted in the past and then moved far apart. According to quantum mechanics, these particles remained "entangled" (though they didn't use that word yet). EPR argued that if you could predict the property of one particle by measuring the other—without ever touching the first—then that property must have been a "pre-existing element of reality". Since quantum mechanics couldn't account for these pre-existing values without allowing for "spooky action at a distance", they concluded that quantum mechanics was incomplete. Thus, an "EPR pair" refers to this specific setup used to challenge the foundations of the theory.

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- Fast forward to 1964: John Bell took the philosophical "EPR Paradox" and turned it into a mathematical test. He showed that the correlations predicted by quantum mechanics for these pairs were actually stronger than any theory involving "pre-existing elements of reality" (Hidden Variables) could ever allow. The four specific, maximally entangled basis states are the mathematical objects that produce these record-breaking correlations. Because Bell used these states to prove EPR wrong (or rather, to prove that "Realism" and "Locality" cannot coexist), they were named the Bell States.

While referring to the same, the choice of name depends on the context:

- EPR Pair: Usually used when discussing the physics and philosophy of entanglement, non-locality, and the nature of reality. It highlights the "spooky" connection between two distant points.

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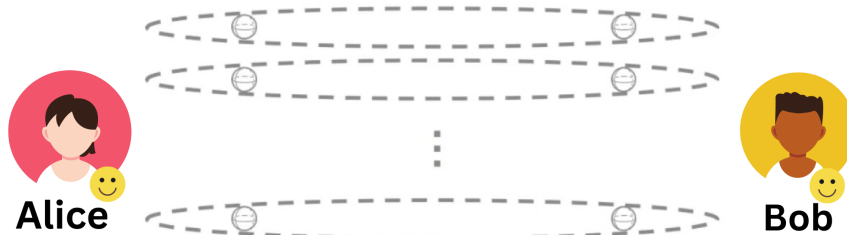
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While referring to the same, the choice of name depends on the context:

- Bell State: Usually used in Quantum Information and Computation. It treats the pair as a mathematical resource used for tasks like teleportation, superdense coding, or error correction.

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They perform measurements on their member of the Bell's pairs, similarly as seen in the Bell experiment: on Alice's side, we have observables Q and R, while on Bob's side, we have S and T.

All is not exactly as in the Bell's experiment, but there is a modification: Alice and Bob are allowed to measure a third observable.

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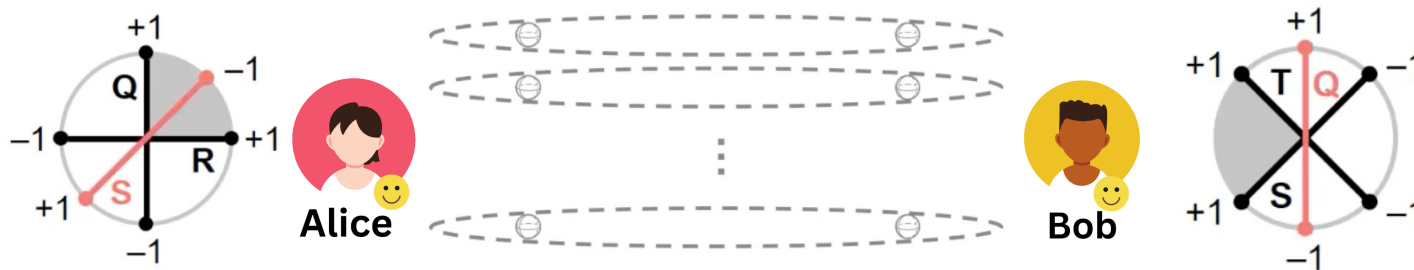


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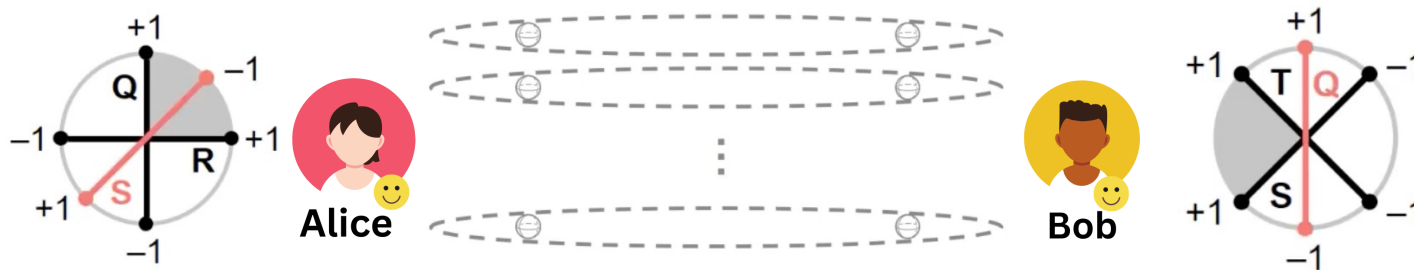


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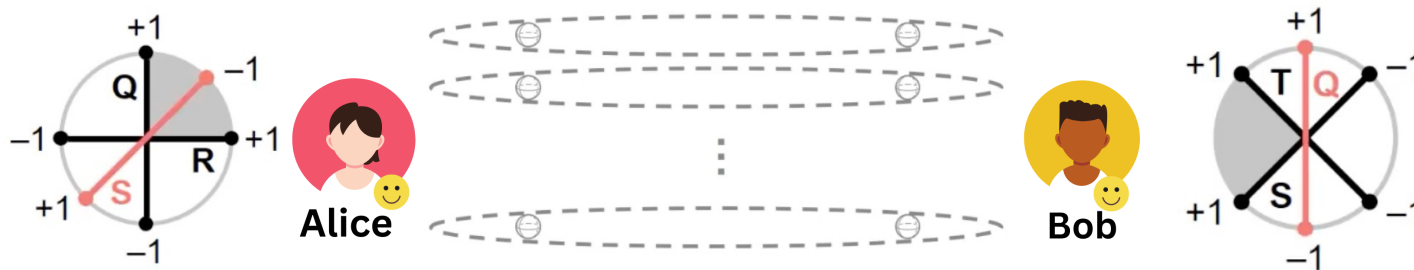


Alice and Bob select randomly one observable and perform the measurements on their side of the Bell's states they collected, without sharing the order in which they want to perform them or which observable they selected. So, they make independent and random measurements.

It may happen that for one Bell's pair, Alice and Bob measure the same observable: by accident, they both measured Q or S for some members of their Bell's pair, while for the others they measured different observables.

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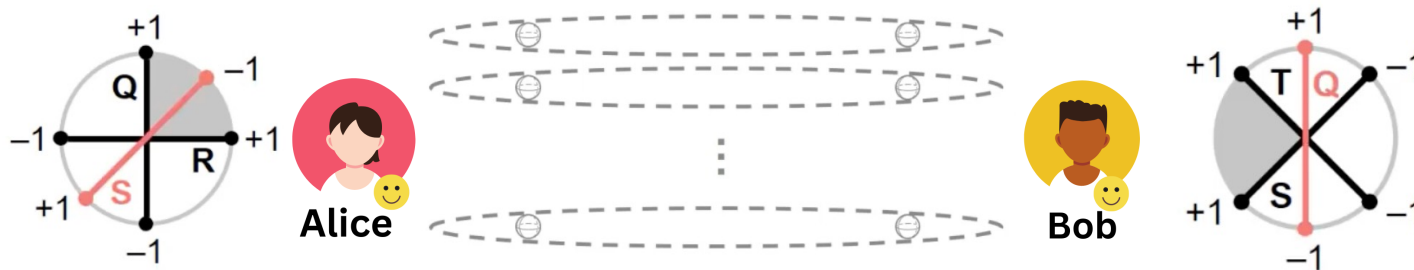
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When they measure the same observables, their outcomes are guaranteed to be opposites of each other, because $|\beta_{11}\rangle$ holds the perfect anti-correlation: if the measurement on one qubit returns 0, then the measurement on the other one is guaranteed to output 1, and vice versa. This property does not hold only in the standard basis, but in any arbitrary one. It does not matter if both measure Q, which corresponds to Pauli-Z and happens to be the standard basis, and its perfectly anti-correlated, but also when they both measure S as it's perfectly anti-correlated.

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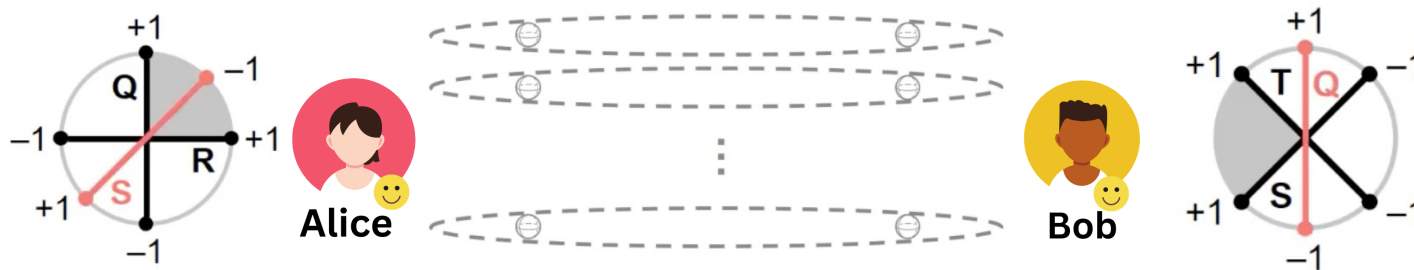
After having performed all the measurements, Alice and Bob communicate with each other over a classical channel about which measurements they performed, and they group the Bell's pairs in two clusters:

- one groups those they performed identical measurements;
- the second contains those on which they performed different measurements.

In the first group, Alice can invert all the measures so they match those of Bob, so that both end up having the same bit strings and the shared key.

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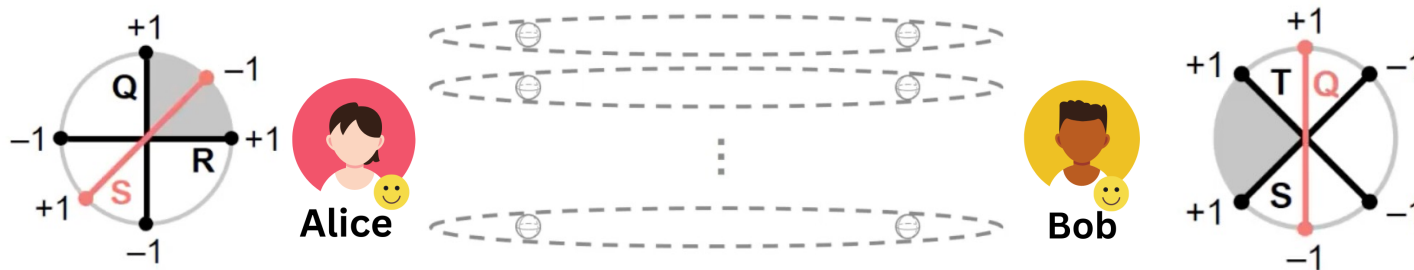


The other group can be used to be sure that nobody eavesdropped on this sharing process. This is done by using the measurements in this group to compute the expected value of the observable $\hat{A}$ and check whenever it is equal to the Tsirelson's bound:

- if it is, Alice and Bob can be sure that the qubits used for the measurements are in the necessary Bell's state $|\beta_{11}\rangle$, then there was no eavesdropper who tempered with the Bell's state.

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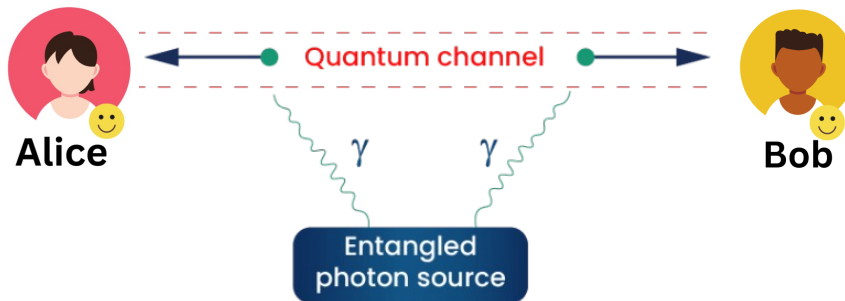
In fact, an eavesdropper would have established a correlation with the Alice and Bob qubits, but the Monogamy of the Entanglement causes a trouble with this operation.

- if it is not, Alice and Bob can suspect the presence of an eavesdropper.

This proves the security of the EPR protocol, as it uses both the Bell's inequality and Tsirelson's bound, as well as the Monogamy of the Entanglement.

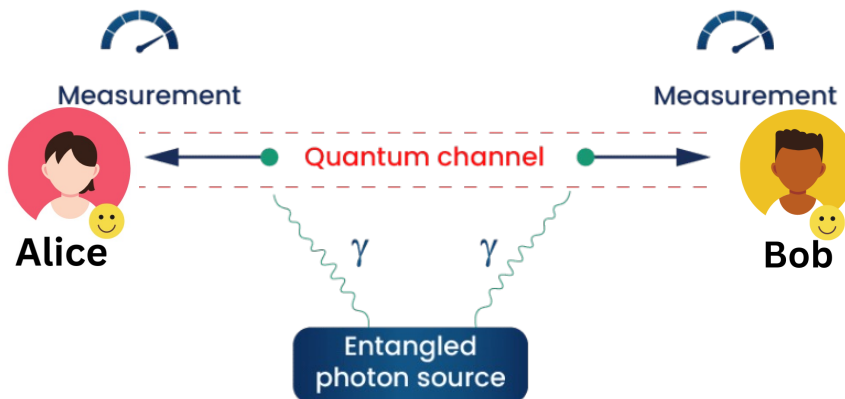
::... E91 Protocol

The most famous "concrete" QKD protocol based on the EPR protocol is the **Ekert91 (E91) protocol**, proposed by Artur Ekert in 1991. The setup requires a source that emits entangled EPR pairs (usually $|\beta_{11}\rangle$) and sends one particle to Alice and one to Bob.



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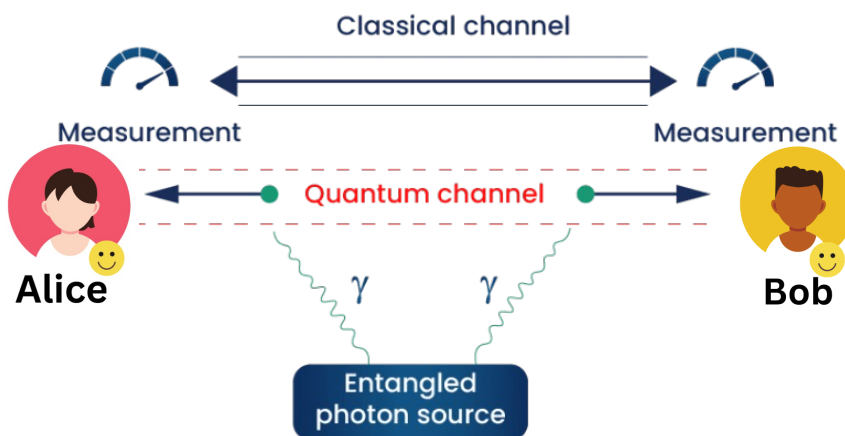
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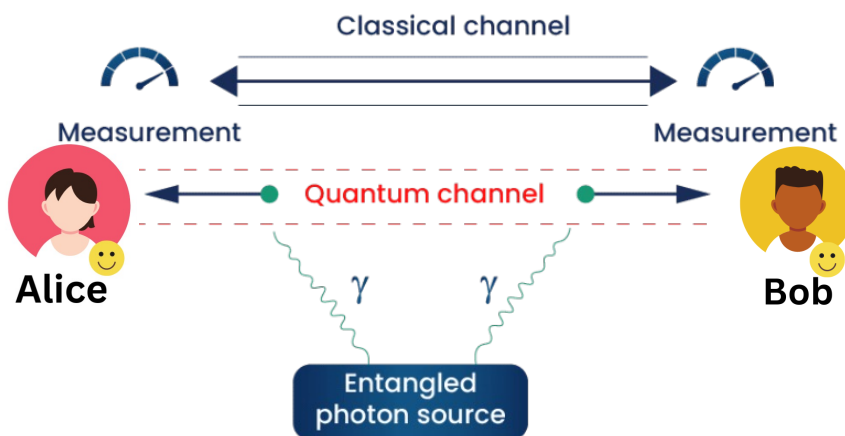


Alice and Bob each have a set of three measurement bases. For every incoming particle, they randomly and independently choose one of their three bases to measure.

After measuring a large number of pairs, Alice and Bob communicate over a public (classical) channel to reveal which bases they used for each measurement, without revealing the actual results. They divide their results into two groups: Matching Bases and Different Bases.

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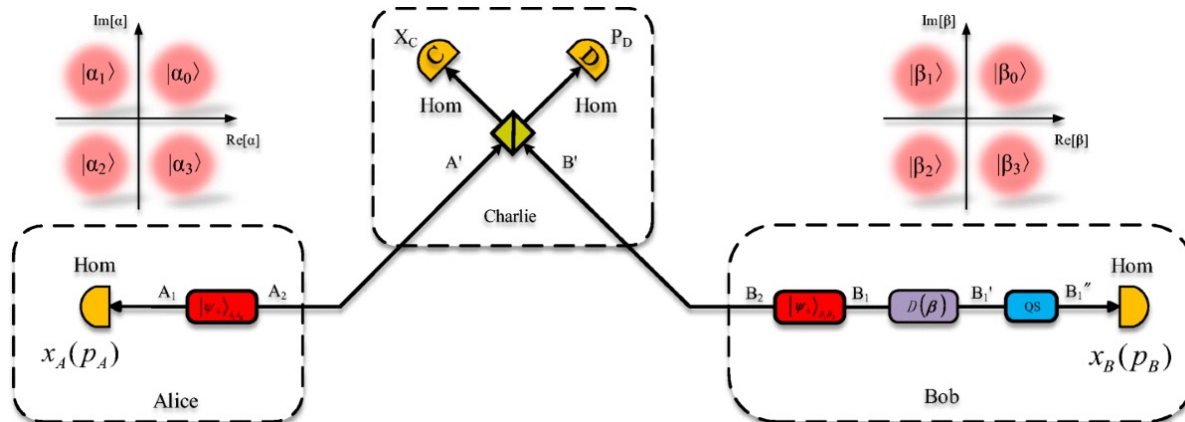
Alice and Bob each have a set of three measurement bases. For every incoming particle, they randomly and independently choose one of their three bases to measure.

In the cases where their bases matched, their results will be perfectly anti-correlated. By flipping his bits, Bob now has a string of data identical to Alice's, and this becomes their Raw Key.

::... E91 Protocol

E91 is the "gold standard" for security, but it is currently too difficult for most commercial applications.

- E91 is rarely used in standard fiber networks because maintaining entangled pairs over long distances is incredibly difficult. Most commercial systems prefer BB84 or its variants because they use single photons, which are easier to handle.
- If E91 is the "theoretical ideal," the following protocols are the "practical successors" that actually run on metropolitan networks:
 - MDI-QKD (Measurement-Device-Independent) - Alice and Bob both send photons to a third party (Charlie) who performs the measurement. Even if Charlie is evil or his equipment is bugged,

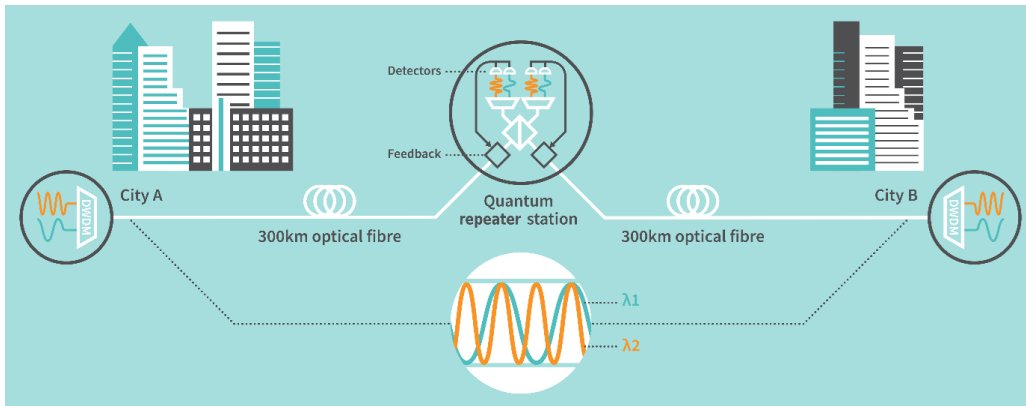


he cannot learn the key. It provides Alice and Bob with "E91-level" security against detector attacks without a high-quality entanglement source.

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 - TF-QKD (Twin-Field QKD) - Instead of sending a whole particle from Alice to Bob, they send two half-pulses that meet in the middle.

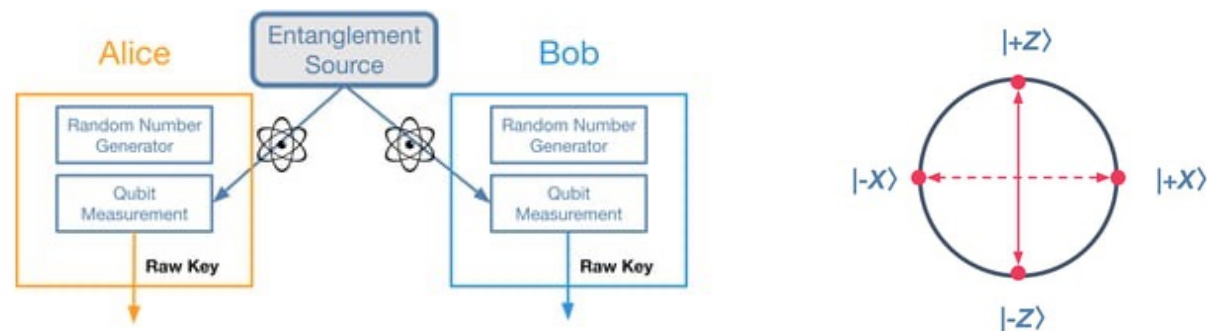


Why it matters: It overcomes the standard loss limits of fiber optic cables that usually kill quantum signals after 100 km.

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 - BBM92 (Bennett-Brassard-Mermin) - While E91 uses three bases to test Bell's Inequality constantly, BBM92 uses only two. It's easier to implement but requires you to trust that your measurement devices haven't been tampered with.



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- If E91 is the "theoretical ideal," the following protocols are the "practical successors" that actually run on metropolitan networks:
 - PQC-QKD Hybrids - QKD (like E91 or MDI) provides the "physical" security, while PQC provides the "mathematical" security. Together, they protect against "Harvest Now, Decrypt Later" attacks.

